The CMB temperature angular power spectrum through fourth order in a radiation-only universe: exact observable, initial data, perturbative kernels and harmonic machinery

(collaborative derivation; see the verification ledger, App. D)

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Abstract

We set up the calculation of the observed temperature angular power spectrum $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}] + O(\epsilon^6)$ in a spatially flat, radiation-only universe governed by general relativity with $\Lambda = 0$, for photons emitted on the physical hypersurface $T_{\rm phys} = T_*$ and observed by a tagged fluid element at fixed proper time. The settled parts are presented here: (i) an exact, non-perturbative reformulation of the measurement, $T_{\rm obs}(\hat{\mathbf{n}}) = T_o\, e^{-S(\hat{\mathbf{n}})}$ with S the fluid shear integrated along the ray, which contains all Sachs–Wolfe, Doppler, integrated Sachs–Wolfe, lensing, time-delay, emission-surface and observer physics; (ii) the definition of the Gaussian non-linear primordial variable and the construction of regular initial data satisfying the constraints at every order; (iii) the perturbative solution of the Einstein–fluid system through third order in Poisson gauge (closed first-order transfer functions, second-order scalar/vector/tensor kernels and Green's functions, third-order contracted kernels); (iv) the exact normalisation bookkeeping and the harmonic-space machinery needed to turn direction-space kernels into $C_\ell^{(22)}$ and $C_\ell^{(13)}$. Sections that assemble the explicit second- and third-order maps, the final C_ℓ, the checks and the monochromatic reduction are marked as placeholders to be supplied. Every result quoted here has been verified symbolically or numerically; the verification ledger lists each check.

Contents

1 Setup and conventions **3**

2 The exact observable **5**
2.1 Conformal frame . . . 5
2.2 Two lemmas . . . 5
2.3 The theorem . . . 6
2.4 Observer, clock, emission event and direction map . . . 7
2.5 Linear order: recovery of the Sachs–Wolfe formula . . . 7

3 Primordial variable and initial data **8**
3.1 The non-linear primordial variable . . . 8
3.2 Regularity and initial data at finite η_i . . . 9

3.3 The conditions at second and third order . . . 9
3.4 The exact dilation family and the zero-mode (tadpole) ambiguity . . . 10

4 Perturbative solution of the Einstein–fluid equations 11
4.1 First order . . . 11
4.2 Second order: operator structure . . . 12
4.3 Third order: contracted kernels . . . 14

5 Normalisation, loop structure and harmonic machinery 15
5.1 Normalisation bookkeeping (exact algebra) . . . 15
5.2 Loop structure . . . 15
5.3 Direction convention for the harmonic rules . . . 16
5.4 Type A: linear line-of-sight maps . . . 16
5.5 Type B: products of maps at the same point . . . 17
5.6 Type C: convolution kernels . . . 17
5.7 The (13) term . . . 18
5.8 Control case: lensed C_ℓ . . . 19
5.9 Monochromatic reduction: definitions . . . 19

6 First order: transfer function $\Delta_\ell(k)$ and $C_\ell^{(11)}$ 20

7 The second-order map $\Theta^{(2)}$ 23
7.1 Ray, triad and direction map . . . 24
7.2 Observer event: clock and displacement . . . 25
7.3 Emission event . . . 26
7.4 Expansion about the background ray . . . 26
7.5 L_2: named pieces . . . 26
7.6 Direct (Sachs–Wolfe) form and equivalence . . . 27
7.7 From L_2 to $\Theta^{(2)}$ . . . 28

8 The third-order map $\Theta^{(3)}$ 28
8.1 L_3: named groups . . . 29
8.2 From L_3 to $\Theta^{(3)}$ . . . 29

9 Assembly of C_ℓ 31

10 Checks of the assembled observable 31

11 Monochromatic primordial spectrum 31

12 Unresolved steps 31

A Explicit kernels (machine generated) 31
A.1 Second-order kernels . . . 32
A.2 Homogeneous solutions and Green's functions . . . 33
A.3 Third-order initial amplitude and intermediate limits . . . 34
A.4 Tadpole (mean-field) functions . . . 34
A.5 Not reproduced here . . . 35

B Line-of-sight term lists of L_2 and L_3 **35**
B.1 $\frac{1}{2}L_2$, shear form, $\hat{\mathbf{n}}=\hat{\mathbf{z}}$: all 34 terms . . . 35
B.2 $\frac{1}{2}L_2$, direct (redshift) form, $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ . . . 36
B.3 $\frac{1}{6}L_3$: group lists . . . 38

C Machine-generated channel kernels of K_2 **45**

D Verification ledger **47**

1 Setup and conventions

Model. Spatially flat FRW background, general relativity with $\Lambda = 0$, signature $(-+++)$, and a single perfect radiation fluid

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + P g^{\mu\nu}, \qquad P = \rho/3, \qquad \rho = \alpha\, T_{\rm phys}^4, \tag{1}$$

with α constant and $u^\mu u_\mu = -1$. The second relation *defines* the physical temperature $T_{\rm phys}$; no chemical potential or particle-number assumption is made. Test photons are emitted isotropically with a Planck spectrum of temperature T_* on the hypersurface $T_{\rm phys} = T_*$, then propagate collisionlessly without backreaction.

Background and units. Conformal time η, $\mathrm{d}s^2 = a^2(\eta)\,\hat{g}_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu$. In radiation domination $a \propto \eta$; we fix the normalisation $a(\eta) = \eta$, so that $\mathcal{H} \equiv a'/a = 1/\eta$ and the Friedmann equation $3\mathcal{H}^2 = 8\pi G a^2 \bar\rho$ reads

$$8\pi G\, a^4 \bar\rho = 3, \qquad \text{i.e.} \qquad 8\pi G\, \bar\rho = 3/\eta^4. \tag{2}$$

Background temperature $\bar T(\eta) = \bar T_i\, \eta_i/\eta$. The *conformal* fluid variables

$$\hat\rho \equiv a^4\rho, \qquad \hat P \equiv a^4 P, \qquad \hat T \equiv a\, T_{\rm phys}, \qquad \hat u^\mu \equiv a\, u^\mu \quad (\hat g_{\mu\nu}\hat u^\mu \hat u^\nu = -1), \tag{3}$$

are constant in the background ($\hat{\bar\rho} = 3/8\pi G$, $\hat{\bar T} = \text{const}$) and are the natural variables of Sec. 2.

Fourier conventions and statistics.

$$f(\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, e^{i\mathbf{k}\cdot\mathbf{x}} f_{\mathbf{k}}, \qquad \langle \zeta_{\mathbf{k}}\zeta_{\mathbf{k}'}\rangle = (2\pi)^3\delta^3(\mathbf{k}+\mathbf{k}')\,P(k), \qquad \langle\zeta^2(\mathbf{x})\rangle = \int \frac{\mathrm{d}^3k}{(2\pi)^3}P(k) = \frac{1}{2\pi^2}\int k^2\mathrm{d}k\,P(k). \tag{4}$$

Products in position space become convolutions, $(fg)_{\mathbf{k}} = \int \mathrm{d}^3q/(2\pi)^3\, f_{\mathbf{q}}\, g_{\mathbf{k}-\mathbf{q}}$, and $\partial_i \to ik_i$, $\nabla^2 \to -k^2$, $\nabla^{-2} \to -1/k^2$. $\langle\cdot\rangle$ denotes the ensemble average over the Gaussian field ζ_1 (Sec. 3); $\langle\cdot\rangle_{\rm sky}$ denotes the solid-angle average over the observer's sky.

Perturbative expansion. Every perturbation $X \in \{\Psi, \Phi, \omega_i, h_{ij}, \delta, v^i, \ldots\}$ is expanded as

$$X = \epsilon X_1 + \frac{\epsilon^2}{2}X_2 + \frac{\epsilon^3}{6}X_3 + O(\epsilon^4), \tag{5}$$

and the same convention is used for every derived quantity (S, Λ, m, Θ, $\zeta_{\rm nl}$). For an equation $E[X] = 0$ we write $E^{(n)} \equiv n!\,[\epsilon^n]E$, so that X_n enters $E^{(n)}$ linearly with unit weight. The bookkeeping parameter ϵ is the amplitude of ζ_1; the final expansion parameter is $\epsilon^2\langle\zeta_1^2\rangle$. Products of lower-order quantities are never dropped.

Poisson gauge. The conformal metric is parametrised *linearly* (no exponentials),

$$\hat{g}_{00} = -(1+2\Psi), \qquad \hat{g}_{0i} = \omega_i, \qquad \hat{g}_{ij} = (1-2\Phi)\delta_{ij} + h_{ij}, \qquad \partial_i\omega_i = 0, \quad h_{ii} = 0 = \partial_i h_{ij}, \tag{6}$$

with ω_i and h_{ij} starting at order ϵ^2 (no primordial vector or tensor modes, Sec. 3). Indices on ω_i, h_{ij}, v^i are raised and lowered with δ_{ij}. The fluid is parametrised by

$$\rho = \bar{\rho}(\eta)\,(1+\delta), \qquad 1+\delta_T \equiv (1+\delta)^{1/4} = \hat{T}/\hat{\bar{T}}, \qquad \hat{u}^\mu = \hat{u}^0\,(1, v^i), \tag{7}$$

with $\hat{u}^0$ fixed by the normalisation $\hat{g}_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = -1$ as a series in ϵ. Irrotationality of v^i is *not* imposed (Sec. 4.2). The Einstein equations are $E_{\mu\nu} \equiv G_{\mu\nu}[g] - 8\pi G\,T_{\mu\nu} = 0$ with $g = \eta^2\hat{g}$ and $8\pi G\,T_{\mu\nu} = (3/\eta^4)(1+\delta)[\frac{4}{3}u_\mu u_\nu + \frac{1}{3}g_{\mu\nu}]$; the conservation laws $\nabla_\mu T^{\mu\nu} = 0$ are implied and are used as checks. Dictionary to the exponential parametrisation of [8, 9, 6] ($\hat{g}_{00} = -e^{2\Phi_{\rm e}}$, $\hat{g}_{ij} = e^{-2\Psi_{\rm e}}\delta_{ij} + \ldots$, note the swapped names): $\Phi_{\rm e}^{(2)} = \Psi_2 - 2\Psi_1^2$, $\Psi_{\rm e}^{(2)} = \Phi_2 + 2\Phi_1^2$ at second order.

Directions, rays and times. $\hat{\mathbf{n}}$ is the direction in which the observer looks (unit vector with respect to the observer's orthonormal triad, Sec. 2.4); the photon propagation direction at the observer is $\hat{\mathbf{e}} = -\hat{\mathbf{n}}$. The background ray through the observer event $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$ is

$$\bar{\mathbf{x}}(\eta) = \bar{\mathbf{x}}_o + \hat{\mathbf{n}}\,\chi(\eta), \qquad \chi(\eta) \equiv \bar{\eta}_o - \eta, \qquad \frac{\mathrm{d}\bar{\mathbf{x}}}{\mathrm{d}\eta} = -\hat{\mathbf{n}}, \tag{8}$$

and the total derivative along it acting on a field is $D \equiv \partial_\eta - \hat{\mathbf{n}}\cdot\nabla$. Four times are kept distinct: the *initial* time η_i at which initial data are posed ($\bar{T}(\eta_i) = T_i$); the background *emission* time η_* defined by $\bar{T}(\eta_*) = T_*$, and the actual emission time $\eta_e(\hat{\mathbf{n}}) = \eta_* + \delta\eta_e(\hat{\mathbf{n}})$ along each ray; the *observation* time $\eta_o = \bar{\eta}_o + \delta\eta_o$ of the observer (direction independent, fixed by proper time); and the dummy time η along the ray. The ordering is $0 < \eta_i < \eta_* < \bar{\eta}_o$. The angular quantities are

$$a_{\ell m}^{(r)} = \int \mathrm{d}\Omega_{\hat{\mathbf{n}}}\, Y_{\ell m}^*(\hat{\mathbf{n}})\,\Theta^{(r)}(\hat{\mathbf{n}}), \qquad C_\ell^{(rs)} = \frac{1}{2\ell+1}\sum_{m=-\ell}^{\ell}\langle a_{\ell m}^{(r)} a_{\ell m}^{(s)*}\rangle_c, \tag{9}$$

with $Y_{\ell m}$ carrying the Condon–Shortley phase, and the target is

$$C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4\Big[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}\Big] + O(\epsilon^6), \qquad \ell \geq 2. \tag{10}$$

Relation to existing work. No complete $O(\epsilon^4)$ (one-loop) computation of the temperature power spectrum exists in the literature: the only (22)-type temperature calculation [23] is for dust in synchronous gauge (ISW terms only) and explicitly leaves out the (13) term for lack of third-order metric perturbations; third-order temperature anisotropies have been studied on large scales in matter domination [24, 25]; second-order Boltzmann codes [20, 22] and the (22)-type B-mode spectrum [21] exist, as do the second-order Sachs–Wolfe analyses [18, 19, 8, 9, 10] which, "following standard practice", drop the observer terms; only [6, 7] keep them, for long-wavelength modes. The lensed C_ℓ at $O(C^\phi)$ [16, 17] is the (22) + (13) structure restricted to lensing and serves as our control case. The radiation-era second-order vector and tensor sectors are treated in [15] and [12, 13, 14]; general second-order perturbation theory is reviewed in [11].

2 The exact observable

2.1 Conformal frame

Write $g_{\mu\nu} = a^2 \hat{g}_{\mu\nu}$ with $a = \eta$. Null geodesics of g and $\hat{g}$ coincide as curves; if $p^\mu = \mathrm{d}x^\mu/\mathrm{d}\lambda$ is affine for g then $\hat{p}^\mu = a^2 p^\mu$ is affine for $\hat{g}$. With the conformal fluid variables (3),

$$E \equiv -u_\mu p^\mu = \hat{E}/a, \qquad \hat{E} \equiv -\hat{g}_{\mu\nu}\hat{u}^\mu \hat{p}^\nu. \tag{11}$$

The radiation stress tensor is traceless, hence conformally covariant: $\hat{T}^{\mu\nu} \equiv a^6 T^{\mu\nu} = (\hat{\rho}+\hat{P})\hat{u}^\mu\hat{u}^\nu + \hat{P}\hat{g}^{\mu\nu}$ with $\hat{\rho} = \alpha\hat{T}^4$, $\hat{P} = \hat{\rho}/3$, and $\hat{\nabla}_\mu \hat{T}^{\mu\nu} = 0$ *exactly*, where $\hat{\nabla}$ is the Levi-Civita connection of $\hat{g}$. Everything in this section therefore holds verbatim for the hatted system, whose background is Minkowski space with a constant fluid; we drop the hats in the lemmas (they hold for any metric) and restore them when we parametrise.

2.2 Two lemmas

Let u^μ be a unit timelike field and decompose its gradient in the standard way,

$$\nabla_\nu u_\mu = \sigma_{\mu\nu} + \omega_{\mu\nu} + \frac{1}{3}\theta\, h_{\mu\nu} - u_\nu a_\mu, \qquad h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu, \quad a_\mu = u^\nu\nabla_\nu u_\mu, \quad \theta = \nabla_\mu u^\mu, \tag{12}$$

with $\sigma_{\mu\nu}$ the symmetric trace-free part and $\omega_{\mu\nu}$ the antisymmetric part of $h_\mu{}^\alpha h_\nu{}^\beta \nabla_\beta u_\alpha$ [2]. For a null vector p^μ write $p^\mu = E(u^\mu + e^\mu)$ with $E = -u\cdot p$, $u\cdot e = 0$, $e\cdot e = 1$, and define the proper-length element $\mathrm{d}l \equiv E\,\mathrm{d}\lambda$ along the ray.

Lemma 2.1 (kinematic redshift identity)**.** *Along an affinely parametrised null geodesic, $p^\nu\nabla_\nu p^\mu = 0$,*

$$\frac{\mathrm{d}\ln E}{\mathrm{d}l} = -\Big(\frac{\theta}{3} + \sigma_{\mu\nu}e^\mu e^\nu + a_\mu e^\mu\Big). \tag{13}$$

Proof. $\mathrm{d}E/\mathrm{d}\lambda = p^\nu\nabla_\nu(-u_\mu p^\mu) = -p^\mu p^\nu \nabla_\nu u_\mu$. Insert (12): the σ term gives $E^2\sigma_{ee}$ ($\sigma \perp u$); the ω term vanishes identically (antisymmetric tensor contracted with $p^\mu p^\nu$), so *vorticity drops out with no assumption*; the θ term gives $\frac{1}{3}\theta E^2 h_{ee} = \frac{1}{3}\theta E^2$; the acceleration term gives $-E^2[u\cdot(u+e)][a\cdot(u+e)] = E^2 a\cdot e$. Divide by E^2. □

Lemma 2.2 (radiation-fluid identities)**.** *For the fluid (1), the conservation law $\nabla_\mu T^{\mu\nu} = 0$ alone (no field equations) is equivalent to*

$$\theta = -3\,u^\mu\partial_\mu \ln T, \qquad a_\mu = -h_\mu{}^\nu \partial_\nu \ln T. \tag{14}$$

Proof. $u_\nu\nabla_\mu T^{\mu\nu} = 0$ gives $u\cdot\partial\rho + (\rho+P)\theta = 0$ and $h^\nu{}_\lambda \nabla_\mu T^{\mu\lambda} = 0$ gives $(\rho+P)a^\nu + h^{\nu\mu}\partial_\mu P = 0$. With $\rho = \alpha T^4$, $P = \rho/3$ one has $\rho + P = \frac{4}{3}\alpha T^4 = T\,\mathrm{d}P/\mathrm{d}T$ and $\mathrm{d}\rho/\mathrm{d}T = 3(\rho+P)/T$, which turn the two equations into (14). Both ingredients are needed: for $P = w\rho$, $\rho \propto T^4$ one finds $a = -\frac{4w}{1+w} h\,\partial\ln T$ and $\theta = -\frac{4}{1+w} u\cdot\partial\ln T$, which reduce to (14) iff $w = \frac{1}{3}$ (App. D, V1-2). □

2.3 The theorem

Theorem 2.3. *Let the photon be emitted at the event e where its past-directed null geodesic last meets the hypersurface $T = T_*$, with an isotropic Planck spectrum in the fluid rest frame, and let it be received at the event o by an observer comoving with the fluid. Then the observed temperature in the direction $\hat{\mathbf{n}}$ is*

$$\boxed{T_{\rm obs}(\hat{\mathbf{n}}) = T_* \frac{E_o}{E_e} = T_o \, \exp[-S(\hat{\mathbf{n}})], \qquad S(\hat{\mathbf{n}}) \equiv \int_e^o \sigma_{\mu\nu} e^\mu e^\nu \, \mathrm{d}l} \tag{15}$$

where T_o is the fluid temperature at o and the integral runs along the ray.

Proof. Since $e^\mu h_\mu{}^\nu = e^\nu$, Lemma 2.2 gives $\theta/3 + a\cdot e = -(u+e)\cdot\partial \ln T = -E^{-1} p\cdot\partial \ln T = -\mathrm{d}\ln T/\mathrm{d}l$. Lemma 2.1 then reads $\mathrm{d}(\ln E - \ln T)/\mathrm{d}l = -\sigma_{ee}$; p is future directed, so λ increases from e to o and $\ln(E_o/E_e) = \ln(T_o/T_e) - S$. With $T_e = T_*$ this is $T_* E_o/E_e = T_o e^{-S}$. Finally, E_o/E_e is independent of the photon energy (the geodesic equation is homogeneous in p), so by Liouville's theorem a Planck spectrum at T_* in the fluid frame at e is received as a Planck spectrum at $T_* E_o/E_e$ in the fluid frame at o; this is what $T_{\rm obs}(\hat{\mathbf{n}})$ means. □

Remark 2.4 (conformal invariance). Theorem 2.3 applies to (g, u, T) and to $(\hat{g}, \hat{u}, \hat{T})$ alike. Since $E = \hat{E}/a$ and $T = \hat{T}/a$ with the same a_e, a_o, the two statements agree iff $S = \hat{S}$: the shear integral is conformally invariant and may be computed entirely in the hatted frame, where the background shear vanishes identically.

Corollary 2.5. *With $\Theta(\hat{\mathbf{n}}) \equiv T_{\rm obs}(\hat{\mathbf{n}})/\langle T_{\rm obs}\rangle_{\rm sky} - 1$ and $\langle\cdot\rangle_{\rm sky} = \dfrac{1}{4\pi}\int \mathrm{d}\Omega$ in the observer's rest frame,*

$$1 + \Theta(\hat{\mathbf{n}}) = \frac{e^{-S(\hat{\mathbf{n}})}}{\langle e^{-S}\rangle_{\rm sky}} \qquad \textit{exactly.} \tag{16}$$

Consequences.

(i) All of the usual emission-surface (intrinsic temperature and $T_{\rm phys} = T_*$ shift), Sachs–Wolfe, Doppler, integrated Sachs–Wolfe, lensing and time-delay physics is contained in S: the only inputs are the fluid shear field $\sigma_{\mu\nu}$, the ray $x^\mu(\lambda)$, and the two endpoints. In particular the perturbation of the emission point along the ray does *not* require a separate "intrinsic δT" term; $T_e = T_*$ has been used exactly.

(ii) T_o is a scalar at the observer event, independent of $\hat{\mathbf{n}}$; it drops out of Θ identically. No expansion of T_o or of $\langle T_{\rm obs}\rangle_{\rm sky}$ in terms of a background temperature is ever needed.

(iii) The observer's tetrad. Fermi–Walker transport of an orthonormal tetrad along the observer's worldline defines the sky directions. The FW tetrad is a realisation-dependent rigid rotation $R = \exp(\epsilon A)$ of the *canonical triad* (Sec. 2.4). A rigid rotation of the sky leaves $\sum_m |a_{\ell m}|^2$ invariant realisation by realisation, hence C_ℓ is independent of the tetrad orientation. However, because R depends on ϵ, $a^{(2)}_{\ell m} \to a^{(2)}_{\ell m} + 2A\, a^{(1)}_{\ell m}$ etc., so $C^{(22)}_\ell$ and $\mathrm{Re}\, C^{(13)}_\ell$ are *not separately* invariant; only $\dfrac{1}{4}C^{(22)}_\ell + \dfrac{1}{3}\mathrm{Re}\, C^{(13)}_\ell$ is (verified explicitly at $\ell = 2$, App. D, V1-5). Consequently $\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}$ must all be computed in one and the same frame; we use the canonical triad throughout. The dipole power $\sum_m |a_{1m}|^2$ is likewise invariant, so "remove the dipole after normalising" is frame independent.

(iv) The emission surface enters only through the endpoint e of the integral: the location of the last crossing of $T = T_*$ along the past ray. Perturbatively there is a single crossing and $\eta_e(\hat{\mathbf{n}}) = \eta_* + \delta\eta_e(\hat{\mathbf{n}})$.

Remark 2.6 (validity). The proof uses the differential form of $\nabla \cdot T = 0$ and C^1 fields along the ray; it is not defined across a shock (irrelevant perturbatively). Isotropic Planck emission in the fluid frame at $T = T_*$ is self-consistent (the photons *are* the fluid there); after emission the "fluid" (u, T) entering S is the model's perfect-fluid continuation, as the problem statement prescribes. The identities also hold for $\rho = 3P + c$, $P = CT^4 - c/4$ (a Λ-like constant), not needed here.

2.4 Observer, clock, emission event and direction map

Observer worldline: a tagged fluid element, $\mathbf{x}_o(\eta) = \mathbf{x}_o^{\rm lab} + \int_{\eta_{\rm init}}^{\eta} v^i(\eta', \mathbf{x}_o(\eta'))\,\mathrm{d}\eta'$. *Clock*: proper time elapsed since the observer crossed the initial-data surface $T_{\rm phys} = T_i$ (Poisson time $\eta_{\rm init}(\mathbf{x}_o)$ with $1 + \delta_T(\eta_{\rm init}, \mathbf{x}_o) = \eta_{\rm init}/\eta_i$, since $\bar{T} \propto 1/\eta$),

$$\Delta\tau = \int_{\eta_{\rm init}}^{\eta_o} \frac{a(\eta)}{\hat{u}^0(\eta, \mathbf{x}_o(\eta))}\,\mathrm{d}\eta \quad \text{held fixed} \qquad \Longrightarrow \qquad \eta_o = \bar{\eta}_o + \delta\eta_o[\zeta], \tag{17}$$

with $\delta\eta_o$ direction independent (it is a functional of the fields on the worldline). *Emission event*: the ray meets $T_{\rm phys} = T_*$, i.e.

$$1 + \delta_T(\eta_e, \mathbf{x}(\eta_e)) = \eta_e/\eta_*. \tag{18}$$

Canonical triad: $\{e_a\}_{a=1,2,3}$ is the Gram–Schmidt orthonormalisation (order $1, 2, 3$) of the coordinate basis ∂_i with respect to $\hat{g}$, orthogonal to $\hat{u}$. *Direction map*: the photon received from the sky direction $\hat{\mathbf{n}}$ has

$$\hat{p}_o = \hat{E}_o(\hat{u}_o - n^a e_a), \qquad k^i(\eta_o) \equiv \frac{\mathrm{d}x^i}{\mathrm{d}\eta}\Big|_o = \frac{\hat{p}_o^i}{\hat{p}_o^0}, \tag{19}$$

so that in the background $k^i = -n^i$ and the ray is (8). With the ray parametrised by η, $k^\mu = \mathrm{d}x^\mu/\mathrm{d}\eta = (1, k^i)$, and using $\sigma \perp u$, $\hat{E}/\hat{p}^0 = -\hat{u}_\mu k^\mu$,

$$S(\hat{\mathbf{n}}) = \int_{\eta_e(\hat{\mathbf{n}})}^{\eta_o} \mathrm{d}\eta\, \frac{\hat{\sigma}_{\mu\nu} k^\mu k^\nu}{-\hat{u}_\mu k^\mu}\Big|_{x^i(\eta)}. \tag{20}$$

The perturbative evaluation of (20) (ray, endpoints, triad, clock, all to third order) is the subject of Secs. 7–8.

2.5 Linear order: recovery of the Sachs–Wolfe formula

At $O(\epsilon)$ the endpoints and the ray may be taken at their background values, since S itself starts at $O(\epsilon)$. With $\hat{g} = \mathrm{diag}(-(1 + 2\epsilon\Psi), (1 - 2\epsilon\Phi)\delta_{ij})$, $\hat{u}^\mu = (1 - \epsilon\Psi, \epsilon v^i)$, $\hat{T} = \bar{\hat{T}}(1 + \epsilon\delta_T)$, the covariant definitions (12) give (App. D, V1-4)

$$\hat{\theta} = \epsilon(\partial\cdot v - 3\Phi'), \qquad \hat{a}_i = \epsilon(v_i' + \partial_i\Psi), \qquad \hat{\sigma}_{ij} = \epsilon[\partial_{(i}v_{j)} - \frac{1}{3}\delta_{ij}\partial\cdot v], \qquad \hat{\sigma}_{0\mu} = 0, \tag{21}$$

and Lemma 2.2 becomes the linearised continuity and Euler equations

$$\delta_T' = \Phi' - \frac{1}{3}\partial\cdot v, \qquad v_i' = -\partial_i(\Psi + \delta_T). \tag{22}$$

With $e = -\hat{\mathbf{n}}$ and $D = \partial_\eta - \hat{\mathbf{n}}\cdot\nabla$ along the background ray one has $n^i n^j \partial_i v_j = \hat{\mathbf{n}}\cdot v' - D(\hat{\mathbf{n}}\cdot v)$, $\hat{\mathbf{n}}\cdot\nabla f = f' - Df$; inserting (22),

$$\hat{\sigma}_{1,ee} = n^i n^j \partial_i v_j - \frac{1}{3}\partial\cdot v = D(\delta_T + \Psi - \hat{\mathbf{n}}\cdot v) - (\Phi' + \Psi') \qquad \text{(on shell)}. \tag{23}$$

Integrating along the ray from $(\eta_*, \bar{\mathbf{x}}(\eta_*))$ to $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$,

$$\boxed{L_1 \equiv -S_1 = [\delta_T + \Psi - \hat{\mathbf{n}}\cdot v]_e - [\delta_T + \Psi - \hat{\mathbf{n}}\cdot v]_o + \int_{\eta_*}^{\bar{\eta}_o} (\Phi' + \Psi')\,\mathrm{d}\eta} \tag{24}$$

which is the standard Sachs–Wolfe [1] + Doppler + integrated Sachs–Wolfe formula with the emission-surface shift (the δ_T at e arises because $\bar{T} \propto 1/\eta$ and the emission surface is $T_{\rm phys} = T_*$, not $\eta = \eta_*$) and with the observer terms: a monopole $-[\delta_T + \Psi]_o$ and the observer dipole $+\hat{\mathbf{n}}\cdot v_o$. Both are removed from $\Theta^{(1)}$ by the normalisation and the dipole subtraction, but they multiply other terms at higher order and are kept (Sec. 5.1). Equation (23) has been verified symbolically and the same identity holds through third order at the integrand level (App. D, C2-1).

3 Primordial variable and initial data

3.1 The non-linear primordial variable

Definition 3.1. In Poisson gauge, with the non-linear δ_T and Φ of (7), (6),

$$\zeta_{\rm nl}(\mathbf{x}) \equiv \lim_{\eta\to 0}\Big[\ln\left(1 + \delta_T(\eta,\mathbf{x})\right) + \frac{1}{2}\ln\left(1 - 2\Phi(\eta,\mathbf{x})\right)\Big] = \lim_{\eta\to 0}\Big[\frac{1}{4}\ln(1+\delta) + \frac{1}{2}\ln(1-2\Phi)\Big]. \tag{25}$$

Why this is the uniform-density curvature perturbation. Since $\rho = \bar{\rho}(\eta)(1+\delta)$ with $\bar{\rho} \propto \eta^{-4}$, the uniform-density hypersurface $\rho = \bar{\rho}(\tilde{\eta})$ is the surface of Poisson time $\eta_P(\mathbf{x}) = \tilde{\eta}\,[1 + \delta_T(\eta_P, \mathbf{x})]$. Its induced metric is $\gamma_{ij} = g_{ij} + g_{00}\partial_i\eta_P\partial_j\eta_P + 2g_{0(i}\partial_{j)}\eta_P$ with $g = \eta_P^2\hat{g}$, i.e.

$$\gamma_{ij} = \tilde{\eta}^2(1+\delta_T)^2\Big[(1-2\Phi)\delta_{ij} + h_{ij} + O(\tilde{\eta}^2)\Big], \tag{26}$$

because $\partial_i\eta_P = \tilde{\eta}\,\partial_i\delta_T = O(\tilde{\eta})$. For regular solutions (below) $h_{ij} \to 0$ as $\tilde{\eta} \to 0$ (the induced tensors are sourced and vanish at $\eta = 0$; there are no free tensor modes), so $\gamma_{ij} \to a(\tilde{\eta})^2 e^{2\zeta_{\rm nl}}\delta_{ij}$: $\zeta_{\rm nl}$ is exactly the quantity ζ of Salopek–Bond and Maldacena, $h_{ij} = a^2 e^{2\zeta}(e^\gamma)_{ij}$ on uniform-density slices with $\gamma = 0$ [3, 4]. (On these slices the fluid is also comoving in the limit, since the regular velocity potential vanishes as $\eta \to 0$, Sec. 4.1.) The spatial coordinates are those of Poisson gauge; the residual freedom of the conformally flat form, a constant rescaling $\mathbf{x} \to B\mathbf{x}$, is the zero-mode ambiguity discussed in Sec. 3.4.

Definition 3.2 (Gaussian initial conditions; absence of higher-order primordial modes)**.**

$$\zeta_{\rm nl} = \epsilon\,\zeta_1 \quad \text{exactly}, \qquad \zeta_1 \text{ a Gaussian random field with spectrum } P(k), \tag{27}$$

i.e. in the expansion (5) of $\zeta_{\rm nl}$, $\zeta^{(2)}(\eta \to 0) = 0$ and $\zeta^{(3)}(\eta \to 0) = 0$ *identically in* $\mathbf{x}$. This is the statement that the exponentiated uniform-density curvature perturbation is Gaussian [4]; in the language of [10] it is $\mathcal{R}^{(2)} = -2\mathcal{R}^{(1)2}$ ($f_{\rm NL} = 0$ for the variable appearing in $e^{2\zeta}$), and it fixes the second- and third-order scalar fields at $\eta \to 0$ completely, with no freedom to add independent second- or third-order scalar, vector or tensor modes.

3.2 Regularity and initial data at finite η_i

Regular means: at every order the solution of the Einstein–fluid system is analytic in η at $\eta = 0$. At first order this discards the decaying mode $\propto (\cos x + x \sin x)/x^3 \sim \eta^{-3}$ ($x = k\eta/\sqrt{3}$); at second and third order it discards the homogeneous scalar decaying modes and all homogeneous vector and tensor modes, and selects the particular solutions that vanish at $\eta = 0$ (Sec. 4.2). The regular solution is uniquely fixed by (27).

Initial data at the finite time η_i ($\bar{T}(\eta_i) = T_i$) are *defined* as the restriction of the regular solution to $\eta = \eta_i$. They satisfy the Hamiltonian and momentum constraints at every order by construction, since the regular solution solves all ten Einstein equations for $\eta > 0$ and the constraints propagate (contracted Bianchi identity, verified to $O(\epsilon^3)$ by the engine, App. D, C1-5). Concretely, every Green's-function integral below may be written either as $\int_0^\eta$ or as $\int_{\eta_i}^\eta$ plus the homogeneous solution matched to the (regular) data at η_i; the two are identical. We keep η_i explicit where the problem statement asks for it and note that $\eta_i \to 0$ reproduces the $\int_0^\eta$ form. This is how the instruction "do not indiscriminately set higher-order perturbations to zero" is implemented: $X_2(\eta_i)$, $X_3(\eta_i)$ are non-zero, non-local functionals of ζ_1 determined by (27) and the constraints.

3.3 The conditions at second and third order

Expanding (25) with (5) (engine output, App. D, C1-8),

$$\zeta^{(1)} = \frac{1}{4}\delta_1 - \Phi_1 = \delta_{T,1} - \Phi_1, \tag{28}$$

$$\zeta^{(2)} = \frac{1}{4}\delta_2 - \frac{1}{4}\delta_1^2 - \Phi_2 - 2\Phi_1^2, \tag{29}$$

$$\zeta^{(3)} = \frac{1}{4}\delta_3 - \frac{3}{4}\delta_1\delta_2 + \frac{1}{2}\delta_1^3 - \Phi_3 - 6\Phi_1\Phi_2 - 8\Phi_1^3. \tag{30}$$

At $\eta \to 0$ the first-order regular solution has (Sec. 4.1) $\Phi_1 = \Psi_1 \to -\frac{2}{3}\zeta_1$, $\delta_1 \to \frac{4}{3}\zeta_1$, $v_1 \to 0$.

Second order. In the kernel notation of Sec. 4.2 ($X_{2,\mathbf{k}} = \int \frac{\mathrm{d}^3q}{(2\pi)^3} K_X(\mathbf{q}, \mathbf{k}-\mathbf{q}; \eta)\zeta_{\mathbf{q}}\zeta_{\mathbf{k}-\mathbf{q}}$, $\mathbf{p} \equiv \mathbf{k} - \mathbf{q}$, local products of first-order fields being constant kernels at $\eta \to 0$), the 00 constraint gives $\delta_2(0) = \frac{1}{3}\lim(\eta^2 S_{00}) - 2\Psi_2(0)$ with $\lim_{\eta\to 0}\eta^2 K_{S_{00}} = \frac{32}{3}$, the traceless ij equation gives $\Psi_2(0) = \Phi_2(0) - K_Q(0)$ with

$$K_Q(\mathbf{q}, \mathbf{p}; 0) = -\frac{13k^4 - 6k^2(p^2+q^2) + 9(p^2-q^2)^2}{9k^4} = -\frac{4}{9}\,\frac{4k^2 - 6kq\mu + 9q^2\mu^2 - 3q^2}{k^2}, \qquad \mu \equiv \hat{\mathbf{k}}\cdot\hat{\mathbf{q}}, \tag{31}$$

and $\zeta^{(2)}(0) = 0$ then yields

$$\boxed{K_{\Phi_2}(\mathbf{q}, \mathbf{p}; 0) = \frac{1}{3}K_Q(\mathbf{q}, \mathbf{p}; 0) - \frac{8}{27} = -\frac{7k^4 - 2k^2(p^2+q^2) + 3(p^2-q^2)^2}{9k^4} = -\frac{4}{9}\,\frac{2k^2 - 2kq\mu + 3q^2\mu^2 - q^2}{k^2}} \tag{32}$$

(the reader may verify: $\frac{1}{4}[\frac{32}{9} + 2K_Q(0) - 2\Phi_2] - \frac{1}{4}\cdot\frac{16}{9} - \Phi_2 - 2\cdot\frac{4}{9} = 0$). In position space, with $\Phi_1 = -\frac{2}{3}\zeta_1$,

$$\Phi_2(0) = -2\Phi_1^2 - \nabla^{-2}(\partial_i\Phi_1\partial_i\Phi_1) + 3\nabla^{-4}\partial_i\partial_j(\partial_i\Phi_1\partial_j\Phi_1), \tag{33}$$

$$\Psi_2(0) - \Phi_2(0) = 4\Phi_1^2 + 3\nabla^{-2}(\partial_i\Phi_1\partial_i\Phi_1) - 9\nabla^{-4}\partial_i\partial_j(\partial_i\Phi_1\partial_j\Phi_1), \tag{34}$$

and $\delta_2(0)$, $v_2(0)$ follow from the constraints (App. A). Equations (33)–(34) agree *exactly* with the radiation-era superhorizon initial conditions of Pitrou–Uzan–Bernardeau [10] (their (3.12b,c) with $\mathcal{R}^{(2)} = -2\mathcal{R}^{(1)2}$, after exchanging their names for the two potentials). Bartolo–Matarrese–Riotto [9] (their (7.30)) agree in the local and gradient terms but differ in the velocity-quadrupole term, which they evaluate with the late-time velocity rather than its $\eta \to 0$ limit (App. D, C1-11).

Third order. Only the q-contracted third-order fields are needed for $C_\ell^{(13)}$ (Sec. 4.3). With $\Phi_{3c}(k,\eta) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, K_{\Phi_{3c}}(k,q,\mu;\eta)$, the regular solution has $\Phi_{3c}(k,0) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, C_3(k;q,\mu)$ and $\zeta^{(3)}(0) = 0$ gives

$$\boxed{C_3(k;q,\mu) = -\frac{16(2k^5 + 12k^4q\mu - 42k^3q^2\mu^2 - 10k^3q^2 + 36k^2q^3\mu^3 + 32k^2q^3\mu + 3kq^4\mu^4 - 42kq^4\mu^2 + kq^4 + 8q^5\mu)}{9\,k\,(k^2 - 2kq\mu + q^2)^2}} \tag{35}$$

($p^2 = k^2 - 2kq\mu + q^2$; invariant form and the intermediate $Q_{3c}(0)$, $\lim \eta^2 K_{3c,S_{00}}$ in App. A). The p^{-4} denominator is the Poisson inversion of the traceless equation; it is integrable over μ at $q \to k$ (the numerator vanishes there to the required order).

3.4 The exact dilation family and the zero-mode (tadpole) ambiguity

The flat FRW solution written in rescaled coordinates $\tilde\eta = A\eta$, $\tilde{\mathbf{x}} = B\mathbf{x}$ (A, B constants) is, in the $(\eta, \mathbf{x})$ chart, the exact Poisson-gauge configuration

$$1 + 2\Psi = A^4, \qquad 1 - 2\Phi = A^2B^2, \qquad 1 + \delta = A^{-4}, \qquad v^i = 0, \qquad \zeta_{\rm nl} = \ln B, \tag{36}$$

which the engine verifies to solve $E_{\mu\nu} = 0$ through $O(\epsilon^3)$ for arbitrary A, B (App. D, C1-6). It is the $k = 0$ member of the solution space: a constant $\zeta_1 = c$ corresponds to $B = e^{\epsilon c}$ (so that $\zeta_{\rm nl} = \epsilon c$ exactly) and, by (28), $a_1 = -c/3$, giving $\Psi_1 = \Phi_1 = -\frac{2}{3}c$, $\delta_1 = \frac{4}{3}c$ in agreement with the $k \to 0$ limit of the first-order transfer functions. At second order, however, the $k = 0$ equations fix only $\delta_2 = -2\Psi_2 + \frac{32}{9}c^2$ and leave Φ_2, Ψ_2 free: this is the residual gauge freedom of Poisson gauge at zero momentum (the time rescaling A, cf. Weinberg's adiabatic modes [5, 6]). Correspondingly, the $k \to 0$ limit of the finite-k kernels (31)–(32) at $\mathbf{p} = -\mathbf{q}$ is direction dependent ($\propto 3\mu^2 - 1$) and its angular average,

$$\langle \Phi_2(0) \rangle_{\rm ang} = -\frac{8}{9}c^2, \qquad \langle \Psi_2(0) \rangle_{\rm ang} = +\frac{8}{9}c^2, \qquad \delta_2 = \frac{16}{9}c^2, \tag{37}$$

lies on the family (36) with $A = e^{-\epsilon c/3}$, $B = e^{\epsilon c}$, i.e. $1 + \delta_T = e^{\zeta/3}$, $1 - 2\Phi = e^{4\zeta/3}$, $1 + 2\Psi = e^{-4\zeta/3}$ (the member with $\Psi_2 = \Phi_2$ instead has $\Phi_2 = \Psi_2 = -\frac{8}{27}c^2$, $\delta_2 = \frac{112}{27}c^2$).

This matters because the third-order map contains *mean-field* (tadpole) terms $\langle X_2 \rangle = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q) K_X(\mathbf{q}, -\mathbf{q}; \eta)$, the $\mathbf{k} = 0$ components of the second-order fields (Sec. 4.3). They are evaluated with the angle-averaged convention (closed forms $\mathrm{tad}_X(q,\eta)$ in App. A; e.g. $\mathrm{tad}_{\Phi_2} \to -\frac{8}{9}$, $\mathrm{tad}_{Q_2} \to -\frac{16}{9}$, $\mathrm{tad}_{\delta_2} \to \frac{16}{9}$ as $\eta \to 0$). Since the whole $k = 0$ family is a coordinate transformation and the observable is defined by physical labels only (fluid worldline, proper time from the $T_{\rm phys} = T_i$ crossing, emission on $T_{\rm phys} = T_*$, sky-monopole normalisation), a constant ζ_1 must be unobservable: the full map must be invariant under c at every order, which fixes the squeezed limits of K_2 and K_3 and forces

object	defined in	meaning
$x = k\eta/\sqrt{3}$	(40)	dimensionless sound-horizon variable
$T_\Phi, T_\delta, T_{\delta_T}, T_V$	(41)	first-order transfer functions ($X_1 = \zeta_1 T_X$)
$\mathcal{L}_\Phi, \mathcal{L}_h, \mathcal{L}_\omega$	(44)	universal linear operators (same at every order)
$S^{(n)}_{\mu\nu}$; $S_{\rm TL}, S^L_0, S^T_{0i}, S^V_{ij}, S^{TT}_{ij}, S_\Phi$	(42), (43)	order-n sources and their SVT projections
$K_X(\mathbf{q},\mathbf{p};\eta)$, $\mathbf{p} = \mathbf{k} - \mathbf{q}$	(46)	second-order kernel of X per $\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}$
$K_Q, K_{S_{00}}, K_{S_{0z}}, K_{S_\Phi}, K_\omega, K_{v^T}, K_h, K^0_{\Phi_2}$	(47), App. A	the explicit second-order kernels
$u_1, u_2, \hat{g}, G_\Phi, G_h$	(48)	homogeneous solutions and Green's functions
$\mathcal{K}_{\Phi_2}, \mathcal{K}_h$	(49)	declared time integrals for Φ_2, h_2
$X_{3c}(k,\eta)$, $K_{3c,X}(k,q,\mu;\eta)$	(50)	q-contracted third-order fields and kernels
$\mathrm{tad}_X(q,\eta)$	(52)	mean-field (tadpole) functions $\langle X_2\rangle$
$C_3(k;q,\mu)$	(35)	third-order superhorizon amplitude of Φ_{3c}

Table 1: Named objects of the perturbative solution.

the tadpole convention to drop out of $\Theta^{(3)}$ and hence of $C_\ell^{(13)}$. This is the dilation check, to be performed on the assembled map (Sec. 10); at the level of the field equations the consistency is already verified by the squeezed limit of (35),

$$\lim_{q\to 0} C_3(k;q,\mu) = -\frac{32}{9}, \tag{38}$$

which is precisely the $c^2\zeta_s$ coefficient $6 \times \frac{1}{2}(\frac{4}{3}c)^2 \times (-\frac{2}{3})$ of the short mode modulated by the long-mode member $A = e^{-\epsilon c/3}$, $B = e^{\epsilon c}$ ($1 - 2\Phi = A^2B^2(1 - 2\Phi_s)$, App. D, C1-14).

4 Perturbative solution of the Einstein–fluid equations

This section summarises the solution of $E_{\mu\nu} = G_{\mu\nu}[\eta^2\hat{g}] - 8\pi G\, T_{\mu\nu} = 0$ through $O(\epsilon^3)$ in the parametrisation of Sec. 1, obtained with a dedicated truncated-polynomial engine (code `cmb4/code/C1_einstein/`, App. D, C1-$*$). The Christoffel symbols, Ricci and Einstein tensors are computed directly from $g = \eta^2\hat{g}$ as series in ϵ; the background part of every component vanishes identically and the contracted Bianchi identity $g^{\mu\alpha}\nabla_\alpha E_{\mu\nu} + g_{\nu\beta}\nabla_\mu T^{\mu\beta} = 0$ holds identically through $O(\epsilon^3)$. Table 1 collects the named objects used in the rest of the paper; each is defined once in the text and the long expressions are in App. A.

4.1 First order

In the frame $\mathbf{k} = k\hat{\mathbf{z}}$ the first-order equations are

$$\begin{aligned}
E^{(1)}_{ij}\ (i \neq j,\ \text{traceless}): &\quad -k^2(\Phi_1 - \Psi_1) = 0 \ \Rightarrow\ \Psi_1 = \Phi_1,\\
E^{(1)}_{0i}: &\quad 2ik_i(\Phi_1' + \Psi_1/\eta) + 4v_{1,i}/\eta^2 = 0 \ \Rightarrow\ v_1^T = 0, \quad v_{1,i} = \partial_i V_1,\ V_1 = -\frac{\eta^2}{2}(\Phi_1' + \Psi_1/\eta),\\
E^{(1)}_{00}: &\quad \delta_1 = -\frac{2}{3}\eta^2k^2\Phi_1 - 2\eta\Phi_1' - 2\Psi_1,\\
\text{trace}: &\quad \Phi_1'' + \frac{4}{\eta}\Phi_1' + \frac{k^2}{3}\Phi_1 = 0.
\end{aligned} \tag{39}$$

The velocity is irrotational and the conservation equations hold identically on this solution (App. D, C1-3). With

$$x \equiv k\eta/\sqrt{3} \tag{40}$$

the regular solution, normalised by $\zeta_1 = \lim_{\eta\to 0}(\delta_{T,1} - \Phi_1)$, is $X_1 = \zeta_1 T_X(x)$ with

$$\begin{aligned}
T_\Phi(x) &= -\frac{2(\sin x - x\cos x)}{x^3} \xrightarrow{x\to 0} -\frac{2}{3}, \qquad \Psi_1 = \Phi_1, \\
T_\delta(x) &= 4T_{\delta_T}(x) = -4\cos x + \frac{8\sin x}{x} + \frac{8\cos x}{x^2} - \frac{8\sin x}{x^3} \;\to\; \frac{4}{3}, \\
T_V(x) &= \eta\,\frac{x^2\sin x + 2x\cos x - 2\sin x}{x^3} \;\to\; \frac{\eta}{3} \qquad (v_{1,i} = \partial_i V_1,\; V_1 = \zeta_1 T_V).
\end{aligned} \tag{41}$$

The second solution $(\cos x + x\sin x)/x^3 \sim \eta^{-3}$ is discarded by regularity. Note $\delta_{T,1} \to \zeta_1/3$, so $\zeta_1 = \delta_{T,1} - \Phi_1 = \frac{1}{3}\zeta_1 + \frac{2}{3}\zeta_1$ as it should. We write $T \equiv T_\Phi$, $\dot T \equiv \partial_\eta T_\Phi$, and $T_q \equiv T_\Phi(q\eta/\sqrt{3})$ etc. in the kernels.

4.2 Second order: operator structure

At order $n \geq 2$ every component of $E^{(n)}_{\mu\nu} = n!\,[\epsilon^n]E_{\mu\nu}$ splits as

$$E^{(n)}_{\mu\nu} = \mathrm{Lin}_{\mu\nu}[X_n] + S^{(n)}_{\mu\nu}, \tag{42}$$

where Lin is the *same* linear operator as in (39) and $S^{(n)}_{\mu\nu}$ is built from lower-order fields ($S^{(2)}$ quadratic in X_1; $S^{(3)}$ contains X_1X_2 and X_1^3 terms; position-space expressions with 6–20 resp. 71–127 terms per component are in `position_sources.tex`). Decomposing $S^{(n)}_{\mu\nu}$ into scalar, vector and tensor parts (frame $\mathbf{k} = k\hat{\mathbf{z}}$, $\mathbf{q}$ in the x–z plane),

$$S_{\rm TL} \equiv S_{zz} - \frac{1}{2}(S_{xx}+S_{yy}), \quad S_0^L \equiv S_{0z}, \quad S_{0i}^T \equiv (S_{0x}, S_{0y}), \quad S^V \equiv S_{xz}, \quad S^{TT} \equiv S_{xx} - S_{yy}, \quad S_{\rm tr} \equiv S_{ii}, \tag{43}$$

the ten equations become (all verified by the engine):

$$\begin{aligned}
\text{traceless scalar:} &\quad Q_n \equiv \Phi_n - \Psi_n = S_{\rm TL}/k^2 \quad \big(= -\frac{3}{2}\nabla^{-2}(\partial_i\partial_j\nabla^{-2} - \frac{1}{3}\delta_{ij})S_{ij} \text{ in position space}\big), \\
00: &\quad \delta_n = \frac{\eta^2}{3}S_{00} - \frac{2}{3}\eta^2k^2\Phi_n - 2\eta\Phi_n' - 2\Psi_n, \\
0i \text{ longitudinal:} &\quad v^L_{n,i} = \partial_i V_n, \qquad V_n = -\frac{\eta^2}{4}\Big[2(\Phi_n' + \Psi_n/\eta) + \frac{S_0^L}{ik}\Big], \\
0i \text{ transverse:} &\quad \frac{1}{2}\nabla^2\omega_{n,i} - \frac{4}{\eta^2}(\omega_{n,i} + v^T_{n,i}) = S^T_{0i} \quad [8\pi G a^2(\rho + P) = 4/\eta^2], \\
ij \text{ vector:} &\quad \mathcal{L}_\omega\,\omega_n = \frac{2}{ik}S^V, \qquad \mathcal{L}_\omega \equiv \partial_\eta + \frac{2}{\eta}, \\
ij \text{ TT:} &\quad \mathcal{L}_h\,h_{n,ij} = -2\Lambda_{ij,kl}(\hat{\mathbf{k}})S_{kl}, \qquad \mathcal{L}_h \equiv \partial_\eta^2 + \frac{2}{\eta}\partial_\eta + k^2, \\
\text{trace:} &\quad \mathcal{L}_\Phi\,\Phi_n = S_\Phi \equiv -\frac{S_{\rm tr} - S_{00}}{6} + \frac{Q_n'}{\eta} - \frac{k^2}{3}Q_n, \qquad \mathcal{L}_\Phi \equiv \partial_\eta^2 + \frac{4}{\eta}\partial_\eta + \frac{k^2}{3},
\end{aligned} \tag{44}$$

where $\Lambda_{ij,kl} = P_{ik}P_{jl} - \frac{1}{2}P_{ij}P_{kl}$, $P_{ij} = \delta_{ij} - \hat k_i\hat k_j$, and in the trace equation δ_n and Ψ_n have been eliminated with the 00 and traceless equations. Thus at every order the scalar sector reduces to *one* forced oscillator $\mathcal{L}_\Phi\Phi_n = S_\Phi$, after which Ψ_n, δ_n, V_n are algebraic; the vector sector to a first-order equation; the tensor sector to a forced wave equation.

Vector sector (relativistic Kelvin theorem). Vorticity-freedom of v^i is *not* implied at second order. What the equations imply (verified: the algebraic ω_2 so obtained satisfies the ij-vector equation identically) is the circulation theorem for the isentropic radiation fluid, whose enthalpy per particle is $\propto T$:

$$\partial_{[i}[(1+\delta_T)\,\hat{u}_{j]}] = 0. \tag{45}$$

Consequently ω_2 is *algebraic* in the first-order fields and v_2^T follows from the transverse $0i$ constraint; no vector Green's function is needed (the vector ODE has $G_\omega = (\eta'/\eta)^2$ and gives the same result).

Kernels. Second-order fields are written as

$$X_{2,\mathbf{k}}(\eta) = \int \frac{\mathrm{d}^3q}{(2\pi)^3}\, K_X(\mathbf{q},\mathbf{p};\eta)\,\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}, \qquad \mathbf{p}\equiv\mathbf{k}-\mathbf{q}, \qquad \mu \equiv \hat{\mathbf{k}}\cdot\hat{\mathbf{q}} = \frac{k^2+q^2-p^2}{2kq}, \tag{46}$$

and all scalar kernels are given in the *invariant form* $K_X(k,q,p;\eta)$ (functions of the three magnitudes), equivalently in the (k,q,μ) form. With $T_q = T_\Phi(q\eta/\sqrt{3})$, $\dot{T}_q = \partial_\eta T_q$, the second-order kernels obtained from (44) with $S^{(2)}$ are (full forms in App. A):

$$\begin{aligned}
K_Q &= T_qT_p\Big[-4+\frac{6\mu q}{k}+\frac{3q^2}{k^2}-\frac{9\mu^2q^2}{k^2}\Big] + \Big[\eta(T_p\dot{T}_q+T_q\dot{T}_p)+\eta^2\dot{T}_q\dot{T}_p\Big]\Big[\frac{2\mu q}{k}+\frac{q^2}{k^2}-\frac{3\mu^2q^2}{k^2}\Big],\\
K_\omega &= \frac{2i}{k^2}\Big[\eta\,(q^2-p^2)\,T_qT_p+\eta^2(q^2T_q\dot{T}_p-p^2T_p\dot{T}_q)\Big] \quad (\omega_{2,i} = \int P_{ij}(\hat{\mathbf{k}})q_j\,K_\omega\,\zeta\zeta;\ \text{antisymmetric}\\
K_h &= 4\Big[3T_qT_p+\eta(\dot{T}_qT_p+T_q\dot{T}_p)+\eta^2\dot{T}_q\dot{T}_p\Big] = 4\Big[2T_qT_p+(\eta\dot{T}+T)_q(\eta\dot{T}+T)_p\Big],
\end{aligned}$$

$$K_{S_{00}},\ K_{S_{0z}},\ K_{S_\Phi},\ K_{v^T}:\ \text{App. A}; \qquad v^T_{2,i} = \int P_{ij}q_jK_{v^T}\zeta\zeta, \qquad K_{v^T} = -\Big(\frac{\eta^2k^2}{8}+1\Big)K_\omega - \frac{\eta^2}{4}K_{S^T_{0i}}. \tag{47}$$

The tensor source K_h is identical to the radiation-era kernel of Kohri–Terada [12] (their eq. (8)) and Baumann et al. [13] (their (11)–(12)) with $w=\frac{1}{3}$, $\Phi=\Psi$ (App. D, C1-9). K_{S_Φ} is finite as $\eta\to0$ (no η^{-2} terms), so the particular integral of the trace equation vanishes at $\eta=0$; $\lim_{\eta\to0}\eta^2K_{S_{00}} = \frac{32}{3}$ and $K_Q(\eta\to0)$ is (31). The vector and tensor sources vanish as $\mathbf{p}\to-\mathbf{q}$ ($k\to0$) because of the projectors, which is what makes the $K\to0$ corner of the (22) loop integrable (Sec. 5.9).

Green's functions and the regular solution. With $x=k\eta/\sqrt{3}$, $x'=k\eta'/\sqrt{3}$,

$$\begin{aligned}
u_1(x) &= \frac{\sin x - x\cos x}{x^3}, \qquad u_2(x) = \frac{\cos x + x\sin x}{x^3}, \qquad \hat{g}(x)\equiv 3u_1(x) = -\frac{3}{2}T_\Phi(x)\ \to 1, \qquad W\equiv u_1u_2' - u\\
G_\Phi(k;\eta,\eta') &= \frac{u_1(x')u_2(x)-u_1(x)u_2(x')}{W(\eta')} = \frac{\sqrt{3}}{k}\,\frac{x'}{x^3}\Big[(1+xx')\sin(x-x')-(x-x')\cos(x-x')\Big] \xrightarrow{k\to0} \frac{\eta'}{3}\Big(1-\frac{\eta'^3}{\eta^3}\Big)\\
G_h(k;\eta,\eta') &= \frac{\eta'}{\eta}\,\frac{\sin k(\eta-\eta')}{k},
\end{aligned} \tag{48}$$

satisfying $\mathcal{L}G=0$, $G(\eta,\eta)=0$, $\partial_\eta G|_{\eta'=\eta}=1$. The regular second-order scalar and tensor fields are then

$$\begin{aligned}
\Phi_{2,\mathbf{k}}(\eta) &= \int\frac{\mathrm{d}^3q}{(2\pi)^3}\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}\,\mathcal{K}_{\Phi_2}(k,q,p;\eta), \qquad \mathcal{K}_{\Phi_2}\equiv\int_0^\eta \mathrm{d}\eta'\,G_\Phi(k;\eta,\eta')K_{S_\Phi}(k,q,p;\eta') + K^0_{\Phi_2}(k,q,p)\,\hat{g}(k\eta),\\
h_{2,ij,\mathbf{k}}(\eta) &= \Lambda_{ij,kl}(\hat{\mathbf{k}})\int\frac{\mathrm{d}^3q}{(2\pi)^3}q_kq_l\,\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}\,\mathcal{K}_h(k,q,p;\eta), \qquad \mathcal{K}_h\equiv\int_0^\eta\mathrm{d}\eta'\,G_h(k;\eta,\eta')K_h(k,q,p;\eta'),
\end{aligned} \tag{49}$$

with $K^0_{\Phi_2}$ the initial amplitude (32) fixed by Gaussianity, and $\Psi_2 = \Phi_2 - Q_2$, δ_2, V_2 algebraic from (44). The lower limit 0 may be replaced by η_i together with the homogeneous solution matched to the regular data at η_i (Sec. 3.2). The time integrals $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$ are [unevaluated integral: $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$; closed forms in terms of Si, Ci of the Kohri–Terada type are being derived separately and will be quoted in Sec. 9].

Checks at second order. Bianchi propagation of the constraints (C1-5); exact dilation family (36) (C1-6); tensor source = Kohri–Terada (C1-9); superhorizon Φ_2, Ψ_2 = Pitrou–Uzan–Bernardeau (C1-10); vorticity structure (45) (C1-7). An independent referee re-derivation (agent V3) is in progress and will be reported in Sec. 10.

4.3 Third order: contracted kernels

The (13) spectrum needs only the correlation of a third-order field with ζ_1, i.e. the *q-contracted* third-order field,

$$\langle X_{3,\mathbf{k}}\,\zeta_{\mathbf{k}'}\rangle = (2\pi)^3\delta^3(\mathbf{k}+\mathbf{k}')\,P(k)\,X_{3c}(k,\eta), \qquad X_{3c}(k,\eta) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\,K_{3c,X}(k,q,\mu;\eta), \quad (50)$$

where the kernel sums the three Wick pairings (it is *not* symmetrised under $\mathbf{q}\to-\mathbf{q}$, so only $p=|\mathbf{k}-\mathbf{q}|$ appears). The contracted source $K_{3c,S}$ is built from $S^{(3)}$ by the rules

$$A_1B_1C_1 \;\to\; A_qB_{-q}C_k + A_qC_{-q}B_k + B_qC_{-q}A_k, \qquad A_1B_2 \;\to\; 2A_q\,K_B(-\mathbf{q},\mathbf{k}) + A_k\,\langle B_2\rangle(\eta), \quad (51)$$

in which $Q_2, \delta_2, V_2, v_2^T, \omega_2$ enter through their explicit kernels and Φ_2, h_2 through the declared integrals $\mathcal{K}_{\Phi_2}(a,b,c;\eta)$, $\mathcal{K}_h(a,b,c;\eta)$ with $(a,b,c) = (|\mathbf{q}_1|,|\mathbf{q}_2|,|\mathbf{q}_1+\mathbf{q}_2|)$ kept as symbols. The mean-field terms

$$\langle X_2\rangle(\eta) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\,\mathrm{tad}_X(q,\eta), \qquad \mathrm{tad}_X(q,\eta) \equiv \langle K_X(\mathbf{q},-\mathbf{q};\eta)\rangle_{\mathrm{ang}}, \quad (52)$$

vanish for vectors, tensors and velocities by isotropy; for the scalars $Q_2, S_{00}, S_\Phi, \Phi_2, \delta_2$ the angle-averaged $\mathbf{k}\to 0$ limits are given in closed form in App. A (e.g. $\mathrm{tad}_{\Phi_2}(q,\eta) = -\frac{8}{9} + \int_0^\eta \frac{\eta'}{3}(1-\frac{\eta'^3}{\eta^3})\mathrm{tad}_{S_\Phi}(q,\eta')\mathrm{d}\eta' = -\frac{8}{9} - \frac{28}{405}q^2\eta^2 + \ldots$), with the zero-mode caveat of Sec. 3.4.

The contracted third-order equations have the same operators (44):

$$Q_{3c} = K_{3c,Q}, \qquad \delta_{3c} = \frac{\eta^2}{3}K_{3c,S_{00}} - \frac{2}{3}\eta^2k^2\Phi_{3c} - 2\eta\Phi'_{3c} - 2\Psi_{3c}, \qquad V_{3c} = -\frac{\eta^2}{4}\Big[2(\Phi'_{3c}+\Psi_{3c}/\eta) + \frac{K_{3c,S_{0z}}}{ik}\Big],$$
$$\mathcal{L}_\Phi\Phi_{3c} = K_{3c,S_\Phi}, \qquad \Phi_{3c}(k,\eta) = \int\frac{\mathrm{d}^3q}{(2\pi)^3}P(q)\Big[\int_0^\eta \mathrm{d}\eta'\,G_\Phi(k;\eta,\eta')K_{3c,S_\Phi}(k,q,\mu;\eta') + C_3(k;q,\mu)\,\hat g(k\eta)\Big], \quad (53)$$

with C_3 the amplitude (35) fixed by $\zeta^{(3)}(0)=0$ and K_{3c,S_Φ} finite at $\eta\to 0$. The contracted vector and tensor sources $(S_{0x}, S_{xz}, S_{xx}-S_{yy})$ are produced as well but integrate to zero over the $\mathbf{q}$ angles by isotropy (S_{0y}, S_{yz}, S_{xy} vanish identically). The kernels $K_{3c,X}$ have 157–830 terms and are not reproduced; they are stored in machine-readable form (`third_order.pkl`, loader `kernels_api.load()['third']`, LaTeX dump `third_order.tex`). The η' integral in (53) is [unevaluated integral: $\int_0^\eta G_\Phi K_{3c,S_\Phi}$, which itself contains $\mathcal{K}_{\Phi_2}$ and $\mathcal{K}_h$ at the lower-order momenta]. The un-contracted kernel $K_3(\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3)$ is not needed for $C_\ell^{(13)}$ and has not been produced.

Summary of what is settled. First order: closed forms (41). Second order: all kernels in closed form except the two time integrals $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$, which are declared. Third order: contracted sources and amplitude C_3 in closed form; Φ_{3c} declared as one further nested time integral. The zero-mode convention for the tadpoles is a declared convention whose irrelevance for the observable is to be shown in Sec. 10.

5 Normalisation, loop structure and harmonic machinery

5.1 Normalisation bookkeeping (exact algebra)

Define $\Lambda \equiv -S$ and $M \equiv e^{\Lambda}$, expanded as in (5):

$$\Lambda = \epsilon L_1 + \frac{\epsilon^2}{2} L_2 + \frac{\epsilon^3}{6} L_3, \qquad M = 1 + \epsilon m_1 + \frac{\epsilon^2}{2} m_2 + \frac{\epsilon^3}{6} m_3, \qquad m_1 = L_1, \quad m_2 = L_2 + L_1^2, \quad m_3 = L_3 + 3L_1L_2 + L_1^3. \tag{54}$$

By Corollary 2.5, $1 + \Theta = M/\bar{M}$ with $\bar{M} \equiv \langle M \rangle_{\rm sky}$. Writing $\langle m \rangle_{\rm sky}$ for the sky monopole of a map ($\langle m \rangle_{\rm sky} = m_{00}/\sqrt{4\pi}$, a random variable correlated with the map) and expanding $1/\bar{M}$, one finds for $\ell \geq 1$

$$\Theta^{(1)}_{\ell m} = m_{1,\ell m}, \qquad \Theta^{(2)}_{\ell m} = m_{2,\ell m} - 2\langle m_1 \rangle_{\rm sky} m_{1,\ell m}, \qquad \Theta^{(3)}_{\ell m} = m_{3,\ell m} - 3\langle m_1 \rangle_{\rm sky} m_{2,\ell m} - 3\langle m_2 \rangle_{\rm sky} m_{1,\ell m} + 6\langle m_1 \rangle^2_{\rm sk} \tag{55}$$

(the monopole $\ell = 0$ of Θ vanishes identically). The measured dipole is removed afterwards, i.e. (10) is used for $\ell \geq 2$ only; by Sec. 2(iii) this is frame independent. All observer terms (the monopole $-[\delta_T + \Psi]_o$ and the dipole $\hat{\mathbf{n}} \cdot v_o$ of L_1, the clock shift $\delta\eta_o$, the observer position) are retained inside L_1, L_2, L_3 and are removed only through (55); products such as $\langle m_1 \rangle_{\rm sky} m_1$ and L_1^2 keep their imprint at $\ell \geq 2$.

5.2 Loop structure

Write each order as a direction-space kernel acting on the Gaussian field,

$$\Theta^{(r)}(\hat{\mathbf{n}}) = \int \prod_{i=1}^{r} \frac{\mathrm{d}^3 k_i}{(2\pi)^3}\, \zeta_{\mathbf{k}_1} \cdots \zeta_{\mathbf{k}_r}\, K_r(\mathbf{k}_1, \ldots, \mathbf{k}_r; \hat{\mathbf{n}}), \qquad K_{r,\ell m}(\mathbf{k}_1, \ldots) \equiv \int \mathrm{d}\Omega_{\hat{\mathbf{n}}} Y^*_{\ell m}(\hat{\mathbf{n}}) K_r, \tag{56}$$

with K_r symmetric in its momenta (symmetrise if necessary). Wick's theorem for $\ell \geq 1$ then gives

$$\begin{aligned} C_\ell^{(11)} &= \frac{1}{2\ell+1} \sum_m \int \frac{\mathrm{d}^3 k}{(2\pi)^3} P(k)\, |K_{1,\ell m}(\mathbf{k})|^2, \\ C_\ell^{(22)} &= \frac{2}{2\ell+1} \sum_m \int \frac{\mathrm{d}^3 k_1\, \mathrm{d}^3 k_2}{(2\pi)^6} P(k_1) P(k_2)\, |K_{2,\ell m}(\mathbf{k}_1, \mathbf{k}_2)|^2, \\ C_\ell^{(13)} &= \frac{3}{2\ell+1} \sum_m \int \frac{\mathrm{d}^3 k}{(2\pi)^3} P(k)\, K_{1,\ell m}(\mathbf{k}) \Big[\int \frac{\mathrm{d}^3 q}{(2\pi)^3} P(q)\, K_{3,\ell m}(\mathbf{k}, \mathbf{q}, -\mathbf{q}) \Big]^*. \end{aligned} \tag{57}$$

The factor 2 in $C^{(22)}$ counts the two connected pairings of a symmetric K_2; the third pairing $\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \rangle$ inside $\Theta^{(2)}$ has $\mathbf{k}_1 + \mathbf{k}_2 = 0$ and feeds only $\ell = 0$. The factor 3 in $C^{(13)}$ counts the three choices of which ζ inside $\Theta^{(3)}$ pairs with $\Theta^{(1)}$; the remaining pair is contracted at $\mathbf{q}_j = -\mathbf{q}_k$, so the "tadpole" configuration $K_3(\mathbf{k}, \mathbf{q}, -\mathbf{q})$ *is* the entire (13) term and must be kept (this is where the mean-field terms of Sec. 4.3 enter). Both prefactors have been re-derived independently and

confirmed on the lensing control case below (App. D, C3-6). Equation (57) uses $\langle a^{(1)}a^{(3)*}\rangle$; only $\mathrm{Re}\,C_\ell^{(13)}$ enters (10). With the contracted fields of (50), $3\int P(q)K_3(\mathbf{k},\mathbf{q},-\mathbf{q})$ is the q-contracted map $\Theta_c^{(3)}(\mathbf{k};\hat{\mathbf{n}})$ built from X_{3c} and the contraction rules (51).

5.3 Direction convention for the harmonic rules

A field on the background ray (8) (observer at $\bar{\mathbf{x}}_o = 0$) reads $f(\eta,\hat{\mathbf{n}}\chi) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} f_{\mathbf{k}}\, e^{+i\mathbf{k}\cdot\hat{\mathbf{n}}\chi} = \int \frac{\mathrm{d}^3k}{(2\pi)^3} f_{\mathbf{k}}\, e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi}$ with $\hat{\mathbf{e}} = -\hat{\mathbf{n}}$ the propagation direction. The rules below (and the code `cmb4/code/C3_harmonic/`) are written for maps expressed through $\hat{\mathbf{e}}$, i.e. with the exponent $e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi}$. Since $Y_{\ell m}(-\hat{\mathbf{n}}) = (-1)^\ell Y_{\ell m}(\hat{\mathbf{n}})$, the multipoles of the same map regarded as a function of $\hat{\mathbf{n}}$ differ by $(-1)^\ell$ at *every* order, so all $C_\ell^{(rs)}$ are unchanged; one convention must simply be used consistently for $\Theta^{(1)},\Theta^{(2)},\Theta^{(3)}$ together. In this subsection and the next, $\hat{\mathbf{n}}$ in the kernels of (56) is to be read as $\hat{\mathbf{e}}$. Rayleigh: $e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi} = 4\pi\sum_{\ell m}(-i)^\ell j_\ell(k\chi)Y_{\ell m}(\hat{\mathbf{e}})Y^*_{\ell m}(\hat{\mathbf{k}})$. Gaunt integrals are

$$\mathcal{G}^{\ell\,\ell_1\ell_2}_{m\,m_1m_2} \equiv \int \mathrm{d}\Omega\, Y^*_{\ell m}Y_{\ell_1m_1}Y_{\ell_2m_2} = (-1)^m\sqrt{\frac{(2\ell+1)(2\ell_1+1)(2\ell_2+1)}{4\pi}}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ -m & m_1 & m_2\end{pmatrix}. \tag{58}$$

5.4 Type A: linear line-of-sight maps

A map of the form

$$\Theta_A(\hat{\mathbf{e}}) = \int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}\int_{\eta_*}^{\bar\eta_o}\mathrm{d}\eta\sum_{s\ge0} g_s(k,\eta)\,(i\,\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})^s\,e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi(\eta)} \tag{59}$$

(endpoint terms are $\delta(\eta-\eta_1)$ pieces of g_s; the monomials are written as $(i\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})^s$ because $i\hat{\mathbf{k}} = \nabla/k$, so a real map has real g_s) has

$$\text{(A-}s\text{)}\qquad a_{\ell m} = 4\pi(-i)^\ell\int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}Y^*_{\ell m}(\hat{\mathbf{k}})\,\Delta_\ell(k),\qquad \Delta_\ell(k) = \int\mathrm{d}\eta\sum_s g_s(k,\eta)\,(-1)^s j_\ell^{(s)}(k\chi(\eta)), \tag{60}$$

i.e. $(i\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})^s e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi} \to (-\mathrm{d}/\mathrm{d}x)^s j_\ell(x)$, $x = k\chi$. Helicity pieces (vectors ω_i, v_i^T contracted with $\hat{\mathbf{e}}$, tensors $h_{ij}\hat{\mathbf{e}}^i\hat{\mathbf{e}}^j$): in the frame $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ with $e^{(\pm1)} = (\hat{\mathbf{x}} \pm i\hat{\mathbf{y}})/\sqrt{2}$, $e^{(\pm2)}_{ij} = e^{(\pm1)}_i e^{(\pm1)}_j$, $Q_{\pm1} = \hat{\mathbf{e}}\cdot e^{(\pm1)}$, $Q_{\pm2} = e^{(\pm2)}_{ij}\hat{\mathbf{e}}^i\hat{\mathbf{e}}^j$,

$$\text{(A-}\lambda\text{)}\qquad Q_\lambda e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi} = \sum_\ell(-i)^\ell\sqrt{4\pi(2\ell+1)}\,\hat\Delta_\ell^{(\lambda)}(x)\,Y_{\ell\lambda}(\hat{\mathbf{e}}),\quad \hat\Delta_\ell^{(\pm1)} = \mp i\sqrt{\frac{\ell(\ell+1)}{2}}\frac{j_\ell(x)}{x},\quad \hat\Delta_\ell^{(\pm2)} = -\frac{1}{2}\sqrt{\frac{(\ell+2)}{(\ell-2)}} \tag{61}$$

and in a general frame $a^{(\lambda)}_{\ell m} = 4\pi(-i)^\ell\int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta^{(\lambda)}_{\mathbf{k}}(-1)^\lambda[{}_{-\lambda}Y_{\ell m}(\hat{\mathbf{k}})]^*\Delta^{(\lambda)}_\ell(k)$, $\Delta^{(\lambda)}_\ell = \int\mathrm{d}\eta\, g^{(\lambda)}\hat\Delta^{(\lambda)}_\ell(k\chi)$. Scalar, vector and tensor pieces of the same $\mathbf{k}$ do not cross-correlate. For two type-A maps

$$C_\ell^{AA'} = \frac{2}{\pi}\int k^2\mathrm{d}k\,P(k)\,\Delta^A_\ell(k)\Delta^{A'}_\ell(k)^*,\qquad \langle A\rangle_{\rm sky} = \int\frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}\Delta^A_0(k),\qquad \langle\langle A\rangle_{\rm sky}\langle A'\rangle_{\rm sky}\rangle = \frac{1}{2\pi^2}\int k^2\mathrm{d}k\,P\,\Delta^A_0\Delta^{A'}_0 \tag{62}$$

and $\langle\langle A\rangle_{\rm sky}\, a^{A'}_{\ell m}\rangle = 0$ for $\ell \ge 1$ by isotropy. The first-order map $m_1 = L_1$ of (24) is of type A (with $s \le 1$ and endpoint terms), so $C_\ell^{(11)}$ is (62) with $A = A' = m_1$ (closed form: Sec. 9).

5.5 Type B: products of maps at the same point

For $B = AA'$ and $B' = CD$ with A, A', C, D type-A maps, $b_{\ell m} = \sum \mathcal{G}^{\ell\ell_1\ell_2}_{mm_1m_2} w(\ell,\ell_1,\ell_2) a_{\ell_1 m_1} a'_{\ell_2 m_2}$, where $w = 1$ for a plain product and $w_\nabla = \frac{1}{2}[\ell_1(\ell_1+1) + \ell_2(\ell_2+1) - \ell(\ell+1)]$ for $\nabla_{\hat{\mathbf{n}}} A \cdot \nabla_{\hat{\mathbf{n}}} A'$ [16]. Wick's theorem with $\langle a^X_{\ell m} a^Y_{\ell' m'}\rangle = (-1)^m C_\ell^{XY} \delta_{\ell\ell'}\delta_{m,-m'}$ gives, for $\ell \geq 1$,

$$\text{(B1)} \qquad C_\ell^{BB'} = \sum_{\ell_1\ell_2} \frac{(2\ell_1+1)(2\ell_2+1)}{4\pi} \begin{pmatrix} \ell & \ell_1 & \ell_2 \\ 0 & 0 & 0 \end{pmatrix}^2 w\, w' \Big[C_{\ell_1}^{AC} C_{\ell_2}^{A'D} + C_{\ell_1}^{AD} C_{\ell_2}^{A'C} \Big], \tag{63}$$

with $\ell + \ell_1 + \ell_2$ even; *both* Wick pairings appear and reduce to a single term with a factor 2 only when $A = A'$ (the third pairing $\langle AA'\rangle\langle CD\rangle$ feeds $\ell = 0$ only). Monopole×map terms, with $\langle X\rangle_{\rm sky}\cdot 1$ treated as an $\ell = 0$ map, give for $\ell \geq 1$

$$\begin{aligned}
&\text{(i)} \quad \frac{1}{2\ell+1}\sum_m \langle |\langle X\rangle_{\rm sky}\, y_{\ell m}|^2\rangle = \langle \langle X\rangle^2_{\rm sky}\rangle\, C_\ell^{YY}, \\
&\text{(ii)} \quad \frac{1}{2\ell+1}\sum_m \langle (AA')_{\ell m} (\langle X\rangle_{\rm sky}\, y_{\ell m})^*\rangle = \langle \langle A\rangle_{\rm sky}\langle X\rangle_{\rm sky}\rangle C_\ell^{A'Y} + \langle \langle A'\rangle_{\rm sky}\langle X\rangle_{\rm sky}\rangle C_\ell^{AY}, \\
&\text{(iii)} \quad \frac{1}{2\ell+1}\sum_m \langle (\langle X\rangle_{\rm sky} y_{\ell m})(\langle Z\rangle_{\rm sky} w_{\ell m})^*\rangle = \langle \langle X\rangle_{\rm sky}\langle Z\rangle_{\rm sky}\rangle C_\ell^{YW}.
\end{aligned} \tag{64}$$

Hence, for the type-B part of m_2 and with (55),

$$\frac{1}{4} C_\ell^{(22)} = \frac{1}{4}\Big[C_\ell^{m_2 m_2} - 4\ \text{Re (ii)}|_{AA'\to m_2,\ X=Y=m_1} + 4\langle \langle m_1\rangle^2_{\rm sky}\rangle C_\ell^{(11)} \Big]. \tag{65}$$

Type D (nested, unequal-time products such as $\int \mathrm{d}\eta_1 \int_{\eta_1} \mathrm{d}\eta_2 W\, (\delta_{ij} - \hat{e}_i\hat{e}_j)\partial_i X(\eta_1)\partial_j Y(\eta_2)$, the lensing-type terms) reduce to (B1) applied to "shell maps" $\tilde{X}(\hat{\mathbf{e}}) = X(\eta_1, -\hat{\mathbf{e}}\chi_1)$ with unequal-time spectra $C_\ell^{xy}(\eta,\eta') = \frac{2}{\pi}\int k^2 \mathrm{d}k\, P\, x(k,\eta) y(k,\eta')^* j_\ell(k\chi) j_\ell(k\chi')$, using the exact identity $(\delta - \hat{e}\hat{e})_{ij}\partial_i X \partial_j Y = (\chi_1\chi_2)^{-1} \nabla_{\hat{\mathbf{e}}}\tilde{X}\cdot\nabla_{\hat{\mathbf{e}}}\tilde{Y}$ (weight w_∇) and $\hat{\mathbf{e}}\cdot\partial X = -\partial_\chi \tilde{X}$ for radial contractions ($j_\ell \to -k j'_\ell$, weight 1).

5.6 Type C: convolution kernels

A genuinely second-order map

$$\Theta_C(\hat{\mathbf{e}}) = \int \frac{\mathrm{d}^3k_1 \mathrm{d}^3k_2}{(2\pi)^6} \zeta_{\mathbf{k}_1}\zeta_{\mathbf{k}_2} \int \mathrm{d}\eta\, K(k_1,k_2,K,\eta)\, \mathcal{Q}(\hat{\mathbf{e}};\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\hat{\mathbf{K}})\, e^{-i\mathbf{K}\cdot\hat{\mathbf{e}}\chi}, \qquad \mathbf{K} = \mathbf{k}_1 + \mathbf{k}_2, \tag{66}$$

with $\mathcal{Q} = \sum q_{abc}(k_1,k_2,\mu)(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}}_1)^a(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}}_2)^b(\hat{\mathbf{e}}\cdot\hat{\mathbf{K}})^c$ a polynomial of degree d (transverse and TT projectors contracted with $\hat{\mathbf{e}}$ are of this form), has

$$K_{2,\ell m}(\mathbf{k}_1,\mathbf{k}_2) = \int \mathrm{d}\eta\, K \sum_{L=|\ell-d|}^{\ell+d} \sum_M 4\pi(-i)^L\, j_L(K\chi)\, Y^*_{LM}(\hat{\mathbf{K}})\, \langle \ell m|\mathcal{Q}|LM\rangle, \qquad \langle \ell m|\mathcal{Q}|LM\rangle = \int \mathrm{d}\Omega\, Y^*_{\ell m}\mathcal{Q}\, Y_{LM}, \tag{67}$$

a finite $3j$ sum generated by repeated use of $(\hat{\mathbf{e}}\cdot\hat{v})Y_{\ell m} = \frac{4\pi}{3}\sum_\mu Y^*_{1\mu}(\hat{v}) \sum_{\ell'=\ell\pm1,m'} \mathcal{G}^{\ell'\ell 1}_{m'm\mu} Y_{\ell' m'}$. In the frame $\hat{\mathbf{K}} = \hat{\mathbf{z}}$, $\mathbf{k}_{1,2}$ in the x–z plane ($\cos\alpha_1 = (k_1 + k_2\mu)/K$, $\cos\alpha_2 = (k_2 + k_1\mu)/K$, $\mu = \hat{\mathbf{k}}_1\cdot\hat{\mathbf{k}}_2$), $K_{2,\ell m} \to \tilde{K}_{\ell m}(k_1,k_2,\mu) = \int \mathrm{d}\eta\, K \sum_L (-i)^L \sqrt{4\pi(2L+1)} j_L(K\chi)\langle \ell m|\mathcal{Q}|L0\rangle$ with $|m| \leq d$,

and isotropy reduces the angular integrals of (57) to

$$\text{(C1)}\quad C_\ell^{(22)} = \frac{1}{4\pi^4(2\ell+1)}\int k_1^2\mathrm{d}k_1\,k_2^2\mathrm{d}k_2\,P(k_1)P(k_2)\int_{-1}^{1}\mathrm{d}\mu\sum_m|\tilde K_{\ell m}(k_1,k_2,\mu)|^2,$$

$$\text{(C2)}\quad \frac{1}{2\ell+1}\sum_m\langle C_{\ell m}B^*_{\ell m}\rangle = \frac{1}{4\pi^4(2\ell+1)}\int k_1^2\mathrm{d}k_1k_2^2\mathrm{d}k_2\,PP\int\mathrm{d}\mu\sum_m\tilde K^C_{\ell m}\tilde K^{B*}_{\ell m}, \tag{68}$$

where for a type-B map $\tilde K^B_{\ell m} = \frac{1}{2}\sum\mathcal{G}^{\ell\ell_1\ell_2}_{mm_1m_2}[\tilde A_{\ell_1m_1}(k_1)\tilde A'_{\ell_2m_2}(k_2) + (A\leftrightarrow A')]$, $\tilde A_{\ell_1m_1}(k) = 4\pi(-i)^{\ell_1}Y^*_{\ell_1m_1}(\hat{\mathbf{k}})\Delta^A_{\ell_1}(k)$ at the frame angles. A non-symmetric kernel must be symmetrised as $\frac{1}{2}(K+K^{\mathrm{T}})$ *before* the m-sum. Two-time kernels use the double Rayleigh expansion instead of (67); (C1) is unchanged.

5.7 The (13) term

From (55),

$$\text{(E0)}\qquad C_\ell^{(13)} = T_{3,\ell} - 3T_{\langle1\rangle_{\rm sky}2,\ell} - 3T_{\langle2\rangle_{\rm sky}1,\ell} + 6\langle\langle m_1\rangle^2_{\rm sky}\rangle C_\ell^{(11)}, \qquad T_{3,\ell}\equiv\frac{1}{2\ell+1}\textstyle\sum_m\langle m_{1,\ell m}m^*_{3,\ell m}\rangle, \tag{69}$$

and $T_{\langle1\rangle_{\rm sky}2}$, $T_{\langle2\rangle_{\rm sky}1}$ the corresponding cross terms with $\langle m_1\rangle_{\rm sky}m_2$, $\langle m_2\rangle_{\rm sky}m_1$. Every q-contracted map is a function of $\hat{\mathbf{e}}\cdot\hat{\mathbf{k}}$ only, hence a Legendre series handled by

$$\text{(A-}L\text{)}\qquad h(k,\eta)P_L(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi}\;\Rightarrow\;\Delta_\ell = \int\mathrm{d}\eta\,h\sum_{\Lambda=|\ell-L|}^{\ell+L}(2\Lambda+1)\begin{pmatrix}\ell & L & \Lambda\\ 0 & 0 & 0\end{pmatrix}^2(-i)^{\Lambda-\ell}j_\Lambda(k\chi), \tag{70}$$

and the $\mathbf{q}$-angular reduction

$$\text{(E1)}\qquad \int\frac{\mathrm{d}\Omega_q}{4\pi}g(\hat{\mathbf{q}}\cdot\hat{\mathbf{k}})(\hat{\mathbf{e}}\cdot\hat{\mathbf{q}})^be^{-iq\cdot\hat{\mathbf{e}}\Delta\chi} = \sum_{L'}(-i)^{L'}(2L'+1)j_{L'}(q\Delta\chi)\sum_{L''}\beta_{L'bL''}\,g_{L''}\,P_{L''}(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}}), \tag{71}$$

with $g_{L''} = \frac{1}{2}\int\mathrm{d}\mu\,P_{L''}(\mu)g(\mu)$ and $x^bP_{L'} = \sum_{L''}\beta_{L'bL''}P_{L''}$; for $\Delta\chi=0$ only $L'=0$ survives. The three structures in m_3 are then:

(iii) genuine third-order kernels: $\Theta^{(3)}_c(\mathbf{k};\hat{\mathbf{e}}) = [\int\frac{q^2\mathrm{d}q}{2\pi^2}P(q)\sum_L h_L(k,q,\eta)P_L(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})]e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi}$ and $T^{(\mathrm{iii})}_{3,\ell} = \frac{2}{\pi}\int k^2\mathrm{d}k\,P\,\Delta^{m_1}_\ell(k)\Delta^{(3c)}_\ell(k)^*$ with $\Delta^{(3c)}_\ell$ from (A-L) (the factor 3 of (57) is already inside the contracted kernel);

(i) products of three type-A maps ABC: $T^{(\mathrm{i})}_{3,\ell} = C^{m_1A}_\ell\xi^{BC}(0) + C^{m_1B}_\ell\xi^{AC}(0) + C^{m_1C}_\ell\xi^{AB}(0)$, $\xi^{XY}(0) = \sum_{\ell'}\frac{2\ell'+1}{4\pi}C^{XY}_{\ell'}$ (with $\xi^{\nabla B\cdot\nabla C}(0) = \sum\frac{\ell'(\ell'+1)(2\ell'+1)}{4\pi}C^{BC}_{\ell'}$ and $\nabla B\nabla C:\nabla\nabla A\to -\ell(\ell+1)R_{BC}C^{m_1A}_\ell$);

(ii) type-A × type-C products $A\cdot C$: $T^{(\mathrm{ii})}_{3,\ell} = C^{m_1A}_\ell\langle\!\langle C\rangle\!\rangle + 2\cdot\frac{2}{\pi}\int k^2\mathrm{d}k\,P\,\Delta^{m_1}_\ell\Delta^{AC*}_\ell$, $\langle\!\langle C\rangle\!\rangle = \int\frac{\mathrm{d}^3k}{(2\pi)^3}P(k)K^C(\mathbf{k},-\mathbf{k};\hat{\mathbf{e}})$ and Δ^{AC}_ℓ the transfer function of $\int\frac{\mathrm{d}^3q}{(2\pi)^3}P(q)K^A(\mathbf{q};\hat{\mathbf{e}})K^C(\mathbf{k},-\mathbf{q};\hat{\mathbf{e}})$ obtained from (E1) with $\Delta\chi=\chi_A-\chi_C$ and (A-L).

The normalisation pieces, with $M_2(\mathbf{q}_1,\mathbf{q}_2) \equiv K_{2,00}(\mathbf{q}_1,\mathbf{q}_2)/\sqrt{4\pi}$ the sky-average kernel of m_2, are

$$\text{(E2)}\quad T_{\langle 2\rangle_{\rm sky}1,\ell} = \langle\!\langle m_2\rangle\!\rangle C_\ell^{(11)} + \frac{1}{\pi^3}\int k^2\mathrm{d}k\, k'^2\mathrm{d}k'\, PP\,\Delta_\ell(k)\Delta_\ell(k')^* \int \mathrm{d}\mu\, P_\ell(\mu) M_2(k\hat{\mathbf{k}}, -k'\hat{\mathbf{k}}')^*, \quad \langle\!\langle m_2\rangle\!\rangle = \frac{1}{2\pi^2}\int k^2$$

$$\text{(E3)}\quad T_{\langle 1\rangle_{\rm sky}2,\ell} = \frac{2}{2\pi^2}\int q^2\mathrm{d}q\, P(q)\Delta_0(q)\,\frac{2}{\pi}\int k^2\mathrm{d}k\, P(k)\Delta_\ell(k)\bar{\Delta}_\ell^{m_2}(k;q)^*, \qquad \bar{\Delta}_\ell(k;q) = \frac{\frac{1}{2}\int \mathrm{d}\mu\, K_{2,\ell 0}(k\hat{\mathbf{z}},\mathbf{q})}{(-i)^\ell\sqrt{4\pi(2\ell+1)}}, \tag{72}$$

and the same objects give the cross term of (65). Formulas (E2), (E3) and $T^{(\mathrm{ii})}$ are derived by Wick counting consistent with the verified ones; (E3) collapses to (64)(ii) for a type-B m_2 (analytic check).

[To be supplied: the explicit h_L, K^C, M_2 of our map are supplied in Secs. 7–8 and assembled in Sec. 9.]

5.8 Control case: lensed C_ℓ

For the remapping $\tilde{\Theta} = \Theta + \nabla\phi\cdot\nabla\Theta + \frac{1}{2}\nabla^i\phi\nabla^j\phi\nabla_i\nabla_j\Theta$ with $\Theta = \epsilon\Theta_1$, $\phi = \epsilon\phi_1$ one has $\Theta^{(2)} = 2\nabla\phi\cdot\nabla\Theta$, $\Theta^{(3)} = 3\nabla\phi\nabla\phi : \nabla\nabla\Theta$, and the bookkeeping (10) with (B1) and the (i)-type pairing reproduces exactly

$$\frac{1}{4}C_\ell^{(22)} = \sum_{\ell_1\ell_2}\frac{F^2_{\ell\ell_1\ell_2}}{2\ell+1}\Big[C_{\ell_1}^{\phi\phi}C_{\ell_2}^{\Theta\Theta} + C_{\ell_1}^{\phi\Theta}C_{\ell_2}^{\Theta\phi}\Big], \qquad \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} = -\ell(\ell+1)R\,C_\ell^{\Theta\Theta}, \qquad R = \frac{1}{2}\langle|\nabla\phi|^2\rangle, \tag{73}$$

with F the coupling of Hu [16]: these are Hu's (62)–(63) [16, 17] (the $C^{\phi\Theta}$ term is the extension to correlated ϕ,Θ, relevant here because the deflection potential correlates with SW/ISW). This was verified algebraically (2×10^{-16}) and by Monte Carlo on the sphere (App. D, C3-5), validating the prefactors of (57), the two-pairing structure of (B1) and the monopole×map rules.

5.9 Monochromatic reduction: definitions

The monochromatic primordial spectrum is defined with the variance $\langle\zeta_1^2\rangle = \sigma^2$ held fixed as k_0 varies:

$$P(k) = \frac{2\pi^2\sigma^2}{k_0^2}\,\delta(k-k_0) \quad\Longleftrightarrow\quad \langle\zeta_1^2(\mathbf{x})\rangle = \sigma^2, \qquad \int k^2\mathrm{d}k\, P\,F(k) \to 2\pi^2\sigma^2F(k_0), \quad \int\frac{\mathrm{d}^3k}{(2\pi)^3}P\,F(\mathbf{k}) \to \sigma^2\langle F(k_0\hat{\mathbf{k}})\rangle \tag{74}$$

The expansion parameter is then $\epsilon^2\sigma^2$. Under (74) the general formulas collapse to (all transfer functions at k_0):

$$C_\ell^{AA'} = 4\pi\sigma^2\Delta_\ell^A\Delta_\ell^{A'*}, \qquad \langle\langle A\rangle_{\rm sky}\langle A'\rangle_{\rm sky}\rangle = \sigma^2\Delta_0^A\Delta_0^{A'}, \qquad \xi^{XY}(0) = \sigma^2\sum_{\ell'}(2\ell'+1)\Delta_{\ell'}^X\Delta_{\ell'}^{Y*};$$

$$\text{(B1)}: \; C_\ell^{BB'} = (4\pi\sigma^2)^2\sum_{\ell_1\ell_2}\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{pmatrix}^2 ww'[\Delta_{\ell_1}^A\Delta_{\ell_1}^C\Delta_{\ell_2}^{A'}\Delta_{\ell_2}^D + \Delta_{\ell_1}^A\Delta_{\ell_1}^D\Delta_{\ell_2}^{A'}\Delta_{\ell_2}^C] \;\text{(effectively fini}$$

$$\text{(C1)}: \; C_\ell^{(22)} = \frac{\sigma^4}{2\ell+1}\int_{-1}^{1}\mathrm{d}\mu\sum_{|m|\le d}|\tilde K_{\ell m}(k_0,k_0,\mu)|^2, \quad K = k_0\sqrt{2(1+\mu)},\; \cos\alpha_1 = \cos\alpha_2 = \sqrt{(1+\mu)/2};$$

$$\text{(13)(iii)}: \; T_{3,\ell} = 12\pi\sigma^2\Delta_\ell\Delta_\ell^{(3c)*} \text{ with the } \frac{1}{3} \text{ of (10) still to be applied, } \Delta_\ell^{(3c)} = \sigma^2\sum_L h_L(k_0,k_0)\sum_\Lambda(2\Lambda+1)\begin{pmatrix}\ell & 1\\ 0 & 0\end{pmatrix}$$

$$h_L = \frac{1}{2}\int \mathrm{d}\mu\, P_L(\mu)[K_3(k_0,k_0,\mu)\times\mathcal{Q}_3\text{-coefficients}], \quad |\mathbf{k}\pm\mathbf{q}| = k_0\sqrt{2(1\pm\mu)};$$

$$\langle\langle m_1\rangle_{\rm sky}^2\rangle = \sigma^2\Delta_0^2, \quad \langle\!\langle m_2\rangle\!\rangle = \sigma^2 M_2(k_0\hat{\mathbf{k}}, -k_0\hat{\mathbf{k}}), \quad \text{(E2) cross} = 4\pi\sigma^4|\Delta_\ell|^2\int\mathrm{d}\mu\, P_\ell(\mu)M_2(k_0\hat{\mathbf{k}}, -k_0\hat{\mathbf{k}}')^*, \quad \text{(E3)} = 8\pi \tag{75}$$

(The factor $12\pi\sigma^2 = 3\times 4\pi\sigma^2$ in (13)(iii) assumes a symmetrised K_3 with the explicit pairing factor 3; with the contracted kernels of Sec. 4.3 the 3 is already included and the prefactor is $4\pi\sigma^2$.) *Delicate points* to be addressed in Sec. 11: (1) $\mu\to-1$ in (C1), where $K\to 0$, $\mathrm{d}\mu = K\mathrm{d}K/k_0^2$ and $j_L(K\chi)\to\delta_{L0}$: a kernel $\sim K^{-p}$ converges only for $p<1$, so every $1/K^2$ (Poisson inversion) must come with a vanishing numerator (the transverse/TT projections of $\mathbf{k}_1\otimes\mathbf{k}_2$ do vanish at $\mathbf{k}_2=-\mathbf{k}_1$; the scalar K_Q-type $1/K^2$ terms require an explicit check of the $K\to 0$ limit); (2) $\mu\to\pm1$ in the (13) loop, where $|\mathbf{k}\mp\mathbf{q}|\to 0$ inside $K_3(\mathbf{k},\mathbf{q},-\mathbf{q})$ and C_3; (3) the $K=0$ tadpoles $\langle\!\langle m_2\rangle\!\rangle$, $\langle\!\langle C\rangle\!\rangle$ require kernels exactly at $\mathbf{K}=0$ and enter only through (55), where the sky-mean normalisation cancels the direction-independent mean of e^Λ: individual tadpoles may be IR-singular while the combination is finite, so they are kept symbolic until combined (cf. Sec. 3.4); (4) $\langle\zeta_{\mathbf{k}_1}\zeta_{\mathbf{k}_2}\rangle$ inside $\Theta^{(2)}$ feeds only $\ell=0$, consistent with the monopole normalisation.

6 First order: transfer function $\Delta_\ell(k)$ and $C_\ell^{(11)}$

Conventions. Observer at $\mathbf{x}_o = 0$, observation direction $\hat{\mathbf{n}}$, background ray $\mathbf{x}(\eta) = \hat{\mathbf{n}}\,\chi(\eta)$, $\chi = \eta_o - \eta$, emission at η_*, $\chi_* = \eta_o - \eta_*$. We write $x = k\eta/\sqrt{3}$, $s = k\eta$, $s_* = k\eta_*$, $S = k\eta_o$, $w = S - s_* = k\chi_*$. The first-order map (SPEC §9a, verified from S_1 in Sec. 2) is

$$\Theta_1(\hat{\mathbf{n}}) = [\delta_{T1} + \Psi_1 - \hat{\mathbf{n}}\cdot\mathbf{v}_1]_e - [\delta_{T1} + \Psi_1 - \hat{\mathbf{n}}\cdot\mathbf{v}_1]_o + \int_{\eta_*}^{\eta_o}(\Phi_1' + \Psi_1')\,\mathrm{d}\eta, \tag{76}$$

with the regular first-order solution ($\Phi_1 = \Psi_1 = \zeta\,T_\Phi$, $\delta_{T1} = \zeta\,T_{\delta T}$, $\mathbf{v}_1 = \nabla V_1$, $V_1 = \zeta T_V$)

$$T_\Phi = -\frac{2j_1(x)}{x}, \quad T_{\delta T} = -\cos x + 2j_0(x) + T_\Phi, \quad kT_V = \sqrt{3}\,(\sin x - 2j_1(x)), \quad 2\partial_\eta T_\Phi = \frac{4k}{\sqrt{3}}\frac{j_2(x)}{x}. \tag{77}$$

Define the emission (Sachs–Wolfe) source $A(x) := T_{\delta T} + T_\Phi = -\cos x + 2j_0(x) - 4j_1(x)/x$.

Rayleigh rules. From $e^{i\mathbf{k}\cdot\hat{\mathbf{n}}\chi} = 4\pi\sum_{\ell m} i^\ell j_\ell(k\chi)Y_{\ell m}(\hat{\mathbf{n}})Y_{\ell m}^*(\hat{\mathbf{k}})$ and $\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}\,e^{ik\chi\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}} = (ik)^{-1}\partial_\chi e^{ik\chi\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}}$: $(i\hat{\mathbf{n}}\cdot\hat{\mathbf{k}})^p e^{ik\chi\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}} \to 4\pi i^\ell\, j_\ell^{(p)}(k\chi)\,Y_{\ell m}^*(\hat{\mathbf{k}})$. Observer terms carry no phase ($\chi=0$): since $j_\ell(0) = \delta_{\ell 0}$

and $j'_\ell(0) = \delta_{\ell 1}/3$, the observer monopole $[\delta_{T1} + \Psi_1]_o$ contributes only to $\ell = 0$ and the observer Doppler term $\hat{\mathbf{n}} \cdot \mathbf{v}_1(\eta_o, 0) = ik\,(\hat{\mathbf{n}} \cdot \hat{\mathbf{k}})\,T_V(x_o)\zeta_{\mathbf{k}}$ only to $\ell = 1$.

Proposition 6.1 (First-order transfer function)**.** $a^{(1)}_{\ell m} = 4\pi i^\ell \int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\zeta_{\mathbf{k}}\,Y^*_{\ell m}(\hat{\mathbf{k}})\,\Delta_\ell(k)$ *with real*

$$\Delta_\ell(k) = A(x_*)\,j_\ell(w) - kT_V(x_*)\,j'_\ell(w) - \delta_{\ell 0}\,A(x_o) + \frac{1}{3}\delta_{\ell 1}\,kT_V(x_o) + I_\ell(k), \qquad I_\ell(k) = \frac{4}{\sqrt{3}}\int_{s_*}^{S} \frac{j_2(s/\sqrt{3})}{s/\sqrt{3}}\,j_\ell(S-s)\,\mathrm{d}s. \tag{78}$$

For $\ell \geq 1$, $\Theta^{(1)}_{\ell m} = m^{(1)}_{\ell m}$ *(SPEC §7) and*

$$C^{(11)}_\ell = \frac{2}{\pi}\int_0^\infty k^2\,\mathrm{d}k\,P(k)\,\Delta_\ell(k)^2, \tag{79}$$

from $\langle \zeta_{\mathbf{k}}\zeta^*_{\mathbf{k}'}\rangle = (2\pi)^3\delta^3(\mathbf{k}-\mathbf{k}')P$, $i^\ell(-i)^\ell = 1$ *and* $\sum_m |Y_{\ell m}|^2 = (2\ell+1)/4\pi$*:* $C_\ell = \frac{(4\pi)^2}{2\ell+1}\frac{2\ell+1}{4\pi}\frac{4\pi}{(2\pi)^3}\int k^2 P \Delta^2_\ell$.

Closed form of the ISW integral. Write $J(s) = j_2(s/\sqrt{3})/(s/\sqrt{3})$ (entire), $Y(s) = y_2(s/\sqrt{3})/(s/\sqrt{3}) = -27/s^4 - 3/(2s^2) + O(1)$, $a_\pm = 1 \pm 1/\sqrt{3}$ and $\mathrm{Cin}(z) = \int_0^z (1-\cos t)\,\mathrm{d}t/t = \gamma + \ln z - \mathrm{Ci}\,z$.

Theorem 6.2 (ISW integral, all ℓ)**.**

$$\begin{aligned} I_\ell = E_\ell(s_*, S) &+ A_\ell(S)[\,\mathrm{Ci}\,a_+S - \mathrm{Ci}\,a_+s_* - \mathrm{Ci}\,a_-S + \mathrm{Ci}\,a_-s_*] + B_\ell(S)[\,\mathrm{Si}\,a_+S - \mathrm{Si}\,a_+s_* - \mathrm{Si}\,a_-S + \mathrm{Si}\,a_-s_*] \\ &+ C_\ell(S)[\mathrm{Cin}(a_+w) - \mathrm{Cin}(a_-w)] + D_\ell(S)[\,\mathrm{Si}\,a_+w + \mathrm{Si}\,a_-w], \end{aligned} \tag{80}$$

$$A_\ell = \frac{2}{\sqrt{3}}\operatorname*{Res}_{s=0} Y(s)\,y_\ell(S-s) = \sqrt{3}\,[3y'''_\ell(S) + y'_\ell(S)], \quad B_\ell = -\frac{2}{\sqrt{3}}\operatorname*{Res}_{s=0} Y(s)\,j_\ell(S-s) = -\sqrt{3}\,[3j'''_\ell(S) + j'_\ell(S)], \tag{81}$$

$$C_\ell = -\frac{2}{\sqrt{3}}\operatorname*{Res}_{u=0} y_\ell(u)\,Y(S-u), \qquad D_\ell = -\frac{2}{\sqrt{3}}\operatorname*{Res}_{u=0} y_\ell(u)\,J(S-u), \tag{82}$$

and E_ℓ *elementary: Laurent polynomials in* s_*, S, w *times* $\sin, \cos$ *of* $a_\pm s_*$, $a_\pm S$, w, $s_*/\sqrt{3}$, $S/\sqrt{3}$.

Derivation. With $g(x) = h^{(1)}_2(x)/x$ and $h = h^{(1)}_\ell$, the integrand is $\frac{1}{\sqrt{3}}(g+\bar{g})(s/\sqrt{3})\,(h+\bar{h})(S-s)$, a sum of four terms $e^{ibs}R(s)$ with $b \in \{\pm a_\pm\}$ (never zero: no resonance) and R rational with poles only at $s = 0$ (order 4, from $y_2(x)/x$) and $s = S$ (order $\ell+1$, from y_ℓ). Partial fractions and the recursion $\int e^{ibs}s^{-m}\mathrm{d}s = -\frac{e^{ibs}}{(m-1)s^{m-1}} + \frac{ib}{m-1}\int e^{ibs}s^{1-m}\mathrm{d}s$ reduce everything to elementary terms plus $\int e^{ibs}\mathrm{d}s/s = \mathrm{Ci}|b|s + i\,\mathrm{sgn}(b)\,\mathrm{Si}|b|s$, whose total coefficient is $\mathrm{Res}[e^{ibs}R]$, i.e. the residue of the product of the *full* Hankel functions. Because j_ℓ and $j_2(x)/x$ are entire, the four products pair up ($A^- = -A^+$, $B^- = -B^+$, $C^- = -C^+$, $D^- = D^+$) and only the y-parts survive, giving the residues above. At the observer end $u = S - s \to 0$ the coefficient of $\ln u$ is $\propto \mathrm{Re}\,\mathrm{Res}_{u=0}[j_\ell(u)g] = 0$, so the endpoint is regular and the Cin combination appears with no finite-part ambiguity. □

Checks (`isw_structure.py`, $\ell = 0, \ldots, 6$): every Ci/Si coefficient of the brute-force antiderivative equals the residue formula to 10^{-169} (40-digit evaluation), E_ℓ contains no $\mathrm{Ci}, \mathrm{Si}, \ln, \gamma$; the closed form agrees with mpmath quadrature to relative 5×10^{-18} (`linear_cl.py`, five parameter sets). Explicit coefficients for $\ell \leq 6$ ($z = S/\sqrt{3}$) are in `isw_coeffs.tex`; e.g.

$$A_2 = \frac{2\sqrt{3}}{S^6}(S^5\sin S + 7S^4\cos S - 36S^3\sin S - 126S^2\cos S + 270S\sin S + 270\cos S),$$

$$B_2 = -\frac{2\sqrt{3}}{S^6}(S^5\cos S - 7S^4\sin S - 36S^3\cos S + 126S^2\sin S + 270S\cos S - 270\sin S),$$

$$C_2 = \frac{4\sqrt{3}}{3z^6}(z^3\sin z + 6z^2\cos z - 15z\sin z - 15\cos z), \qquad D_2 = \frac{4\sqrt{3}}{3z^6}(z^3\cos z - 6z^2\sin z - 15z\cos z + 15\sin z);$$

the elementary parts E_ℓ (56–464 terms for $\ell = 0..6$) are machine-generated in `isw_closed_forms.tex`.

Monochromatic spectrum. For $P = 2\pi^2\sigma^2\delta(k-k_0)/k_0^2$ ($\langle\zeta^2\rangle = \sigma^2$),

$$C_\ell^{(11)} = 4\pi\sigma^2\Big[A(x_{0*})\,j_\ell(k_0\chi_*) - k_0T_V(x_{0*})\,j_\ell'(k_0\chi_*) - \delta_{\ell0}A(x_{0o}) + \frac{1}{3}\delta_{\ell1}k_0T_V(x_{0o}) + I_\ell(k_0)\Big]^2, \quad (83)$$

$x_{0*} = k_0\eta_*/\sqrt{3}$, $x_{0o} = k_0\eta_o/\sqrt{3}$, with $I_\ell(k_0)$ from Theorem 6.2. A real-space check (two-point function of the map built directly from the kernel vs $\sum_\ell \frac{2\ell+1}{4\pi}C_\ell P_\ell$, including $\ell = 0, 1$ observer terms) agrees to 6×10^{-15}. Example ($\sigma^2 = 1$, $\eta_* = 0.5$, $\eta_o = 2$, $k_0 = 4$): $\Delta_{0..4} = 0.44006$, -0.33896, 0.27562, 0.22303, 0.118 $C_{2,3,4} = 0.95460$, 0.62508, 0.17507.

Limit $k_0\eta_* \to 0$ (Sachs–Wolfe). Superhorizon the constraints give $\Phi_1' = 0$, $\delta_1 = -2\Psi_1$, i.e. $\delta_{T1} = -\Psi_1/2$; with $\zeta = \delta_{T1} - \Phi_1$ (SPEC §5, linearised) and $\Phi_1 = \Psi_1$: $\Phi_1 = -\frac{2}{3}\zeta$ and

$$[\delta_{T1} + \Psi_1]_e \to \frac{1}{2}\Psi_1 = -\frac{1}{3}\,\zeta \qquad \text{(RD; cf. } -\frac{1}{5}\zeta \text{ in matter domination)}, \quad (84)$$

while the emission Doppler term vanishes ($kT_V \sim x$). Hence at fixed S, for $\ell \geq 2$, $\Delta_\ell \to -\frac{1}{3}j_\ell(S) + I_\ell^0(S)$, $I_\ell^0 = I_\ell|_{s_*=0}$ (closed form: Theorem 6.2 with $s_* \to 0$: the poles of E_ℓ in s_* cancel against the small-argument expansions of the s_*-dependent terms, $\mathrm{Ci}\,a_+s_* - \mathrm{Ci}\,a_-s_* \to \ln(a_+/a_-)$; implemented as `I_l_closed_eta0`, checked against quadrature). For a scale-invariant $P = 2\pi^2A_s/k^3$ this is an exact plateau, $\ell(\ell+1)C_\ell/2\pi A_s = 2\ell(\ell+1)\int_0^\infty \frac{\mathrm{d}S}{S}\Delta_\ell(S)^2$, independent of η_o. The SW term alone gives $1/9$ ($\int \mathrm{d}y\, j_\ell^2/y = 1/[2\ell(\ell+1)]$, the RD analogue of $1/25$). In RD, however, Φ decays after horizon entry and every observed mode has entered by η_o ($S \gtrsim \ell$), so the ISW term is not small: $\int 2T_\Phi'\,\mathrm{d}\eta = +\frac{4}{3}$. Numerically (`plateau_isw.py`)

ℓ	2	3	4	6	10	20	40	100	∞
SW only	.1111	.1111	.1111	.1111	.1111	.1111	.1111	.1111	1/9
SW+ISW	.3772	.4474	.5003	.5757	.6649	.7629	.8290	.8770	.91377

The large-ℓ value follows from the flat-sky limit: with the line-of-sight source $K(s) = -\frac{1}{3}\delta(s) + 2\,\mathrm{d}T_\Phi/\mathrm{d}s$, $\ell(\ell+1)C_\ell/2\pi A_s \to \int_0^1 \mathrm{d}\mu\,|\tilde{K}(\mu)|^2$, and using $j_1(x)/x = \frac{1}{4}\int_{-1}^1(1-t^2)e^{ixt}\mathrm{d}t$,

$$\tilde{K}(\mu) = \int K e^{i\mu s}\mathrm{d}s = 1 - 2\beta^2 - \beta(1-\beta^2)\ln\Big|\frac{1+\beta}{1-\beta}\Big| + i\pi\beta(1-\beta^2)\,\theta(1-\beta), \qquad \beta = \sqrt{3}\,\mu, \quad (85)$$

($\tilde{K}(0) = -\frac{1}{3}+\frac{4}{3} = 1$), giving 0.913769 (ISW alone 0.865111); verified against a numerical transform.

Limit $k_0 \to \infty$. $A(x) = -\cos x + O(1/x)$, $kT_V = \sqrt{3}\sin x + O(1/x)$ and $I_\ell = O(1/(k\eta_*))$ (no resonance, $a_\pm \neq 0$), so

$$\Delta_\ell \to -\cos x_{0*}\,j_\ell(k_0\chi_*) - \sqrt{3}\sin x_{0*}\,j_\ell'(k_0\chi_*) + \delta_{\ell0}\cos x_{0o} + \frac{1}{\sqrt{3}}\delta_{\ell1}\sin x_{0o}, \quad (86)$$

(acoustic oscillation with Doppler/SW amplitude ratio $1/c_s = \sqrt{3}$). For fixed $\ell \geq 2$, $C_\ell^{(11)} \to \frac{4\pi\sigma^2}{(k_0\chi_*)^2}[\cos x_{0*}\sin(k_0\chi_* - \frac{\ell\pi}{2}) + \sqrt{3}\sin x_{0*}\cos(k_0\chi_* - \frac{\ell\pi}{2})]^2 \to 0$; the observer terms keep $C_0 \to 4\pi\sigma^2\cos^2 x_{0o}$, $C_1 \to \frac{4\pi}{3}\sigma^2\sin^2 x_{0o}$. Checked at $k = 300, 3000$.

Small k (gauge modes). Expanding in k (ISW integrated term by term): $\Delta_0 = -\frac{k^4\chi_*^3(\eta_o+3\eta_*)}{810} + O(k^6)$, $\Delta_1 = \frac{2k^3\chi_*^2(\eta_o+2\eta_*)}{405} + O(k^5)$: the k^0, k^2 terms of Δ_0 and the k term of Δ_1 cancel between emission, observer and ISW, as required by the invisibility of constant and gradient ζ modes (SPEC §9d at first order).

Sky monopole. $\langle m_1\rangle_{\rm sky} = \frac{1}{4\pi}\int \mathrm{d}\Omega\, m_1 = a_{00}^{(1)}/\sqrt{4\pi}$, so

$$\langle m_1\rangle_{\rm sky} = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}} M(k), \qquad M(k) = \Delta_0(k) = A(x_*)\, j_0(w) + kT_V(x_*)\, j_1(w) - A(x_o) + I_0(k), \tag{87}$$

observer monopole included; $M = O(k^4)$. Then $\langle\langle m_1\rangle_{\rm sky}^2\rangle = \frac{1}{2\pi^2}\int k^2 P\, M^2\, \mathrm{d}k = C_0^{(11)}/4\pi$, monochromatic $\sigma^2 M(k_0)^2$ (example above: 0.193652). For $\ell \geq 1$,

$$\langle\langle m_1\rangle_{\rm sky}\, a_{\ell m}^{(1)*}\rangle = 4\pi(-i)^\ell \int \frac{k^2\mathrm{d}k}{(2\pi)^3} P\, M\, \Delta_\ell \int \mathrm{d}\Omega_k\, Y_{\ell m}(\hat{\mathbf{k}}) = 0, \tag{88}$$

since $\int Y_{\ell m}\mathrm{d}\Omega = \sqrt{4\pi}\,\delta_{\ell 0}\delta_{m0}$ (statistical isotropy; any P). This is a statement about the Gaussian first-order fields only; at higher order $\langle m_1\rangle_{\rm sky}$ multiplies $m_{\ell m}$ in $\Theta_{\ell m}^{(2)}$ (SPEC §7) and its connected correlations with products of fields do not vanish. Numerical check: the sky average of the kernel equals $M(k_0)$ for every $\hat{\mathbf{k}}$ to 5×10^{-15}, and $|\langle\langle m_1\rangle_{\rm sky} a_{\ell m}^*\rangle| < 10^{-14}$ for $\ell = 1..3$.

7 The second-order map $\Theta^{(2)}$

This section turns the exact observable (20) into explicit line-of-sight functionals L_1, L_2 (this section) and L_3 (Sec. 8) of the Poisson-gauge fields of Sec. 4, with the normalisation of Sec. 5.1. All results are from agent C2 (code `cmb4/code/C2_observable/`, term lists `L_terms.pkl`, reproduced in App. B); the definitions of the ray functionals are stated here with a sign correction found by agent A1 (Remark 7.1). The whole chain has been tested end to end against a non-perturbative solver (Sec. 10).

Write the integrand of (20) as a function of the spacetime point and of the tangent $k^\mu = (1, k^i)$,

$$\mathcal{I}(x;k) \equiv \frac{\hat\sigma_{\mu\nu}k^\mu k^\nu}{-\hat u_\mu k^\mu}, \qquad S(\hat{\mathbf{n}}) = \int_{\eta_e(\hat{\mathbf{n}})}^{\eta_o} \mathrm{d}\eta\, \mathcal{I}(\eta, \mathbf{x}(\eta); k(\eta)), \qquad \Lambda \equiv -S. \tag{89}$$

Since $\hat\sigma$ vanishes on the background, $\mathcal{I} = O(\epsilon)$, and at first order on the background ray ($k = (1, -\hat{\mathbf{n}})$)

$$I_1 \equiv \mathcal{I}_1|_{k^i=-n^i} = n^i n^j \partial_i v_1^j - \frac{1}{3}\partial_i v_1^i = \hat\sigma_{1,ee} \tag{90}$$

(cf. (23)). The *expansion scheme* has three ingredients, each expanded in ϵ as in (5): (i) the perturbed ray $x^i(\eta) = \bar x^i(\eta) + \Delta x^i(\eta)$ with perturbed tangent $k^i(\eta)$; (ii) the perturbed endpoints, $\eta_o = \bar\eta_o + \delta\eta_o$ (clock) with the observer at $\bar{\mathbf{x}}_o + \delta\mathbf{x}_o$, and $\eta_e = \eta_* + \delta\eta_e(\hat{\mathbf{n}})$ (emission surface); (iii) Taylor expansion of every field about the background ray (8), so that the final functionals involve only fields at $(\eta, \bar{\mathbf{x}}(\eta))$, $\eta_* \leq \eta \leq \bar\eta_o$, at the background observer event $(\bar\eta_o, \bar{\mathbf{x}}_o)$ and at the background emission point $(\eta_*, \bar{\mathbf{x}}(\eta_*))$.

7.1 Ray, triad and direction map

Geodesic equation. With the ray parametrised by η, the null geodesic equation of $\hat{g}$ is, exactly,

$$\frac{\mathrm{d}k^i}{\mathrm{d}\eta} = G^i \equiv -\hat{\Gamma}^i_{\mu\nu}k^\mu k^\nu + \hat{\Gamma}^0_{\mu\nu}k^\mu k^\nu\, k^i, \qquad \frac{\mathrm{d}x^i}{\mathrm{d}\eta} = k^i. \tag{91}$$

The background $\hat{g}$ is Minkowski, so the background ray is straight, $k^i = -n^i$. We write $k^i = -n^i + \epsilon k_1^i + \frac{\epsilon^2}{2}k_2^i + \ldots$, $\Delta x^i = \epsilon \Delta x_1^i + \frac{\epsilon^2}{2}\Delta x_2^i + \ldots$, and use the decomposition along $\hat{\mathbf{n}}$: $\Delta x_\parallel \equiv \hat{\mathbf{n}}\cdot\Delta\mathbf{x}$ (radial: time delay) and $\Delta\mathbf{x}_\perp \equiv \Delta\mathbf{x} - \hat{\mathbf{n}}\,\Delta x_\parallel$ (transverse: lensing), likewise for k. At first order ($|\hat{\mathbf{n}}|=1$)

$$G_1^i = -\partial_i(\Phi_1+\Psi_1) + 2n^i\,\hat{\mathbf{n}}\cdot\nabla(\Phi_1+\Psi_1) - n^i(\Phi_1'+\Psi_1'), \qquad \hat{\mathbf{n}}\cdot\vec{G}_1 = -D(\Phi_1+\Psi_1), \tag{92}$$

whose transverse part $-\nabla_\perp(\Phi_1+\Psi_1)$ is the lensing deflection. $G_2^i/2$ consists of the second-order Christoffel symbols on the background ray plus $\Gamma_1 \times k_1$ cross terms (82–85 terms per component, stored in `ray_cache.pkl`); the positional shift $\Delta x_1^j \partial_j G_1^i$ is added explicitly in (98).

Canonical triad. The Gram–Schmidt triad of Sec. 2.4 (order $1,2,3$) is needed through second order, since $\mathcal{I} = O(\epsilon)$ (in the coefficient convention $X = \epsilon X_1 + \frac{\epsilon^2}{2}X_2$):

$$\begin{aligned}
e_a{}^0 &= \epsilon v_1^a + \epsilon^2[\frac{1}{2}(v_2^a + \omega_{2,a}) - (\Phi_1 + 2\Psi_1)v_1^a], \\
e_a{}^i &= \delta_a^i\{1 + \epsilon\Phi_1 + \epsilon^2[\frac{3}{2}\Phi_1^2 + \frac{1}{2}\Phi_2 - \frac{1}{4}h_{2,aa} + \frac{1}{2}(v_1^a)^2]\} + \epsilon^2 \begin{cases} v_1^a v_1^i, & i > a, \\ -\frac{1}{2}h_{2,ai}, & i < a, \end{cases}
\end{aligned} \tag{93}$$

(no sum over a; the asymmetry between $i > a$ and $i < a$ is that of the Gram–Schmidt order). Verified: $\hat{g}(e_a, e_b) = \delta_{ab}$, $\hat{g}(e_a, \hat{u}) = 0$ through third order (App. D, C2-5).

Direction map. From (19), $k^i(\eta_o) = (\hat{u}^i - n^a e_a{}^i)/(\hat{u}^0 - n^a e_a{}^0)$ with all fields at the actual observer event. Expanding, with fields at that event,

$$k^i_{1,\mathrm{init}} = v_1^i - n^i(\hat{\mathbf{n}}\cdot\mathbf{v}_1) - n^i(\Phi_1+\Psi_1), \tag{94}$$

i.e. aberration $\mathbf{v}_{1\perp}$ in the transverse direction and $|\vec{k}| = 1 + \Phi_1 + \Psi_1$ radially (the null condition $\hat{\mathbf{n}}\cdot\vec{k}_1 = -(\Phi_1+\Psi_1)$ holds at every η, by (92)). At second order, displayed for $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ ($\perp = x, y$; general $\hat{\mathbf{n}}$: 23–26 terms per component, `ray.k_init`),

$$\begin{aligned}
\frac{1}{2}k^\perp_{2,\mathrm{init}} &= \frac{1}{2}v_2^\perp + v_1^\perp v_1^z + \frac{1}{2}h_{2,\perp z}, \\
\frac{1}{2}k^z_{2,\mathrm{init}} &= -\frac{1}{2}\Psi_2 - \frac{1}{2}\Phi_2 - \frac{3}{2}\Phi_1^2 - \Phi_1\Psi_1 + \frac{1}{2}\Psi_1^2 + \frac{1}{4}h_{2,zz} - \frac{1}{2}\omega_{2,z} + \frac{1}{2}|\mathbf{v}_{1\perp}|^2.
\end{aligned} \tag{95}$$

The map $\hat{\mathbf{n}} \mapsto k^\mu(\eta_o)$ is null through third order (App. D, C2-6).

Ray functionals. Integrating (91) backwards from the observer and transporting the observer event to $(\bar\eta_o, \bar{\mathbf{x}}_o)$ gives, with all fields on the background ray,

$$k_1^i(\eta) = k_{1,\mathrm{init}}^i|_{\bar o} - \int_\eta^{\bar\eta_o} G_1^i \,\mathrm{d}\eta', \tag{96}$$

$$\Delta x_1^i(\eta) = \delta x_{o1}^i + n^i\,\delta\eta_{o1} - \int_\eta^{\bar\eta_o} k_1^i \,\mathrm{d}\eta', \tag{97}$$

$$\frac{1}{2}k_2^i(\eta) = \frac{1}{2}k_{2,\mathrm{init}}^i|_{\bar o} + \delta\eta_{o1}\,\partial_\eta k_{1,\mathrm{init}}^i|_{\bar o} + \delta x_{o1}^j\,\partial_j k_{1,\mathrm{init}}^i|_{\bar o} - \delta\eta_{o1}G_1^i(\bar\eta_o) - \int_\eta^{\bar\eta_o}[\frac{1}{2}G_2^i + \Delta x_1^j\partial_j G_1^i]\mathrm{d}\eta', \tag{98}$$

$$\frac{1}{2}\Delta x_2^i(\eta) = \frac{1}{2}\delta x_{o2}^i + \frac{1}{2}n^i\,\delta\eta_{o2} - \delta\eta_{o1}\,k_1^i(\bar\eta_o) - \int_\eta^{\bar\eta_o}\frac{1}{2}k_2^i\,\mathrm{d}\eta', \tag{99}$$

where $|_{\bar o}$ means fields at $(\bar\eta_o, \bar{\mathbf{x}}_o)$, the derivative terms in (98) transport the initial direction from the actual to the background observer event, and G_2^i contains $k_1(\eta)$. The $n^i\delta\eta_o$ terms in Δx express that the photon reaching the later observer event passes $\bar\eta_o$ displaced by $-k\,\delta\eta_o$.

Remark 7.1 (sign of the tangent integrals). Since $\mathrm{d}k/\mathrm{d}\eta = G$ and the integrals run from η to the observer, $k(\eta) = k(\bar\eta_o) - \int_\eta^{\bar\eta_o} G$: the *minus* signs in (96) and (98) are correct. The definitions printed by C2 (`ray.definitions`, `L_terms.pkl['definitions']`, report C2) carry a + sign; this is a documentation slip only: the term lists were generated with the correct derivative rules ($\mathrm{d}k_1/\mathrm{d}\eta \to G_1$, $\mathrm{d}\Delta x_1/\mathrm{d}\eta \to k_1$). With the + sign the null condition fails at $O(1)$ and the shear and direct forms of Sec. 7.6 disagree; with the − sign the null condition holds to 10^{-15} and the two forms agree (agent A1, App. D, A1-4). Equations (97), (99) are as printed by C2.

7.2 Observer event: clock and displacement

The observer is the fluid element whose Poisson position tends to the label $\bar{\mathbf{x}}_o$ as $\eta \to 0$,

$$\mathbf{x}_o(\eta) = \bar{\mathbf{x}}_o + \int_0^\eta \mathbf{v}(\eta', \mathbf{x}_o(\eta'))\mathrm{d}\eta', \qquad \delta x_{o1}^i = \int_0^{\bar\eta_o} v_1^i(\eta, \bar{\mathbf{x}}_o)\,\mathrm{d}\eta,$$
$$\frac{1}{2}\delta x_{o2}^i = \int_0^{\bar\eta_o}[\frac{1}{2}v_2^i + \delta x_{o1}^j(\eta)\partial_j v_1^i]\mathrm{d}\eta + \delta\eta_{o1}v_1^i|_{\bar o}, \tag{100}$$

with $\delta x_{o1}^j(\eta)$ the first integral with upper limit η and $\delta\mathbf{x}_o \equiv \mathbf{x}_o(\eta_o) - \bar{\mathbf{x}}_o$. (Starting the worldline integral at η_{init} instead, as in Sec. 2.4, selects a different fluid element, offset by $\int_0^{\eta_i}\mathbf{v} = O(\eta_i^2)$; this is irrelevant for C_ℓ but must be matched when comparing with the planar solver at finite η_i, Sec. 10.) Expanding the clock condition (17) with $1/\hat u^0 = 1 + \epsilon\Psi_1 + \epsilon^2[\frac{1}{2}\Psi_2 - \frac{1}{2}\Psi_1^2 - \frac{1}{2}|\mathbf{v}_1|^2] + O(\epsilon^3)$ and $\eta_{\mathrm{init}} = \eta_i(1+\delta_T)$, and dropping terms $O(\eta_i^2)$,

$$\delta\eta_{o1} = -\frac{1}{\bar\eta_o}\int_0^{\bar\eta_o}\eta\,\Psi_1(\eta, \bar{\mathbf{x}}_o)\,\mathrm{d}\eta, \tag{101}$$

$$\delta\eta_{o2} = -\frac{2}{\bar\eta_o}\Big\{\frac{1}{2}\delta\eta_{o1}^2 + \bar\eta_o\,\delta\eta_{o1}\Psi_1|_{\bar o} + \int_0^{\bar\eta_o}\eta[\frac{1}{2}\Psi_2 - \frac{1}{2}\Psi_1^2 - \frac{1}{2}|\mathbf{v}_1|^2 + \delta x_{o1}^j(\eta)\partial_j\Psi_1]_{(\eta,\bar{\mathbf{x}}_o)}\mathrm{d}\eta\Big\}, \tag{102}$$

both integrals running along the observer worldline. (The $O(\eta_i^2)$ terms, from the initial event, are kept in the numerical test of Sec. 10.) For the constant mode of Sec. 3.4 these give $\eta_o = \bar\eta_o/\lambda$ with λ the time rescaling of the dilation family, as they must. The clock and the displacement are direction independent random variables; they affect the map only through (97)–(99), the transport terms of (98) and the endpoint terms below.

7.3 Emission event

Solving (18), $1+\delta_T(\eta_e,\mathbf{x}(\eta_e))=\eta_e/\eta_*$, with $\mathbf{x}(\eta_e)=\bar{\mathbf{x}}(\eta_*)-\hat{\mathbf{n}}\,\delta\eta_e+\Delta\mathbf{x}(\eta_*)+\dots$,

$$\delta\eta_{e1}=\eta_*\,\delta_{T,1}|_*,\qquad \delta\eta_{e2}=\eta_*\Big[\delta_{T,2}+2\,\delta\eta_{e1}\,\delta'_{T,1}+2(\Delta\mathbf{x}_1(\eta_*)-\hat{\mathbf{n}}\,\delta\eta_{e1})\cdot\nabla\delta_{T,1}\Big]_*, \tag{103}$$

with $|_*$ the background emission point $(\eta_*,\bar{\mathbf{x}}(\eta_*))$. $\delta\eta_{e1}$ is the first-order emission-surface shift ("temperature at fixed time" bookkeeping of standard derivations); $\delta\eta_{e2}$ contains the second-order temperature perturbation and the lensed and time-delayed position of the first-order one.

7.4 Expansion about the background ray

Let $F(\eta)\equiv\mathcal{I}(\eta,\mathbf{x}(\eta);k(\eta))$. The three steps are: (1) *position*: $F=\mathcal{I}+\Delta x^j\partial_j\mathcal{I}+\frac{1}{2}\Delta x^j\Delta x^l\partial_j\partial_l\mathcal{I}$, all at $(\eta,\bar{\mathbf{x}}(\eta))$, with $k=k(\eta)$ inserted in $\mathcal{I}$; (2) *endpoints*:

$$\int_{\eta_e}^{\eta_o}F\,\mathrm{d}\eta=\int_{\eta_*}^{\bar\eta_o}F\,\mathrm{d}\eta+\sum_{j\ge1}\Big[\frac{\delta\eta_o^j}{j!}D^{j-1}F\Big]_{\bar\eta_o}-\sum_{j\ge1}\Big[\frac{\delta\eta_e^j}{j!}D^{j-1}F\Big]_{\eta_*}, \tag{104}$$

where D is the total derivative along the ray in the background representation ($D=\partial_\eta-\hat{\mathbf{n}}\cdot\nabla$ on fields, $\mathrm{d}k_1/\mathrm{d}\eta\to G_1$, $\mathrm{d}\Delta x_1/\mathrm{d}\eta\to k_1$, …); (3) at the observer endpoint the functionals take their initial values ($k_1\to k_{1,\mathrm{init}}|_{\bar o}$, $\Delta\mathbf{x}_1\to\delta\mathbf{x}_{o1}+\hat{\mathbf{n}}\,\delta\eta_{o1}$, …), while at the emission endpoint they appear as $k_1(\eta_*)$, $\Delta x_1(\eta_*)$. The outcome is stored as lists of terms, each labelled *ray* (integrated over $[\eta_*,\bar\eta_o]$ on the background ray), *obs* (at $(\bar\eta_o,\bar{\mathbf{x}}_o)$) or *emi* (at $(\eta_*,\bar{\mathbf{x}}(\eta_*))$). No equation of motion is used in building the lists; $|\hat{\mathbf{n}}|=1$ is imposed only in the $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ "split" versions.

First order. $L_1=-\int_{\eta_*}^{\bar\eta_o}I_1\,\mathrm{d}\eta$ (3 terms for $\hat{\mathbf{n}}=\hat{\mathbf{z}}$, 15 for general $\hat{\mathbf{n}}$), which on shell equals the Sachs–Wolfe form (24) including the observer monopole and dipole (App. D, C2-4).

7.5 L_2: named pieces

Collecting the ϵ^2 terms,

$$\frac{1}{2}L_2=-\int_{\eta_*}^{\bar\eta_o}\mathrm{d}\eta\Big[\frac{1}{2}\mathcal{I}_2|_{k=(1,-\hat{\mathbf{n}})}+k_1^i\frac{\partial\mathcal{I}_1}{\partial k^i}\Big|_{k=(1,-\hat{\mathbf{n}})}+\Delta x_1^j\partial_jI_1\Big]-\delta\eta_{o1}\,I_1|_{\bar o}+\delta\eta_{e1}\,I_1|_*, \tag{105}$$

where $\frac{1}{2}\mathcal{I}_2$ is the ϵ^2 part of $\mathcal{I}$ at fixed point and fixed background tangent and $k_1^i\partial\mathcal{I}_1/\partial k^i=-2k_1^i\hat\sigma_{1,ij}n^j$. Table 2 lists the pieces. Two simplifications occur in the $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ form: the radial direction perturbation $k_1^z=-(\Phi_1+\Psi_1)$, the $(1-2\Phi)$ of $\hat\sigma_{ij}$ and the factor $1/(-\hat u\cdot k)=1-\Psi-\hat{\mathbf{n}}\cdot\mathbf{v}+\dots$ cancel exactly (as they must for σ_{ee} with e a unit vector), so no undifferentiated Φ_1,Ψ_1 survive; and ω_2,Φ_2,Ψ_2 drop out of the second-order shear (ω cancels between $\partial_{(i}\hat u_{j)}$ and $\hat\Gamma^0_{ij}\hat u_0$). These fields *do* appear in L_3 and in the direct form below.

Explicitly, for $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ (all fields first order unless indexed, $\perp=x,y$, $\partial_\perp\cdot\mathbf{v}_\perp=\partial_xv^x+\partial_yv^y$,

piece	where	physical meaning	$\hat{\mathbf{n}}=\hat{\mathbf{z}}$	general $\hat{\mathbf{n}}$
(a) second-order shear	ray	intrinsic second-order shear of the fluid: v_2 and the tensor shear $\frac{1}{2}h_2'$ (ISW-like tensor term)	5	22
(b) quadratic local	ray	products of first-order fields at the same point: non-linearity of $\hat{\sigma}_{\mu\nu}k^\mu k^\nu/(-\hat{u}\cdot k)$	10	111
(c) direction (k_1)	ray	aberration at the observer ($\mathbf{v}_{1\perp}$ in $k_{1,\mathrm{init}}$) and accumulated deflection, changing the projection σ_{kk}	4	21
(d) lensing ($\Delta\mathbf{x}_{1\perp}$)	ray	first-order shear sampled at the transversely displaced point	6	45†
(e) time delay ($\Delta x_{1\parallel}$)	ray	first-order shear sampled at the radially displaced point (Shapiro delay, clock shift, radial observer displacement)	3	†
(f) clock ($\delta\eta_{o1}$)	obs	observation at a perturbed time: shift of the upper limit	3	15
(g) emission surface ($\delta\eta_{e1}$)	emi	emission on $T_{\mathrm{phys}}=T_*$ rather than $\eta=\eta_*$: shift of the lower limit	3	15
total			34	229

Table 2: Pieces of $\frac{1}{2}L_2$ (shear form, C2's groups); term counts as stored in `L_terms.pkl` (†(d) and (e) form one group for general $\hat{\mathbf{n}}$). The observer displacement $\delta\mathbf{x}_{o1}$ enters through $\Delta\mathbf{x}_1$ in (d), (e) (and through k_2 only at third order). Full expressions: App. B.1.

$I_1=\partial_z v_1^z-\frac{1}{3}\partial_i v_1^i$, and $I_1[\mathbf{v}_2]$ the same with $\mathbf{v}_2$):

$$\frac{1}{2}L_2=\int_{\eta_*}^{\bar{\eta}_o}\mathrm{d}\eta\,\Big\{\underbrace{-\frac{1}{4}h_{2,zz}'-\frac{1}{2}I_1[\mathbf{v}_2]}_{\text{(a)}}\;\underbrace{+\mathbf{v}_\perp\cdot(\frac{1}{3}\mathbf{v}_\perp'-\partial_z\mathbf{v}_\perp-\nabla_\perp v^z)+v^z(\frac{1}{3}\partial_\perp\cdot\mathbf{v}_\perp-\frac{2}{3}v^{z\prime}-\frac{2}{3}\partial_z v^z)}_{\text{(b)}}$$
$$\underbrace{+\vec{k}_{1\perp}\cdot(\partial_z\mathbf{v}_\perp+\nabla_\perp v^z)}_{\text{(c)}}\;\underbrace{-\Delta\mathbf{x}_{1\perp}\cdot\nabla_\perp I_1}_{\text{(d)}}\;\underbrace{-\Delta x_1^z\,\partial_z I_1}_{\text{(e)}}\Big\}\;\underbrace{-\delta\eta_{o1}I_1|_{\bar{o}}}_{\text{(f)}}\;\underbrace{+\delta\eta_{e1}I_1|_*}_{\text{(g)}},\tag{106}$$

with $h_{2,zz}=-h_{2,xx}-h_{2,yy}$, k_1, Δx_1 from (96)–(97), $\delta\eta_{o1}$ from (101), $\delta\eta_{e1}=\eta_*\delta_{T,1}|_*$. For general $\hat{\mathbf{n}}$, (a) reads $-\frac{1}{2}I_1[\mathbf{v}_2]-\frac{1}{4}n^i n^j h_{2,ij}'$ and (d)+(e) $=-\Delta\mathbf{x}_1\cdot\nabla I_1$. No equation of motion has been used; $\mathbf{v}'$ may be eliminated with the Euler equation (22).

7.6 Direct (Sachs–Wolfe) form and equivalence

The alternative organisation integrates the photon energy directly. Since $\hat{E}=\hat{p}^0(-\hat{u}_\mu k^\mu)$ and $\mathrm{d}\ln\hat{p}^0/\mathrm{d}\eta=-\hat{\Gamma}^0_{\mu\nu}k^\mu k^\nu$,

$$\Lambda=\ln\frac{T_{\mathrm{obs}}}{T_o}=-\int_{\eta_e}^{\eta_o}\hat{\Gamma}^0_{\mu\nu}k^\mu k^\nu\,\mathrm{d}\eta+\Big[\ln(-\hat{u}_\mu k^\mu)-\ln(1+\delta_T)\Big]_e^o,\tag{107}$$

expanded with the same ray, endpoints and Taylor rules. The boundary function is $\Psi+\hat{\mathbf{n}}\cdot\mathbf{v}-\delta_T+O(\epsilon^2)$, and at first order (107) gives (24) after one integration by parts. At second order, $\hat{\mathbf{n}}=\hat{\mathbf{z}}$,

with $\mathcal{G}_1 \equiv \Phi_1' - \Psi_1' + 2\partial_z\Psi_1 = -(\hat{\Gamma}^0_{kk})_1$ and $B_1 \equiv \Psi_1 - \delta_{T,1} + v_1^z$:

$$\begin{aligned}
\frac{1}{2}L_2^{\rm dir} = \int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta \Big\{ & \frac{1}{2}(\Phi_2' - \Psi_2') + \partial_z\Psi_2 - \frac{1}{4}h_{2,zz}' + \frac{1}{2}\partial_z\omega_{2,z} \\
& + 2\Phi_1\Phi_1' + 2\Phi_1\partial_z\Psi_1 + 2\Psi_1\Psi_1' - 2\Psi_1\partial_z\Psi_1 \\
& - 2\vec{k}_{1\perp}\cdot\nabla_\perp\Psi_1 + \Delta\mathbf{x}_1\cdot\nabla\mathcal{G}_1 \Big\} \\
+ \Big[& \frac{1}{2}(\Psi_2 - \delta_{T,2} + \omega_{2,z} + v_2^z) - \Phi_1 v^z - \Psi_1^2 - \Psi_1 v^z + \frac{1}{2}\delta_{T,1}^2 - \frac{1}{2}|\mathbf{v}_\perp|^2 \\
& + \delta\eta_{o1}(\mathcal{G}_1 + B_1') + \delta\mathbf{x}_{o1}\cdot\nabla B_1 \Big]_{\bar{o}} \\
+ \Big[& -\frac{1}{2}(\Psi_2 - \delta_{T,2} + \omega_{2,z} + v_2^z) + \Phi_1 v^z + \Psi_1^2 + \Psi_1 v^z - \frac{1}{2}\delta_{T,1}^2 - \frac{1}{2}|\mathbf{v}_\perp|^2 \\
& - \delta\eta_{e1}(\mathcal{G}_1 + DB_1) - \Delta\mathbf{x}_1(\eta_*)\cdot\nabla B_1 + \vec{k}_{1\perp}(\eta_*)\cdot\mathbf{v}_\perp \Big]_*,
\end{aligned} \tag{108}$$

76 terms (179 for general $\hat{\mathbf{n}}$; App. B.2). Its pieces have the familiar names: second-order ISW/Rees–Sciama, vector and tensor integrand $\frac{1}{2}(\Phi_2' + \Psi_2' + \ldots)$ after integration by parts, quadratic ISW corrections, transverse deflection, lensing and Shapiro delay of the ISW integrand, second-order Sachs–Wolfe and Doppler at both ends with the clock, displacement and emission-surface shifts.

Equivalence. The two forms are equal, not merely equivalent on average: (i) at the integrand level along an arbitrary perturbed ray, $-\hat{\Gamma}^0_{kk} + \mathrm{d}_\eta[\ln(-\hat{u}\cdot k) - \ln\hat{T}] + \hat{\sigma}_{kk}/(-\hat{u}\cdot k) + \hat{\theta}\hat{g}_{kk}/(3(-\hat{u}\cdot k)) \equiv 0$ modulo $\hat{\nabla}_\mu\hat{T}^{\mu\nu} = 0$ at orders 1, 2, 3 (App. D, C2-3); (ii) at the assembled level, $\Lambda^{\rm dir} - \Lambda^{\rm shear}$ vanishes piece by piece (ray, obs, emi) at orders 1, 2 *and* 3 modulo the conservation equations and their derivatives (C2-7); (iii) numerically, for the plane wave of Sec. 10, the two forms agree to 4×10^{-10} (L_2) and 9×10^{-7} (L_3), the level of the quadrature (A1-4). With $\mathbf{v} = 0$, emission at fixed coordinate time and the observer at the origin, (108) reduces term by term to the second-order Sachs–Wolfe result of Boubekeur et al. [18], the only remainder being $\int[\Delta\mathbf{x}_1\cdot\nabla(\Phi_1' + \Psi_1') + 2\Phi_1\Phi_1' - 2\Psi_1\Psi_1']$, which vanishes under their matter-era assumption $\Phi_1' = \Psi_1' = 0$ (C2-10).

7.7 From L_2 to $\Theta^{(2)}$

By (54) and (55),

$$m_2 = L_2 + L_1^2, \qquad \Theta^{(2)}_{\ell m} = m_{2,\ell m} - 2\langle m_1\rangle_{\rm sky}\, m_{1,\ell m} \quad (\ell \geq 1), \tag{109}$$

with L_2 twice the expression (106) (or (108)) and the second-order fields of Sec. 4.2 inserted. In the language of Sec. 5: L_1^2 and the emission-surface piece (g) are products of type-A maps on the sky (type B); the local pieces (a), (b) are line-of-sight integrals of fields at one spacetime point (type C, with the second-order kernels of (47) in (a)); the direction, lensing and time-delay pieces (c)–(e) are nested unequal-time products (type D); the clock piece (f) is a direction-independent random variable times a quadratic polynomial in $\hat{\mathbf{n}}$ built from fields at the observer, and contributes to $\ell \leq 2$ only. The kernels $K_2(\mathbf{k}_1, \mathbf{k}_2; \hat{\mathbf{n}})$ and their harmonic transforms are assembled in Sec. 9.

8 The third-order map $\Theta^{(3)}$

The third-order map is built with exactly the machinery of Sec. 7: the integrand $\mathcal{I}$ of (89) expanded to $O(\epsilon^3)$ with $k^i = -n^i + \epsilon k_1^i + \frac{\epsilon^2}{2}k_2^i$ (the k_3 and Δx_3 terms do not enter, since $\mathcal{I} = O(\epsilon)$), the

position expansion to second order in $\Delta\mathbf{x} = \epsilon\Delta\mathbf{x}_1 + \frac{\epsilon^2}{2}\Delta\mathbf{x}_2$, the endpoint formula (104) with $\delta\eta_o = \epsilon\delta\eta_{o1} + \frac{\epsilon^2}{2}\delta\eta_{o2}$ and $\delta\eta_e = \epsilon\delta\eta_{e1} + \frac{\epsilon^2}{2}\delta\eta_{e2}$, and the functionals (96)–(99), (101)–(103). The new ingredients relative to L_2 are the second-order functionals k_2, $\Delta\mathbf{x}_2$, $\delta\eta_{o2}$, $\delta\eta_{e2}$, the triad and direction map at second order (93), (95), the second-order fields (including ω_2, Φ_2, Ψ_2, which now appear in products), and the third-order fields in the local shear.

8.1 L_3: named groups

C2's result, $\frac{1}{6}L_3$, has 359 terms in 32 groups for $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ (2484 terms in 24 groups for general $\hat{\mathbf{n}}$, without $|\hat{\mathbf{n}}|= 1$). Table 3 organises the 32 groups by the three endpoints and by physical content; the machine-generated lists, with the explicit expressions of all groups of at most ten terms, are in App. B.3. The structure is that of L_2 promoted by one order: every second-order piece (a)–(g) of Table 2 reappears (1) with one first-order field replaced by a second-order field, (2) multiplied by one more first-order factor (a field, a functional k_1, $\Delta\mathbf{x}_1$, or a shift $\delta\eta_{o1}, \delta\eta_{e1}$), or (3) with a first-order functional replaced by its second-order counterpart (k_2, $\Delta\mathbf{x}_2$, $\delta\eta_{o2}$, $\delta\eta_{e2}$). The one genuinely new piece is the third-order shear,

$$\frac{1}{6}L_3 \supset -\frac{1}{6}\int_{\eta_*}^{\bar{\eta}_o}\Big(I_1[\mathbf{v}_3] + \frac{1}{2}n^i n^j h'_{3,ij}\Big)\mathrm{d}\eta \qquad (5 \text{ terms for } \hat{\mathbf{n}} = \hat{\mathbf{z}}), \tag{110}$$

and, as at second order, ω_3, Φ_3, Ψ_3 do not appear in the shear form (they do in the direct form). The second-order functionals carry the second-order physics of the ray: k_2 contains the second-order aberration (95), the transport of the initial direction to the background event, and the second-order deflection $\int[\frac{1}{2}G_2 + \Delta\mathbf{x}_1\cdot\nabla G_1]$ (lens–lens coupling and the deflection by second-order potentials, ω_2, h_2); $\Delta\mathbf{x}_2$ is the second-order displacement (post-Born lensing and time delay); $\delta\eta_{e2}$ the emission surface at second order including the lensed position of $\delta_{T,1}$ (103); $\delta\eta_{o2}$ the second-order clock (102).

8.2 From L_3 to $\Theta^{(3)}$

By (54) and (55),

$$m_3 = L_3 + 3L_1L_2 + L_1^3, \qquad \Theta^{(3)}_{\ell m} = m_{3,\ell m} - 3\langle m_1\rangle_{\rm sky} m_{2,\ell m} - 3\langle m_2\rangle_{\rm sky} m_{1,\ell m} + 6\langle m_1\rangle^2_{\rm sky} m_{1,\ell m} \quad (\ell \geq 1), \tag{111}$$

which C2 re-derived from the exponential and ratio series (App. D, C2-8). Only the correlation of $\Theta^{(3)}$ with $\Theta^{(1)}$ is needed, $C_\ell^{(13)}$ of (57), decomposed as in (E0) of Sec. 5.7.

What survives the self-contraction. In $\langle m_1\, m_3^*\rangle$ one ζ of m_3 is paired with m_1 and the remaining two are contracted with each other. Qualitatively the groups of Table 3 and the products $3L_1L_2$, L_1^3 fall into four classes (their explicit evaluation is the subject of Sec. 9):

(i) *Variance × map.* Products of three first-order objects (cubic local terms; k_1^2, $(\Delta\mathbf{x}_1)^2$, $k_1\Delta\mathbf{x}_1$, $\delta\eta_{o1}^2$, $\delta\eta_{e1}^2$, …; first-order parts of L_1L_2 and L_1^3): contracting two of the three factors leaves a one-point variance or an equal-sky-point correlation times a first-order map, structure (i) of Sec. 5.7. The $(\Delta\mathbf{x}_{1\perp})^2\partial\partial I_1$ group, for instance, produces the convergence-type term $-\ell(\ell+1)R\,C_\ell$ of the lensing control case (Sec. 5.8), now with the radiation-era deflection correlated with the source.

group (fields)	meaning	terms
ray (integrated over $[\eta_*, \bar{\eta}_o]$)		
local (3)	third-order shear (110)	5
local (2+1), local (1+1+1)	non-linearity of $\mathcal{I}$: second×first, cubic	57, 55
$k_{1\perp}$ (1+1), (2); $k_{1\perp}^2$ (1)	direction perturbation acting on quadratic / second-order shear; quadratic in the deflection	22, 6, 8
$k_{2\perp}$ (1), k_2^z (1)	second-order direction (aberration, transport, post-Born deflection) acting on I_1	4, 3
lensing $\Delta\mathbf{x}_{1\perp}$ (1+1), (2); $(\Delta\mathbf{x}_{1\perp})^2$ (1)	quadratic / second-order shear at the lensed point; second-order Taylor term (convergence-type)	50, 10, 9
lensing $\Delta\mathbf{x}_{2\perp}$ (1)	second-order (post-Born) transverse displacement	6
time delay Δx_1^z (1+1), (2); $(\Delta x_1^z)^2$ (1)	as lensing, radial	26, 5, 3
time delay Δx_2^z (1)	second-order radial displacement	3
mixed $\Delta\mathbf{x}_{1\perp}\Delta x_1^z$, $k_{1\perp}\Delta\mathbf{x}_{1\perp}$, $k_{1\perp}\Delta x_1^z$ (1)	cross terms of displacement and direction	6, 8, 4
obs (at $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$)		
$\delta\eta_{o2}$ (1)	second-order clock	3
$\delta\eta_{o1}$ (1+1), (2)	clock acting on quadratic / second-order shear at the observer	6, 5
$\delta\eta_{o1}\delta x_{o1}$ (1)	clock × observer displacement	9
$\delta\eta_{o1}^2$ (1)	second-order Taylor term of the upper limit (two groups)	3+6
emi (at $(\eta_*, \bar{\mathbf{x}}(\eta_*))$)		
$\delta\eta_{e2}$ (1)	second-order emission surface	3
$\delta\eta_{e1}$ (1+1), (2)	emission shift acting on quadratic / second-order shear	10, 5
$\delta\eta_{e1}k_{1\perp}$, $\delta\eta_{e1}\Delta\mathbf{x}_{1\perp}$, $\delta\eta_{e1}\Delta x_1^z$ (1)	emission shift × direction / lensing / time delay at η_*	4, 6, 3
$\delta\eta_{e1}^2$ (1)	second-order Taylor term of the lower limit	6
total	32 groups	359

Table 3: The groups of $\frac{1}{6}L_3$ (shear form, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$), from `L_terms.pkl['L_split'][3]`. "(1+1)" etc. gives the orders of the explicit fields; functionals and shifts are counted separately. The direct form has 480 terms (69 groups) for $\hat{\mathbf{n}} = \hat{\mathbf{z}}$.

(ii) *Tadpole (mean field).* A second-order field X_2 times a first-order factor, with the first-order factor paired with m_1: the contraction rule $A_1B_2 \to A_k\langle B_2\rangle$ of (51) gives the mean field $\langle X_2\rangle$ times a first-order map. This includes the Ψ_2 in $\delta\eta_{o2}$, the $\delta_{T,2}$ in $\delta\eta_{e2}$ and the zero-mode parts of the local (2+1) terms. Individual tadpoles depend on the zero-mode convention of Sec. 3.4; only their sum with the clock, emission-surface and normalisation terms is physical (Sec. 10).

(iii) *Cross-contracted.* The same products with the first-order factor contracted with one leg of X_2 (rule $A_1B_2 \to 2A_qK_B(-\mathbf{q}, \mathbf{k})$), and the cross-pairings of three first-order factors at different points: a loop convolution, structure (ii) of Sec. 5.7.

(iv) *Third-order fields.* The local shear (110) (and $\Phi_3, \Psi_3, \delta_{T,3}$ in the direct form), which enter only through the contracted fields X_{3c} of (50), including the amplitude C_3 (35): structure (iii).

The normalisation terms $-3\langle m_1\rangle_{\rm sky}m_2 - 3\langle m_2\rangle_{\rm sky}m_1 + 6\langle m_1\rangle_{\rm sky}^2 m_1$ of (111) supply the remaining pieces of (E0). The *obs* groups are polynomials in $\hat{\mathbf{n}}$ of bounded degree (all fields at the observer event, direction-independent shifts), so their direct contribution to $C_\ell^{(13)}$ is confined to a few low

multipoles; through the products $3L_1L_2$, L_1^3 and the normalisation the observer quantities nevertheless affect all ℓ.

9 Assembly of C_ℓ

[**To be supplied:** Assembly agent: $C_\ell^{(11)}$ (closed form, agent C5), $\frac{1}{4}C_\ell^{(22)}$ and $\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}$ as fully specified integrals using the rules of Sec. 5; explicit statement of which pieces are evaluated in closed form and which remain as declared integrals (Sec. 12).]

10 Checks of the assembled observable

[**To be supplied:** Dilation check (constant ζ_1 unobservable: tadpole independence, Sec. 3.4), gradient check (Mirbabayi–Zaldarriaga), comparison of the assembled $\Theta^{(1,2,3)}$ with the non-perturbative planar solver (agent N1, reference $a_{\ell 0}$ for a single plane wave), independent referee of the second/third-order kernels (agent V3), closed-form time integrals (agent C4). The checks already performed on the settled parts are listed in App. D.]

11 Monochromatic primordial spectrum

[**To be supplied:** Specialisation of the assembled C_ℓ to $P(k) = 2\pi^2\sigma^2\delta(k-k_0)/k_0^2$ (Sec. 5.9) with $\sigma^2 = \langle\zeta^2\rangle$ held fixed; the single μ-integral forms of Sec. 5.9; treatment of the $\mu \to -1$ ($K \to 0$) and $\mu \to \pm 1$ endpoints; exchange of limits justified analytically.]

12 Unresolved steps

[**To be supplied:** Complete list of declared-but-unevaluated integrals (time integrals over Green's functions, η-line-of-sight integrals, k-, q-, μ-loop integrals), the tadpole zero-mode convention and its validation, any exchange of limit and integration not yet justified. Items flagged [unevaluated integral: ...] or [To be supplied] in the settled sections must be collected here.]

A Explicit kernels (machine generated)

All expressions below are exported directly from the engine output of agent C1 (`cmb4/code/C1_einstein/second_o` `third_order_ic.pkl`; loader `kernels_api.load()`, `closed_form()` inserts T_Φ). Notation: $T_q \equiv T_\Phi(q\eta/\sqrt{3})$, $\dot{T}_q \equiv \partial_\eta T_q$, $p = |\mathbf{k}-\mathbf{q}|$, $\mu = \hat{\mathbf{k}}\cdot\hat{\mathbf{q}} = (k^2+q^2-p^2)/(2kq)$; every second-order kernel is per $\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}$ in the sense of (46). Sources are those of $E^{(2)}_{\mu\nu} = 2[\epsilon^2]E_{\mu\nu}$ in the frame $\mathbf{k} = k\hat{\mathbf{z}}$, $\mathbf{q}$ in the x–z plane. The transfer functions (41) and the dictionary $\dot{T}_q = \partial_\eta T_\Phi$ with $\partial_\eta = (q/\sqrt{3})\,\mathrm{d}/\mathrm{d}x$ convert everything to closed form in $\sin, \cos$ of $q\eta/\sqrt{3}$, $p\eta/\sqrt{3}$.

A.1 Second-order kernels

K_Q **(traceless scalar,** $Q_2 = \Phi_2 - \Psi_2$**)**

$$\begin{aligned}
&-\frac{T_p T_q \left(13k^4 - 6k^2p^2 - 6k^2q^2 + 9p^4 - 18p^2q^2 + 9q^4\right)}{4k^4} \\
&- \frac{T_p \dot{T}_q \eta \left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4} \\
&- \frac{T_q \dot{T}_p \eta \left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4} \\
&- \frac{\dot{T}_p \dot{T}_q \eta^2 \left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4}
\end{aligned} \tag{112}$$

$K_{S_{00}}$ **(source of the 00 constraint)**

$$\begin{aligned}
&-\frac{2T_p T_q \left(\eta^2k^2 + 3\eta^2p^2 + 3\eta^2q^2 - 12\right)}{\eta^2} - T_p \dot{T}_q \eta \left(-k^2 + p^2 + q^2\right) \\
&- T_q \dot{T}_p \eta \left(-k^2 + p^2 + q^2\right) + \dot{T}_p \dot{T}_q \left(\eta^2k^2 - \eta^2p^2 - \eta^2q^2 + 6\right)
\end{aligned} \tag{113}$$

$K_{S_{0z}}$ **(longitudinal** $0i$ **source,** S_0^L**)**

$$\begin{aligned}
&-\frac{2iT_p T_q \left(-\eta^2k^2p^2 - \eta^2k^2q^2 + \eta^2p^4 - 2\eta^2p^2q^2 + \eta^2q^4 - 6k^2\right)}{3\eta k} \\
&- \frac{iT_p \dot{T}_q \left(-2\eta^2k^2p^2 + 2\eta^2p^4 - 2\eta^2p^2q^2 - 27k^2 + 9p^2 - 9q^2\right)}{3k} \\
&+ \frac{iT_q \dot{T}_p \left(2\eta^2k^2q^2 + 2\eta^2p^2q^2 - 2\eta^2q^4 + 27k^2 + 9p^2 - 9q^2\right)}{3k} + 4i\dot{T}_p \dot{T}_q \eta k
\end{aligned} \tag{114}$$

K_{S_Φ} **(source of** $\mathcal{L}_\Phi \Phi_2 = S_\Phi$**)**

$$\begin{aligned}
&\frac{T_p T_q \left(-7k^6 - 19k^4p^2 - 19k^4q^2 + 7k^2p^4 - 22k^2p^2q^2 + 7k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6\right)}{12k^4} \\
&+ \frac{T_p \dot{T}_q \eta \left(k^2 + p^2\right) \left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4} \\
&+ \frac{T_q \dot{T}_p \eta \left(k^2 + q^2\right) \left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4} \\
&+ \frac{\dot{T}_p \dot{T}_q \left(-\eta^2k^6 - 2\eta^2k^4p^2 - 2\eta^2k^4q^2 + 3\eta^2k^2p^4 - 6\eta^2k^2p^2q^2 + 3\eta^2k^2q^4 + 12k^4 - 24k^2p^2 - 24k^2q^2 + 36p^4 - 72p^2q^2 -}{12k^4}
\end{aligned} \tag{115}$$

K_ω **(vector metric,** $\omega_{2,i} = \int P_{ij} q_j K_\omega \zeta\zeta$**)**

$$-\frac{2iT_p T_q \eta \left(p - q\right) \left(p + q\right)}{k^2} - \frac{2iT_p \dot{T}_q \eta^2 p^2}{k^2} + \frac{2iT_q \dot{T}_p \eta^2 q^2}{k^2} \tag{116}$$

K_{v^T} **(transverse velocity,** $v_{2,i}^T = \int P_{ij} q_j K_{v^T} \zeta\zeta$**)**

$$\begin{aligned}
&-\frac{iT_p T_q \eta \left(p - q\right) \left(p + q\right) \left(\eta^2k^2 - 24\right)}{12k^2} - \frac{iT_p \dot{T}_q \eta^2 \left(\eta^2k^2p^2 + 18k^2 - 24p^2\right)}{12k^2} \\
&+ \frac{iT_q \dot{T}_p \eta^2 \left(\eta^2k^2q^2 + 18k^2 - 24q^2\right)}{12k^2}
\end{aligned} \tag{117}$$

K_h **(tensor source,** $h_{2,ij} = \Lambda_{ij,kl} q_k q_l \int_0^\eta G_h K_h$**)**

$$12T_pT_q + 4T_p\dot{T}_q\eta + 4T_q\dot{T}_p\eta + 4\dot{T}_p\dot{T}_q\eta^2 \tag{118}$$

$K^0_{\Phi_2}$ **(initial amplitude of** Φ_2**, Eq. (32))**

$$-\frac{7k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{9k^4} \tag{119}$$

$K_Q(\eta \to 0)$

$$-\frac{13k^4 - 6k^2p^2 - 6k^2q^2 + 9p^4 - 18p^2q^2 + 9q^4}{9k^4} \tag{120}$$

kernel of the local quadratic part $-\frac{1}{4}\delta_1^2 - 2\Phi_1^2$ **of** $\zeta^{(2)}$**, Eq. (29)** $(\to -\frac{4}{3}$ **as** $\eta \to 0)$

$$-\frac{T_pT_q\left(\eta^4p^2q^2 + 3\eta^2p^2 + 3\eta^2q^2 + 27\right)}{9} - \frac{T_p\dot{T}_q\eta\left(\eta^2p^2 + 3\right)}{3} - \frac{T_q\dot{T}_p\eta\left(\eta^2q^2 + 3\right)}{3} - \dot{T}_p\dot{T}_q\eta^2 \tag{121}$$

Algebraic second-order scalars (with S_{00}, S_{0z} the corresponding sources)

$$\delta_2 = \frac{S_{00}\eta^2}{3} - \frac{2\eta^2k^2\Phi_2(\eta)}{3} - 2\eta\frac{d}{d\eta}\Phi_2(\eta) - 2\Psi_2(\eta) \tag{122}$$

$$v_{2,z} = \frac{\eta\left(-S0z\eta - 2i\eta k\frac{d}{d\eta}\Phi_2(\eta) - 2ik\Psi_2(\eta)\right)}{4} \tag{123}$$

A.2 Homogeneous solutions and Green's functions

u_1

$$\frac{3\sqrt{3}\left(-\frac{\sqrt{3}\eta k\cos\left(\frac{\sqrt{3}\eta k}{3}\right)}{3} + \sin\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{124}$$

u_2

$$\frac{3\sqrt{3}\left(\frac{\sqrt{3}\eta k\sin\left(\frac{\sqrt{3}\eta k}{3}\right)}{3} + \cos\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{125}$$

$\hat{g} = 3u_1$

$$\frac{9\sqrt{3}\left(-\frac{\sqrt{3}\eta k\cos\left(\frac{\sqrt{3}\eta k}{3}\right)}{3} + \sin\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{126}$$

$G_\Phi(k;\eta,\eta')$

$$\frac{\sqrt{3}\eta'\left(\eta'\eta k^2\sin\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right) + \sqrt{3}\eta' k\cos\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right) - \sqrt{3}\eta k\cos\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right) + 3\sin\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)\right)}{\eta^3k^3} \tag{127}$$

$G_h(k;\eta,\eta')$

$$\frac{\eta' \sin\left(k\left(-\eta'+\eta\right)\right)}{\eta k} \tag{128}$$

$G_\omega(\eta,\eta')$ **(for the ODE form of the vector equation)**

$$\frac{\eta'^2}{\eta^2} \tag{129}$$

A.3 Third-order initial amplitude and intermediate limits

Per $P(q)\,\zeta_{\mathbf{k}}$ in the contracted sense of (50); $p^2 = k^2 - 2kq\mu + q^2$.

$C_3(k;q,\mu)$**, Eq.** (35)

$$-\frac{16\left(2k^5+12k^4\mu q-42k^3\mu^2q^2-10k^3q^2+36k^2\mu^3q^3+32k^2\mu q^3+3k\mu^4q^4-42k\mu^2q^4+kq^4+8\mu q^5\right)}{9k\left(k^2-2k\mu q+q^2\right)^2} \tag{130}$$

C_3 **in invariant form** (k,q,p)

$$-\frac{35k^8+12k^6p^2-84k^6q^2+66k^4p^4-52k^4p^2q^2+66k^4q^4-84k^2p^6+84k^2p^4q^2+20k^2p^2q^4-20k^2q^6+3p^8-12p^6q^2}{9k^4p^4} \tag{131}$$

$Q_{3c}(\eta\to 0)$

$$-\frac{16q\left(44k^4\mu-142k^3\mu^2q-18k^3q+108k^2\mu^3q^2+104k^2\mu q^2+9k\mu^4q^3-126k\mu^2q^3-3kq^3+24\mu q^4\right)}{9k\left(k^2-2k\mu q+q^2\right)^2} \tag{132}$$

$\lim_{\eta\to 0}\eta^2 K_{3c,S_{00}}$

$$-\frac{256kq\left(2k^2\mu-k\mu^2q-3kq+2\mu q^2\right)}{3\left(k^2-2k\mu q+q^2\right)^2} \tag{133}$$

A.4 Tadpole (mean-field) functions

$\mathrm{tad}_X(q,\eta)=\langle K_X(\mathbf{q},-\mathbf{q};\eta)\rangle_{\mathrm{ang}}$, Eq. (52), angle-averaged $\mathbf{k}\to 0$ limits of the second-order kernels (convention of Sec. 3.4); small-η behaviour in brackets.

$\mathrm{tad}_{Q_2}\ [-\frac{16}{9}+\frac{88}{405}q^2\eta^2+\ldots]$

$$\frac{8\left(\eta^6q^6\sin^2\left(\frac{\sqrt{3}\eta q}{3}\right)+\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+6\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-6\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+18\eta^2q^2\left(16\sin^2\left(\frac{\sqrt{3}\eta q}{3}\right)-\right.\right.}{15\eta^6q^6} \tag{134}$$

$\mathrm{tad}_{S_{00}}\ [\frac{32}{3\eta^2}-\frac{272}{45}q^2+\ldots]$

$$\frac{24\left(\frac{\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)}{2}-\frac{\eta^6q^6}{2}-2\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-21\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-6\eta^4q^4+36\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+126\eta^2q^2\right.}{\eta^8q^6} \tag{135}$$

$\mathrm{tad}_{S_\Phi}\ [-\frac{56}{81}q^2+\ldots]$

$$\frac{4\left(2\sqrt{3}\eta^7q^7\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+12\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-30\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-684\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-270\eta^4q^4+1764\sqrt{3}\right.}{45\eta^8q^6} \tag{136}$$

$\mathrm{tad}_{\Phi_2}\ [-\frac{8}{9}-\frac{28}{405}q^2\eta^2+\ldots]$

$$\frac{-116\eta^6q^6+3\eta^3q^3\left(-2\sqrt{3}\eta^2q^2\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-18\eta q\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)+180\eta q+33\sqrt{3}\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)\right)+108\eta^2q^2\left(17\cos\left(\frac{2\sqrt{}\right.\right.}{45\eta^6q^6} \tag{137}$$

$\mathrm{tad}_{\delta_2}\ [\frac{16}{9}-\frac{472}{405}q^2\eta^2+\ldots]$

$$\frac{4\left(45\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)+19\eta^6q^6-186\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-1944\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-270\eta^4q^4+3879\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{}\right.}{45\eta^6q^6} \tag{138}$$

A.5 Not reproduced here

The contracted third-order source kernels $K_{3c,X}$ for $X\in\{Q,S_{00},S_{0z},S_\Phi,S_{0x},S_{xz},S_{xx}-S_{yy}\}$ (157–830 terms each, containing the declared integrals $\mathcal{K}_{\Phi_2}(a,b,c;\eta)$, $\mathcal{K}_h(a,b,c;\eta)$ as symbols) are in `third_order.pkl`/`third_order.tex`; the position-space sources $S^{(2)}_{\mu\nu}$, $S^{(3)}_{\mu\nu}$ in `position_sources.tex`; the full $E^{(n)}_{\mu\nu}$ in `E.tex`.

B Line-of-sight term lists of L_2 and L_3

This appendix is generated mechanically from the deliverable `code/C2_observable/L_terms.pkl` of agent C2 (Secs. 7–8). The stored expressions are the coefficients of ϵ^m in $\Lambda=-S$, i.e. L_1, $\frac{1}{2}L_2$, $\frac{1}{6}L_3$. Notation: v^i_n, Ψ_n, Φ_n, $\delta_{T,n}$, $\omega_{n,i}$, $h_{n,ij}$ are the order-n Poisson-gauge fields of (6)–(7) with $h_{n,zz}\equiv -h_{n,xx}-h_{n,yy}$ substituted; k^i_1, k^i_2, Δx^i_1, Δx^i_2 are the ray functionals (96)–(99) (functions of η on the ray, evaluated at η_* where marked); $\delta\eta_{o1},\delta\eta_{o2}$, δx^i_{o1} the observer-event shifts (101)–(100); $\delta\eta_{e1},\delta\eta_{e2}$ the emission shifts (103). For $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ the radial direction functional has been eliminated with the null condition $k^z_1=-(\Phi_1+\Psi_1)$ and $|\hat{\mathbf{n}}|=1$ used; the general-$\hat{\mathbf{n}}$ lists keep n^i symbolic without $|\hat{\mathbf{n}}|=1$ and are therefore longer. Counts: L_1: 3 terms ($\hat{\mathbf{n}}=\hat{\mathbf{z}}$) / 15 (general $\hat{\mathbf{n}}$); $\frac{1}{2}L_2$: 34 / 229; $\frac{1}{6}L_3$: 359 / 2484 (shear form); direct form $\frac{1}{2}L_2$: 76 / 179, $\frac{1}{6}L_3$: 480 / 1402.

B.1 $\frac{1}{2}L_2$, shear form, $\hat{\mathbf{n}}=\hat{\mathbf{z}}$: all 34 terms

Each group is listed as `[where] name | fields:orders` exactly as stored in `L_terms.pkl['L_split'][2]`; "ray" groups are integrated $\int_{\eta_*}^{\bar\eta_o}\mathrm{d}\eta$ with all fields at $(\eta,\bar{\mathbf{x}}(\eta))$, "obs" groups are evaluated at $(\bar\eta_o,\bar{\mathbf{x}}_o)$, "emi" groups at $(\eta_*,\bar{\mathbf{x}}(\eta_*))$. Field orders refer to the explicit fields; the ray functionals $k^i_1,\Delta x^i_1$ and the shifts $\delta\eta_{o1},\delta\eta_{e1}$ are first order. Primes are ∂_η; all v are v^i_1 unless indexed otherwise.

`[ray] local | fields:2` (5 terms)

$$\frac{1}{4}h_{2,xx}{}' + \frac{1}{4}h_{2,yy}{}' + \frac{1}{6}\partial_x v_2^x + \frac{1}{6}\partial_y v_2^y - \frac{1}{3}\partial_z v_2^z$$

`[ray] transverse direction k1 (aberration+deflection) | fields:1` (4 terms)

$$k_1^x\partial_z v_1^x + k_1^x\partial_x v_1^z + k_1^y\partial_z v_1^y + k_1^y\partial_y v_1^z$$

`[ray] local | fields:1+1` (10 terms)

$$\frac{1}{3}v_1^x v_1^{x\prime} - v_1^x\partial_z v_1^x - v_1^x\partial_x v_1^z + \frac{1}{3}v_1^y v_1^{y\prime} - v_1^y\partial_z v_1^y - v_1^y\partial_y v_1^z + \frac{1}{3}v_1^z\partial_x v_1^x + \frac{1}{3}v_1^z\partial_y v_1^y - \frac{2}{3}v_1^z v_1^{z\prime} - \frac{2}{3}v_1^z\partial_z v_1^z$$

`[ray] time delay (radial Dx1) | fields:1` (3 terms)

$$-\frac{2}{3}\Delta x_1^z\partial_z^2 v_1^z + \frac{1}{3}\Delta x_1^z\partial_x\partial_z v_1^x + \frac{1}{3}\Delta x_1^z\partial_y\partial_z v_1^y$$

`[ray] lensing (transverse Dx1) | fields:1` (6 terms)

$$\frac{1}{3}\Delta x_1^x\partial_x^2 v_1^x + \frac{1}{3}\Delta x_1^x\partial_x\partial_y v_1^y - \frac{2}{3}\Delta x_1^x\partial_x\partial_z v_1^z + \frac{1}{3}\Delta x_1^y\partial_y^2 v_1^y + \frac{1}{3}\Delta x_1^y\partial_x\partial_y v_1^x - \frac{2}{3}\Delta x_1^y\partial_y\partial_z v_1^z$$

`[obs] observer clock | fields:1` (3 terms)

$$\frac{1}{3}\delta\eta_{o1}\partial_x v_1^x + \frac{1}{3}\delta\eta_{o1}\partial_y v_1^y - \frac{2}{3}\delta\eta_{o1}\partial_z v_1^z$$

`[emi] emission-surface shift | fields:1` (3 terms)

$$-\frac{1}{3}\delta\eta_{e1}\partial_x v_1^x - \frac{1}{3}\delta\eta_{e1}\partial_y v_1^y + \frac{2}{3}\delta\eta_{e1}\partial_z v_1^z$$

Total: 34 terms ($\hat{\mathbf{n}} = \hat{\mathbf{z}}$). The general-$\hat{\mathbf{n}}$ list `L[2]` (no $|\hat{\mathbf{n}}|=1$ imposed) has 229 terms in 6 groups:

B.2 $\frac{1}{2}L_2$, direct (redshift) form, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$

`[ray] local | fields:2` (6 terms)

$$\frac{1}{2}\Phi_2{}' - \frac{1}{2}\Psi_2{}' + \partial_z\Psi_2 + \frac{1}{4}h_{2,xx}{}' + \frac{1}{4}h_{2,yy}{}' + \frac{1}{2}\partial_z\omega_{2,z}$$

Table 4: Groups of $\frac{1}{2}L_2$ for general $\hat{\mathbf{n}}$ (`L_terms.pkl['L'][2]`).

#	where	content	fields	terms
1	ray	local	2	22
2	ray	direction perturbation k1 (aberration + deflection propagated) [`k1`]	1	21
3	ray	local	1+1	111
4	ray	path displacement Dx1 (lensing [transverse] + time delay [radial]) [`Dx1`]	1	45
5	obs	observer clock shift [`deta_o1`]	1	15
6	emi	emission-surface shift [`deta_e1`]	1	15
		total		229

`[ray] lensing (transverse Dx1) | fields:1` (6 terms)

$$\begin{aligned}&\Delta x_1^x\partial_\eta\partial_x\Phi_1 - \Delta x_1^x\partial_\eta\partial_x\Psi_1 + 2\Delta x_1^x\partial_x\partial_z\Psi_1\\&\quad + \Delta x_1^y\partial_\eta\partial_y\Phi_1 - \Delta x_1^y\partial_\eta\partial_y\Psi_1 + 2\Delta x_1^y\partial_y\partial_z\Psi_1\end{aligned}$$

`[ray] time delay (radial Dx1) | fields:1` (3 terms)

$$2\Delta x_1^z\partial_z^2\Psi_1 + \Delta x_1^z\partial_\eta\partial_z\Phi_1 - \Delta x_1^z\partial_\eta\partial_z\Psi_1$$

`[ray] transverse direction k1 (aberration+deflection) | fields:1` (2 terms)

$$-2k_1^x\partial_x\Psi_1 - 2k_1^y\partial_y\Psi_1$$

`[ray] local | fields:1+1` (4 terms)

$$\begin{aligned}&2\Phi_1{\Phi_1}' + 2\Phi_1\partial_z\Psi_1 + 2\Psi_1{\Psi_1}'\\&\quad - 2\Psi_1\partial_z\Psi_1\end{aligned}$$

`[obs] observer clock | fields:1` (3 terms)

$$\delta\eta_{o1}{\Phi_1}' - \delta\eta_{o1}{\Psi_1}' + 2\delta\eta_{o1}\partial_z\Psi_1$$

`[emi] emission-surface shift | fields:1` (3 terms)

$$-\delta\eta_{e1}{\Phi_1}' + \delta\eta_{e1}{\Psi_1}' - 2\delta\eta_{e1}\partial_z\Psi_1$$

`[obs] local | fields:1+1` (6 terms)

$$\begin{aligned}&-\Phi_1 v_1^z - \Psi_1 v_1^z - (\Psi_1)^2\\&\quad + \frac{1}{2}(\delta_{T,1})^2 - \frac{1}{2}(v_1^x)^2 - \frac{1}{2}(v_1^y)^2\end{aligned}$$

`[obs] local | fields:2` (4 terms)

$$\begin{aligned}&\frac{1}{2}\Psi_2 - \frac{1}{2}\delta_{T,2} + \frac{1}{2}\omega_{2,z}\\&\quad + \frac{1}{2}v_2^z\end{aligned}$$

`[obs] observer clock | fields:1` (3 terms)

$$\delta\eta_{o1}\Psi_1{}' - \delta\eta_{o1}\delta_{T,1}{}' + \delta\eta_{o1}v_1^{z\prime}$$

`[obs] observer displacement | fields:1` (9 terms)

$$\begin{aligned}&\delta x^x_{o1}\partial_x\Psi_1 - \delta x^x_{o1}\partial_x\delta_{T,1} + \delta x^x_{o1}\partial_x v_1^z\\ &+ \delta x^y_{o1}\partial_y\Psi_1 - \delta x^y_{o1}\partial_y\delta_{T,1} + \delta x^y_{o1}\partial_y v_1^z\\ &+ \delta x^z_{o1}\partial_z\Psi_1 - \delta x^z_{o1}\partial_z\delta_{T,1} + \delta x^z_{o1}\partial_z v_1^z\end{aligned}$$

`[emi] local | fields:1+1` (6 terms)

$$\begin{aligned}&\Phi_1 v_1^z + \Psi_1 v_1^z + (\Psi_1)^2\\ &- \frac{1}{2}(\delta_{T,1})^2 - \frac{1}{2}(v_1^x)^2 - \frac{1}{2}(v_1^y)^2\end{aligned}$$

`[emi] local | fields:2` (4 terms)

$$\begin{aligned}&- \frac{1}{2}\Psi_2 + \frac{1}{2}\delta_{T,2} - \frac{1}{2}\omega_{2,z}\\ &- \frac{1}{2}v_2^z\end{aligned}$$

`[emi] emission-surface shift | fields:1` (6 terms)

$$\begin{aligned}&- \delta\eta_{e1}\Psi_1{}' + \delta\eta_{e1}\partial_z\Psi_1 + \delta\eta_{e1}\delta_{T,1}{}'\\ &- \delta\eta_{e1}\partial_z\delta_{T,1} - \delta\eta_{e1}v_1^{z\prime} + \delta\eta_{e1}\partial_z v_1^z\end{aligned}$$

`[emi] lensing (transverse Dx1) | fields:1` (6 terms)

$$\begin{aligned}&- \Delta x_1^x(\eta_*)\partial_x\Psi_1 + \Delta x_1^x(\eta_*)\partial_x\delta_{T,1} - \Delta x_1^x(\eta_*)\partial_x v_1^z\\ &- \Delta x_1^y(\eta_*)\partial_y\Psi_1 + \Delta x_1^y(\eta_*)\partial_y\delta_{T,1} - \Delta x_1^y(\eta_*)\partial_y v_1^z\end{aligned}$$

`[emi] time delay (radial Dx1) | fields:1` (3 terms)

$$- \Delta x_1^z(\eta_*)\partial_z\Psi_1 + \Delta x_1^z(\eta_*)\partial_z\delta_{T,1} - \Delta x_1^z(\eta_*)\partial_z v_1^z$$

`[emi] transverse direction k1 (aberration+deflection) | fields:1` (2 terms)

$$k_1^x(\eta_*)v_1^x + k_1^y(\eta_*)v_1^y$$

Total: 76 terms ($\hat{\mathbf{n}} = \hat{\mathbf{z}}$); general $\hat{\mathbf{n}}$: 179 terms in 15 groups.

B.3 $\frac{1}{6}L_3$: group lists

Explicit short groups. The groups of Table 5 with at most ten terms ($\hat{\mathbf{n}} = \hat{\mathbf{z}}$):

Table 5: Groups of $\frac{1}{6}L_3$, shear form, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ (`L_split[3]`); bracketed tags as stored.

#	where	content	fields	terms
1	ray	local	3	5
2	ray	local	2+1	57
3	ray	transverse direction k1 (aberration+deflection) [`k1perp`]	2	6
4	ray	transverse k2 [`k2perp`]	1	4
5	ray	time delay (radial Dx1) [`tdelay`]	1+1	26
6	ray	local	1+1+1	55
7	ray	transverse direction k1 (aberration+deflection)^2 [`k1perp^2`]	1	8
8	ray	lensing (transverse Dx1)^2 [`lens^2`]	1	9
9	ray	time delay (radial Dx1)^2 [`tdelay^2`]	1	3
10	ray	radial k2 [`k2rad`]	1	3
11	ray	time delay 2nd (radial Dx2) [`tdelay2`]	1	3
12	ray	time delay (radial Dx1) [`tdelay`]	2	5
13	ray	lensing 2nd (transverse Dx2) [`lens2`]	1	6
14	ray	lensing (transverse Dx1) [`lens`]	2	10
15	ray	transverse direction k1 (aberration+deflection) [`k1perp`]	1+1	22
16	ray	transverse direction k1 (aberration+deflection); time delay (radial Dx1) [`k1perp * tdelay`]	1	4
17	ray	transverse direction k1 (aberration+deflection); lensing (transverse Dx1) [`k1perp * lens`]	1	8
18	ray	lensing (transverse Dx1) [`lens`]	1+1	50
19	ray	lensing (transverse Dx1); time delay (radial Dx1) [`lens * tdelay`]	1	6
20	obs	observer clock^2 [`deta_o1^2`]	1	3
21	obs	observer clock [`deta_o1`]	2	5
22	obs	observer clock 2nd [`deta_o2`]	1	3
23	obs	observer clock; observer displacement [`deta_o1 * dxo1`]	1	9
24	obs	observer clock [`deta_o1`]	1+1	6
25	obs	observer clock^2 [`deta_o1^2`]	1	6
26	emi	emission-surface shift [`deta_e1`]	2	5
27	emi	emission-surface shift 2nd [`deta_e2`]	1	3
28	emi	emission-surface shift [`deta_e1`]	1+1	10
29	emi	emission-surface shift; transverse direction k1 (aberration+deflection) [`deta_e1 * k1perp`]	1	4
30	emi	emission-surface shift; lensing (transverse Dx1) [`deta_e1 * lens`]	1	6
31	emi	emission-surface shift; time delay (radial Dx1) [`deta_e1 * tdelay`]	1	3
32	emi	emission-surface shift^2 [`deta_e1^2`]	1	6
		total		359

Table 6: Groups of $\frac{1}{6}L_3$, shear form, general $\hat{\mathbf{n}}$ (`L[3]`).

#	where	content	fields	terms
1	ray	direction perturbation k1 (aberration + deflection propagated) [`k1^2`]	1	15
2	ray	local	1+1+1	411
3	ray	local	3	22
4	ray	direction perturbation k1 (aberration + deflection propagated) [`k1`]	1+1	171
5	ray	direction perturbation k1 (aberration + deflection propagated) [`k1`]	2	31
6	ray	second-order direction k2 [`k2`]	1	21
7	ray	local	2+1	404
8	ray	path displacement Dx1 (lensing [transverse] + time delay [radial]) [`Dx1`]	1+1	651
9	ray	path displacement Dx1 (lensing [transverse] + time delay [radial]) [`Dx1^2`]	1	90
10	ray	second-order path displacement [`Dx2`]	1	45
11	ray	path displacement Dx1 (lensing [transverse] + time delay [radial]) [`Dx1`]	2	66
12	ray	path displacement Dx1 (lensing [transverse] + time delay [radial]); direction perturbation k1 (aberration + deflection propagated) [`Dx1 * k1`]	1	63
13	obs	observer clock shift [`deta_o1^2`]	1	36
14	obs	observer clock shift [`deta_o1`]	2	22
15	obs	observer clock shift (2nd) [`deta_o2`]	1	15
16	obs	observer clock shift; observer displacement [`deta_o1 * dxo1`]	1	45
17	obs	observer clock shift [`deta_o1`]	1+1	60
18	obs	observer clock shift [`deta_o1^2`]	1	51
19	emi	emission-surface shift [`deta_e1`]	2	22
20	emi	emission-surface shift (2nd) [`deta_e2`]	1	15
21	emi	emission-surface shift [`deta_e1`]	1+1	111
22	emi	emission-surface shift; direction perturbation k1 (aberration + deflection propagated) [`deta_e1 * k1`]	1	21
23	emi	path displacement Dx1 (lensing [transverse] + time delay [radial]); emission-surface shift [`Dx1 * deta_e1`]	1	45
24	emi	emission-surface shift [`deta_e1^2`]	1	51
		total		2484

`[ray] local | fields:3` (5 terms)

$$\begin{aligned}&\frac{1}{12}h_{3,xx}{}' + \frac{1}{12}h_{3,yy}{}' \\ &\quad + \frac{1}{18}\partial_x v_3^x + \frac{1}{18}\partial_y v_3^y \\ &\quad - \frac{1}{9}\partial_z v_3^z\end{aligned}$$

`[ray] transverse direction k1 (aberration+deflection) | fields:2` (6 terms)

$$\begin{aligned}&\frac{1}{2}k_1^x h_{2,xz}{}' + \frac{1}{2}k_1^x \partial_z v_2^x \\ &\quad + \frac{1}{2}k_1^x \partial_x v_2^z + \frac{1}{2}k_1^y h_{2,yz}{}' \\ &\quad + \frac{1}{2}k_1^y \partial_z v_2^y + \frac{1}{2}k_1^y \partial_y v_2^z\end{aligned}$$

`[ray] transverse k2 | fields:1` (4 terms)

$$\begin{aligned}&\frac{1}{2}k_2^x \partial_z v_1^x + \frac{1}{2}k_2^x \partial_x v_1^z \\ &\quad + \frac{1}{2}k_2^y \partial_z v_1^y + \frac{1}{2}k_2^y \partial_y v_1^z\end{aligned}$$

`[ray] transverse direction k1 (aberration+deflection)^2 | fields:1` (8 terms)

$$\begin{aligned}&-\frac{2}{3}(k_1^x)^2 \partial_x v_1^x + \frac{1}{3}(k_1^x)^2 \partial_y v_1^y \\ &\quad + \frac{1}{3}(k_1^x)^2 \partial_z v_1^z + \frac{1}{3}(k_1^y)^2 \partial_x v_1^x \\ &\quad - \frac{2}{3}(k_1^y)^2 \partial_y v_1^y + \frac{1}{3}(k_1^y)^2 \partial_z v_1^z \\ &\quad - k_1^x k_1^y \partial_y v_1^x - k_1^x k_1^y \partial_x v_1^y\end{aligned}$$

`[ray] lensing (transverse Dx1)^2 | fields:1` (9 terms)

$$\begin{aligned}&\frac{1}{6}(\Delta x_1^x)^2 \partial_x^3 v_1^x + \frac{1}{6}(\Delta x_1^x)^2 \partial_x^2 \partial_y v_1^y \\ &\quad - \frac{1}{3}(\Delta x_1^x)^2 \partial_x^2 \partial_z v_1^z + \frac{1}{6}(\Delta x_1^y)^2 \partial_y^3 v_1^y \\ &\quad + \frac{1}{6}(\Delta x_1^y)^2 \partial_x \partial_y^2 v_1^x - \frac{1}{3}(\Delta x_1^y)^2 \partial_y^2 \partial_z v_1^z \\ &\quad + \frac{1}{3}\Delta x_1^x \Delta x_1^y \partial_x^2 \partial_y v_1^x + \frac{1}{3}\Delta x_1^x \Delta x_1^y \partial_x \partial_y^2 v_1^y \\ &\quad - \frac{2}{3}\Delta x_1^x \Delta x_1^y \partial_x \partial_y \partial_z v_1^z\end{aligned}$$

`[ray] time delay (radial Dx1)^2 | fields:1` (3 terms)

$$\begin{aligned}&-\frac{1}{3}(\Delta x_1^z)^2 \partial_z^3 v_1^z + \frac{1}{6}(\Delta x_1^z)^2 \partial_x \partial_z^2 v_1^x \\ &\quad + \frac{1}{6}(\Delta x_1^z)^2 \partial_y \partial_z^2 v_1^y\end{aligned}$$

`[ray] radial k2 | fields:1` (3 terms)

$$
\begin{aligned}
&- \frac{1}{3} k_2^z \partial_x v_1^x - \frac{1}{3} k_2^z \partial_y v_1^y \\
&+ \frac{2}{3} k_2^z \partial_z v_1^z
\end{aligned}
$$

`[ray] time delay 2nd (radial Dx2) | fields:1` (3 terms)

$$
\begin{aligned}
&- \frac{1}{3} \Delta x_2^z \partial_z^2 v_1^z + \frac{1}{6} \Delta x_2^z \partial_x \partial_z v_1^x \\
&+ \frac{1}{6} \Delta x_2^z \partial_y \partial_z v_1^y
\end{aligned}
$$

`[ray] time delay (radial Dx1) | fields:2` (5 terms)

$$
\begin{aligned}
&- \frac{1}{3} \Delta x_1^z \partial_z^2 v_2^z + \frac{1}{4} \Delta x_1^z \partial_\eta \partial_z h_{2,xx} \\
&+ \frac{1}{4} \Delta x_1^z \partial_\eta \partial_z h_{2,yy} + \frac{1}{6} \Delta x_1^z \partial_x \partial_z v_2^x \\
&+ \frac{1}{6} \Delta x_1^z \partial_y \partial_z v_2^y
\end{aligned}
$$

`[ray] lensing 2nd (transverse Dx2) | fields:1` (6 terms)

$$
\begin{aligned}
&\frac{1}{6} \Delta x_2^x \partial_x^2 v_1^x + \frac{1}{6} \Delta x_2^x \partial_x \partial_y v_1^y \\
&- \frac{1}{3} \Delta x_2^x \partial_x \partial_z v_1^z + \frac{1}{6} \Delta x_2^y \partial_y^2 v_1^y \\
&+ \frac{1}{6} \Delta x_2^y \partial_x \partial_y v_1^x - \frac{1}{3} \Delta x_2^y \partial_y \partial_z v_1^z
\end{aligned}
$$

`[ray] lensing (transverse Dx1) | fields:2` (10 terms)

$$
\begin{aligned}
&\frac{1}{6} \Delta x_1^x \partial_x^2 v_2^x + \frac{1}{4} \Delta x_1^x \partial_\eta \partial_x h_{2,xx} \\
&+ \frac{1}{4} \Delta x_1^x \partial_\eta \partial_x h_{2,yy} + \frac{1}{6} \Delta x_1^x \partial_x \partial_y v_2^y \\
&- \frac{1}{3} \Delta x_1^x \partial_x \partial_z v_2^z + \frac{1}{6} \Delta x_1^y \partial_y^2 v_2^y \\
&+ \frac{1}{4} \Delta x_1^y \partial_\eta \partial_y h_{2,xx} + \frac{1}{4} \Delta x_1^y \partial_\eta \partial_y h_{2,yy} \\
&+ \frac{1}{6} \Delta x_1^y \partial_x \partial_y v_2^x - \frac{1}{3} \Delta x_1^y \partial_y \partial_z v_2^z
\end{aligned}
$$

`[ray] transverse direction k1 (aberration+deflection); time delay (radial Dx1) | fields:1` (4 terms)

$$
\begin{aligned}
&\Delta x_1^z k_1^x \partial_z^2 v_1^x + \Delta x_1^z k_1^x \partial_x \partial_z v_1^z \\
&+ \Delta x_1^z k_1^y \partial_z^2 v_1^y + \Delta x_1^z k_1^y \partial_y \partial_z v_1^z
\end{aligned}
$$

`[ray] transverse direction k1 (aberration+deflection); lensing (transverse Dx1) | fields:1` (8 terms)

$$\begin{aligned}&\Delta x_1^x k_1^x \partial_x^2 v_1^z + \Delta x_1^x k_1^x \partial_x \partial_z v_1^x \\ &\quad + \Delta x_1^x k_1^y \partial_x \partial_z v_1^y + \Delta x_1^x k_1^y \partial_x \partial_y v_1^z \\ &\quad + \Delta x_1^y k_1^x \partial_y \partial_z v_1^x + \Delta x_1^y k_1^x \partial_x \partial_y v_1^z \\ &\quad + \Delta x_1^y k_1^y \partial_y^2 v_1^z + \Delta x_1^y k_1^y \partial_y \partial_z v_1^y\end{aligned}$$

`[ray] lensing (transverse Dx1); time delay (radial Dx1) | fields:1` (6 terms)

$$\begin{aligned}&\frac{1}{3}\Delta x_1^x \Delta x_1^z \partial_x^2 \partial_z v_1^x - \frac{2}{3}\Delta x_1^x \Delta x_1^z \partial_x \partial_z^2 v_1^z \\ &\quad + \frac{1}{3}\Delta x_1^x \Delta x_1^z \partial_x \partial_y \partial_z v_1^y + \frac{1}{3}\Delta x_1^y \Delta x_1^z \partial_y^2 \partial_z v_1^y \\ &\quad - \frac{2}{3}\Delta x_1^y \Delta x_1^z \partial_y \partial_z^2 v_1^z + \frac{1}{3}\Delta x_1^y \Delta x_1^z \partial_x \partial_y \partial_z v_1^x\end{aligned}$$

`[obs] observer clock^2 | fields:1` (3 terms)

$$\begin{aligned}&-\frac{2}{3}\delta\eta_{o1}^2 \partial_z^2 v_1^z + \frac{1}{3}\delta\eta_{o1}^2 \partial_x \partial_z v_1^x \\ &\quad + \frac{1}{3}\delta\eta_{o1}^2 \partial_y \partial_z v_1^y\end{aligned}$$

`[obs] observer clock | fields:2` (5 terms)

$$\begin{aligned}&\frac{1}{4}\delta\eta_{o1} h_{2,xx}{}' + \frac{1}{4}\delta\eta_{o1} h_{2,yy}{}' \\ &\quad + \frac{1}{6}\delta\eta_{o1} \partial_x v_2^x + \frac{1}{6}\delta\eta_{o1} \partial_y v_2^y \\ &\quad - \frac{1}{3}\delta\eta_{o1} \partial_z v_2^z\end{aligned}$$

`[obs] observer clock 2nd | fields:1` (3 terms)

$$\begin{aligned}&\frac{1}{6}\delta\eta_{o2} \partial_x v_1^x + \frac{1}{6}\delta\eta_{o2} \partial_y v_1^y \\ &\quad - \frac{1}{3}\delta\eta_{o2} \partial_z v_1^z\end{aligned}$$

`[obs] observer clock; observer displacement | fields:1` (9 terms)

$$\begin{aligned}&\frac{1}{3}\delta\eta_{o1}\delta x_{o1}^x \partial_x^2 v_1^x + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^x \partial_x \partial_y v_1^y \\ &\quad - \frac{2}{3}\delta\eta_{o1}\delta x_{o1}^x \partial_x \partial_z v_1^z + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^y \partial_y^2 v_1^y \\ &\quad + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^y \partial_x \partial_y v_1^x - \frac{2}{3}\delta\eta_{o1}\delta x_{o1}^y \partial_y \partial_z v_1^z \\ &\quad - \frac{2}{3}\delta\eta_{o1}\delta x_{o1}^z \partial_z^2 v_1^z + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^z \partial_x \partial_z v_1^x \\ &\quad + \frac{1}{3}\delta\eta_{o1}\delta x_{o1}^z \partial_y \partial_z v_1^y\end{aligned}$$

`[obs] observer clock | fields:1+1` (6 terms)

$$\begin{aligned}&\frac{1}{3}\delta\eta_{o1}v_1^x v_1^{x\prime}+\frac{1}{3}\delta\eta_{o1}v_1^y v_1^{y\prime}\\&\quad+\frac{1}{3}\delta\eta_{o1}v_1^z\partial_x v_1^x+\frac{1}{3}\delta\eta_{o1}v_1^z\partial_y v_1^y\\&\quad-\frac{2}{3}\delta\eta_{o1}v_1^z v_1^{z\prime}-\frac{2}{3}\delta\eta_{o1}v_1^z\partial_z v_1^z\end{aligned}$$

`[obs] observer clock^2 | fields:1` (6 terms)

$$\begin{aligned}&\frac{1}{3}\delta\eta_{o1}^2\partial_z^2 v_1^z+\frac{1}{6}\delta\eta_{o1}^2\partial_\eta\partial_x v_1^x\\&\quad-\frac{1}{6}\delta\eta_{o1}^2\partial_x\partial_z v_1^x+\frac{1}{6}\delta\eta_{o1}^2\partial_\eta\partial_y v_1^y\\&\quad-\frac{1}{6}\delta\eta_{o1}^2\partial_y\partial_z v_1^y-\frac{1}{3}\delta\eta_{o1}^2\partial_\eta\partial_z v_1^z\end{aligned}$$

`[emi] emission-surface shift | fields:2` (5 terms)

$$\begin{aligned}&-\frac{1}{4}\delta\eta_{e1}{h_{2,xx}}'-\frac{1}{4}\delta\eta_{e1}{h_{2,yy}}'\\&\quad-\frac{1}{6}\delta\eta_{e1}\partial_x v_2^x-\frac{1}{6}\delta\eta_{e1}\partial_y v_2^y\\&\quad+\frac{1}{3}\delta\eta_{e1}\partial_z v_2^z\end{aligned}$$

`[emi] emission-surface shift 2nd | fields:1` (3 terms)

$$\begin{aligned}&-\frac{1}{6}\delta\eta_{e2}\partial_x v_1^x-\frac{1}{6}\delta\eta_{e2}\partial_y v_1^y\\&\quad+\frac{1}{3}\delta\eta_{e2}\partial_z v_1^z\end{aligned}$$

`[emi] emission-surface shift | fields:1+1` (10 terms)

$$\begin{aligned}&-\frac{1}{3}\delta\eta_{e1}v_1^x v_1^{x\prime}+\delta\eta_{e1}v_1^x\partial_z v_1^x\\&\quad+\delta\eta_{e1}v_1^x\partial_x v_1^z-\frac{1}{3}\delta\eta_{e1}v_1^y v_1^{y\prime}\\&\quad+\delta\eta_{e1}v_1^y\partial_z v_1^y+\delta\eta_{e1}v_1^y\partial_y v_1^z\\&\quad-\frac{1}{3}\delta\eta_{e1}v_1^z\partial_x v_1^x-\frac{1}{3}\delta\eta_{e1}v_1^z\partial_y v_1^y\\&\quad+\frac{2}{3}\delta\eta_{e1}v_1^z v_1^{z\prime}+\frac{2}{3}\delta\eta_{e1}v_1^z\partial_z v_1^z\end{aligned}$$

`[emi] emission-surface shift; transverse direction k1 (aberration+deflection) | fields:1` (4 terms)

$$\begin{aligned}&-\delta\eta_{e1}k_1^x(\eta_*)\partial_z v_1^x-\delta\eta_{e1}k_1^x(\eta_*)\partial_x v_1^z\\&\quad-\delta\eta_{e1}k_1^y(\eta_*)\partial_z v_1^y-\delta\eta_{e1}k_1^y(\eta_*)\partial_y v_1^z\end{aligned}$$

`[emi] emission-surface shift; lensing (transverse Dx1) | fields:1` (6 terms)

$$
\begin{aligned}
&-\frac{1}{3}\delta\eta_{e1}\Delta x_1^x(\eta_*)\partial_x^2 v_1^x-\frac{1}{3}\delta\eta_{e1}\Delta x_1^x(\eta_*)\partial_x\partial_y v_1^y\\
&+\frac{2}{3}\delta\eta_{e1}\Delta x_1^x(\eta_*)\partial_x\partial_z v_1^z-\frac{1}{3}\delta\eta_{e1}\Delta x_1^y(\eta_*)\partial_y^2 v_1^y\\
&-\frac{1}{3}\delta\eta_{e1}\Delta x_1^y(\eta_*)\partial_x\partial_y v_1^x+\frac{2}{3}\delta\eta_{e1}\Delta x_1^y(\eta_*)\partial_y\partial_z v_1^z
\end{aligned}
$$

`[emi] emission-surface shift; time delay (radial Dx1) | fields:1` (3 terms)

$$
\begin{aligned}
&\frac{2}{3}\delta\eta_{e1}\Delta x_1^z(\eta_*)\partial_z^2 v_1^z-\frac{1}{3}\delta\eta_{e1}\Delta x_1^z(\eta_*)\partial_x\partial_z v_1^x\\
&-\frac{1}{3}\delta\eta_{e1}\Delta x_1^z(\eta_*)\partial_y\partial_z v_1^y
\end{aligned}
$$

`[emi] emission-surface shift^2 | fields:1` (6 terms)

$$
\begin{aligned}
&-\frac{1}{3}\delta\eta_{e1}^2\partial_z^2 v_1^z-\frac{1}{6}\delta\eta_{e1}^2\partial_\eta\partial_x v_1^x\\
&+\frac{1}{6}\delta\eta_{e1}^2\partial_x\partial_z v_1^x-\frac{1}{6}\delta\eta_{e1}^2\partial_\eta\partial_y v_1^y\\
&+\frac{1}{6}\delta\eta_{e1}^2\partial_y\partial_z v_1^y+\frac{1}{3}\delta\eta_{e1}^2\partial_\eta\partial_z v_1^z
\end{aligned}
$$

The direct form of $\frac{1}{6}L_3$ has 480 terms ($\hat{\mathbf{n}}=\hat{\mathbf{z}}$, 69 groups) and 1402 terms for general $\hat{\mathbf{n}}$.

C Machine-generated channel kernels of K_2

This appendix reproduces the channel kernels of $K_2(\mathbf{k}_1,\mathbf{k}_2;\hat{\mathbf{n}})$ exactly as generated by `cmb4/code/A2_second_order` (output `symbolic_kernels.pkl`, `symbolic_kernels.tex`) from the direct-form term list of $L_2/2$ (`L_terms.pkl`, Sec. 7), with the symbol names of Sec. **??** substituted typographically (no algebra was changed; the script that performs the substitution also verifies symbolically that every expression below equals the compact form quoted in Sec. **??**: residual 0 for all fifteen channels). Each expression is already the contribution to K_2, i.e. it includes the factor 2 of $K_2=2\,\mathrm{Sym}[K_{L_2/2}]+\dots$ and the symmetrisation $1\leftrightarrow 2$. Notation (Sec. **??**): $\mu=\hat{\mathbf{k}}_1\cdot\hat{\mathbf{k}}_2$, $\mu_a=\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}_a$, $\mathbf{K}=\mathbf{k}_1+\mathbf{k}_2$, $K_n=\hat{\mathbf{n}}\cdot\mathbf{K}=k_1\mu_1+k_2\mu_2$, $P_\perp=k_1k_2(\mu-\mu_1\mu_2)$, $E_a=e^{ik_a\mu_a\chi}$, $E_K=E_1E_2$; first-order transfer functions $T_{X,a}\equiv T_X(k_a,\eta)$ and their η-derivatives $\dot T_{X,a}$; ray functionals $\mathcal{F}_a$, $\mathcal{D}_{\perp,a}$, $\mathcal{D}_{r,a}$ (**??**); observer and emission objects $X_{o,a}$, $\delta\eta_{o,a}$, $\delta\eta_{e,a}$ (**??**)–(**??**); second-order pair kernels $\Phi_2,\dot\Phi_2,\Psi_2,\dot\Psi_2,\delta_{T2},V_2$ and $K_\omega^{(ab)}\equiv K_\omega(\mathbf{k}_a,\mathbf{k}_b)$, $K_{v^T}^{(ab)}$, $\mathcal{H}$, $\dot{\mathcal{H}}$ (**??**). The numerical value of every channel agrees with the independent numerical engine (`kernel2.py`) to $\le 1.8\times10^{-15}$ (`check_symbolic.log`). The type-B products are $K_1(\mathbf{k}_1)K_1(\mathbf{k}_2)-\Delta_0(k_1)K_1(\mathbf{k}_2)-\Delta_0(k_2)K_1(\mathbf{k}_1)$ (**??**).

[ray] second-order fields (C: Φ_2,Ψ_2 ISW, ω_2, h_2). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$
\begin{aligned}
&E_K\Big(\dot{\mathcal{H}}K_n^2k_1^2\left(k_1^2+2k_1k_2\mu+k_2^2\mu^2\right)-K^4\Big(-2\dot{\mathcal{H}}k_1^2\mu_1^2+\dot{\mathcal{H}}k_1^2+2iK_nK_\omega^{(12)}k_1\mu_1+2iK_nK_\omega^{(21)}k_2\mu_2+8iK_n\Psi_2+4\dot\Phi_2
\end{aligned}
$$

[ray] lensing, transverse $\Delta\mathbf{x}_1$ (D). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$-2iP_\perp\left(\mathcal{D}_{\perp,1}E_2T_{\Phi,2}k_2\mu_2+\mathcal{D}_{\perp,2}E_1T_{\Phi,1}k_1\mu_1\right)$$

[ray] time delay, radial $\Delta\mathbf{x}_1$ (D). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$-2\mathcal{D}_{r,1}E_2T_{\Phi,2}k_2^2\mu_2^2-2\mathcal{D}_{r,2}E_1T_{\Phi,1}k_1^2\mu_1^2$$

[ray] aberration + deflection, transverse $\mathbf{k}_1$ (D). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$2P_\perp\left(E_1\mathcal{F}_2T_{\Phi,1}+E_2\mathcal{F}_1T_{\Phi,2}\right)$$

[ray] local 1×1 (C). $\int_{\eta_*}^{\eta_o}\mathrm{d}\eta$ of

$$4E_1E_2\left(T_{\Phi,1}\dot{T}_{\Phi,2}+T_{\Phi,2}\dot{T}_{\Phi,1}\right)$$

[obs] observer clock (D). at $\eta=\eta_o$, $\chi=0$ ($E_a=1$):

$$\begin{aligned}&2iE_1T_{\Phi,1}\delta\eta_{o,2}k_1\mu_1+E_1\dot{T}_{\Phi,1}\delta\eta_{o,2}+iE_1\dot{T}_{V,1}\delta\eta_{o,2}k_1\mu_1-E_1\dot{T}_{\delta_T,1}\delta\eta_{o,2}\\&+2iE_2T_{\Phi,2}\delta\eta_{o,1}k_2\mu_2+E_2\dot{T}_{\Phi,2}\delta\eta_{o,1}+iE_2\dot{T}_{V,2}\delta\eta_{o,1}k_2\mu_2-E_2\dot{T}_{\delta_T,2}\delta\eta_{o,1}\end{aligned}$$

[emi] emission-surface shift (D). at $\eta=\eta_*$, $\chi=\chi_*$:

$$\begin{aligned}&-iE_1T_{\Phi,1}\delta\eta_{e,2}k_1\mu_1-E_1\dot{T}_{\Phi,1}\delta\eta_{e,2}-E_1T_{V,1}\delta\eta_{e,2}k_1^2\mu_1^2-iE_1\dot{T}_{V,1}\delta\eta_{e,2}k_1\mu_1\\&-iE_1T_{\delta_T,1}\delta\eta_{e,2}k_1\mu_1+E_1\dot{T}_{\delta_T,1}\delta\eta_{e,2}-iE_2T_{\Phi,2}\delta\eta_{e,1}k_2\mu_2-E_2\dot{T}_{\Phi,2}\delta\eta_{e,1}\\&-E_2T_{V,2}\delta\eta_{e,1}k_2^2\mu_2^2-iE_2\dot{T}_{V,2}\delta\eta_{e,1}k_2\mu_2-iE_2T_{\delta_T,2}\delta\eta_{e,1}k_2\mu_2+E_2\dot{T}_{\delta_T,2}\delta\eta_{e,1}\end{aligned}$$

[obs] local 1×1 (B, endpoint). at $\eta=\eta_o$, $\chi=0$ ($E_a=1$):

$$E_1E_2\left(P_\perp T_{V,1}T_{V,2}-2T_{\Phi,1}T_{\Phi,2}-2iT_{\Phi,1}T_{V,2}k_2\mu_2-2iT_{\Phi,2}T_{V,1}k_1\mu_1+T_{\delta_T,1}T_{\delta_T,2}\right)$$

[obs] second-order fields (C: $\Psi_2,\delta_{T2},V_2,v_2^T+\omega_2$). at $\eta=\eta_o$, $\chi=0$ ($E_a=1$):

$$-\frac{E_K\left(-K^2\left(2iK_nV_2+K_\omega^{(12)}k_1\mu_1+K_\omega^{(21)}k_2\mu_2+K_{v^T}^{(12)}k_1\mu_1+K_{v^T}^{(21)}k_2\mu_2+2\Psi_2-2\delta_{T2}\right)+K_nK_\omega^{(12)}k_1^2+K_nK_\omega^{(12)}k\right.}{2K^2}$$

[obs] observer displacement (D). at $\eta=\eta_o$, $\chi=0$ ($E_a=1$):

$$\begin{aligned}&-E_1P_\perp T_{\Phi,1}X_{o,2}-iE_1P_\perp T_{V,1}X_{o,2}k_1\mu_1+E_1P_\perp T_{\delta_T,1}X_{o,2}-E_1T_{\Phi,1}X_{o,2}k_1k_2\mu_1\mu_2\\&-iE_1T_{V,1}X_{o,2}k_1^2k_2\mu_1^2\mu_2+E_1T_{\delta_T,1}X_{o,2}k_1k_2\mu_1\mu_2-E_2P_\perp T_{\Phi,2}X_{o,1}-iE_2P_\perp T_{V,2}X_{o,1}k_2\mu_2\\&+E_2P_\perp T_{\delta_T,2}X_{o,1}-E_2T_{\Phi,2}X_{o,1}k_1k_2\mu_1\mu_2-iE_2T_{V,2}X_{o,1}k_1k_2^2\mu_1\mu_2^2+E_2T_{\delta_T,2}X_{o,1}k_1k_2\mu_1\mu_2\end{aligned}$$

[emi] local 1×1 (B, endpoint). at $\eta=\eta_*$, $\chi=\chi_*$:

$$E_1E_2\left(P_\perp T_{V,1}T_{V,2}+2T_{\Phi,1}T_{\Phi,2}+2iT_{\Phi,1}T_{V,2}k_2\mu_2+2iT_{\Phi,2}T_{V,1}k_1\mu_1-T_{\delta_T,1}T_{\delta_T,2}\right)$$

[emi] second-order fields (C: $\Psi_2, \delta_{T2}, V_2, v_2^T + \omega_2$). at $\eta = \eta_*$, $\chi = \chi_*$:

$$E_K \left(K^2 \left(-2iK_n V_2 - K_\omega^{(12)} k_1\mu_1 - K_\omega^{(21)} k_2\mu_2 - K_{v^T}^{(12)} k_1\mu_1 - K_{v^T}^{(21)} k_2\mu_2 - 2\Psi_2 + 2\delta_{T2} \right) + K_n K_\omega^{(12)} k_1^2 + K_n K_\omega^{(12)} k_1 k \right.$$
$$\overline{\qquad 2K^2}$$

[emi] lensing at emission (D). at $\eta = \eta_*$, $\chi = \chi_*$:

$$P_\perp \left(\mathcal{D}_{\perp,1} E_2 T_{\Phi,2} + i\mathcal{D}_{\perp,1} E_2 T_{V,2} k_2\mu_2 - \mathcal{D}_{\perp,1} E_2 T_{\delta_T,2} + \mathcal{D}_{\perp,2} E_1 T_{\Phi,1} + i\mathcal{D}_{\perp,2} E_1 T_{V,1} k_1\mu_1 - \mathcal{D}_{\perp,2} E_1 T_{\delta_T,1} \right)$$

[emi] time delay at emission (D). at $\eta = \eta_*$, $\chi = \chi_*$:

$$\begin{aligned} &-i\mathcal{D}_{r,1} E_2 T_{\Phi,2} k_2\mu_2 + \mathcal{D}_{r,1} E_2 T_{V,2} k_2^2\mu_2^2 + i\mathcal{D}_{r,1} E_2 T_{\delta_T,2} k_2\mu_2 \\ &- i\mathcal{D}_{r,2} E_1 T_{\Phi,1} k_1\mu_1 + \mathcal{D}_{r,2} E_1 T_{V,1} k_1^2\mu_1^2 + i\mathcal{D}_{r,2} E_1 T_{\delta_T,1} k_1\mu_1 \end{aligned}$$

[emi] aberration + deflection at emission (D). at $\eta = \eta_*$, $\chi = \chi_*$:

$$-P_\perp \left(E_1 \mathcal{F}_2 T_{V,1} + E_2 \mathcal{F}_1 T_{V,2} \right)$$

D Verification ledger

Every check performed on the settled material, with its outcome. "Exact" means symbolic identity (sympy); numbers are maximal relative or absolute deviations. Code locations: V1 `code/V1_referee/`, C1 `code/C1_einstein/`, C2 `code/C2_observable/`, C3 `code/C3_harmonic/` (`run_all_tests.py`, log `test_log.txt`), N1 `code/N1_planar/`.

Exact observable (referee V1; Sec. 2)

V1-1 Lemma 2.1 in a generic Lorentzian metric $g = G_0 + G_{1\alpha} x^\alpha$ (random rational coefficients, no symmetry), generic unit $u(x)$, generic null p: geodesic-derived $\mathrm{d}E/\mathrm{d}\lambda$ vs $-pp\nabla u$, decomposition, $\omega_{\mu\nu} p^\mu p^\nu = 0$, final identity: exact.

V1-2 Lemma 2.2: energy and Euler equations from $\nabla \cdot T = 0$ for general ρ, P; equivalence to (14) iff $\rho = 3P + c$, $P = CT^4 - c/4$; for $P = w\rho$, $\rho \propto T^4$ the identities hold iff $w = 1/3$: exact.

V1-3 Off-shell identity $\mathrm{d}\ln E/\mathrm{d}l - \mathrm{d}\ln T/\mathrm{d}l + \sigma_{ee} = -[(\theta + 3u\cdot\partial \ln T)/3 + (a + h\partial \ln T)\cdot e]$ at a generic point: exact (residual -0.93 off shell in a random instance, 0 on shell).

V1-4a FRW radiation: $\sigma = 0$, $T_* E_o/E_e/T_o = 1$ symbolically for all p_i.

V1-4b Bianchi I $\mathrm{d}s^2 = -\mathrm{d}t^2 + \sum t^{2q_i} \mathrm{d}x_i^2$ with a radiation test fluid $u = \partial_t$: $\mathrm{d}/\mathrm{d}t[\ln E - \ln T] + \sigma_{ee} = 0$ symbolically; integrated for Kasner $(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ and non-Kasner $(\frac{1}{2}, \frac{1}{3}, \frac{1}{5})$, three directions, $t_e = 1 \to t_o = 5$: $|T_{\rm obs}^{\rm exact} - T_o e^{-S}| \lesssim 2 \times 10^{-31}$ with $O(1)$ direction dependence.

V1-4c Milne flow in Minkowski: $\theta = 3/\tau$, $\sigma = a = 0$, $T_{\rm obs} = T(o)$ isotropic: exact.

V1-5 Linear Poisson gauge: $\hat\theta, \hat a, \hat\sigma$ of (21); direct geodesic $\mathrm{d}\ln\hat E/\mathrm{d}\eta$ vs kinematic: agree; identity (23) off shell non-zero, on shell 0: exact. Sachs–Wolfe + Doppler + ISW with observer terms recovered.

V1-6 Tetrad rotation $R = \exp(\epsilon A)$ at $\ell = 2$ (exact spin-2 generators): $\sum_m |a_{\ell m}|^2$ invariant through $O(\epsilon^4)$ and $a^{(1)}$ invariant; $a^{(2)} \to a^{(2)} + 2Aa^{(1)}$; $C^{(22)}$ and $\mathrm{Re}\, C^{(13)}$ separately *not* invariant ($\sum |a_2|^2$: $321/25$ vs $329107/1225 - 8798\sqrt{6}/1225$) while $\frac{1}{4} C^{(22)} + \frac{1}{3} \mathrm{Re}\, C^{(13)}$ is $(107/60$ both). Dipole power invariant.

Exact observable, independent route (agent C2, milestone 1; Sec. 2.5)

C2-1 Kinematic identity $p^a p^b \nabla_a u_b = \sigma_{pp} + \theta(p\cdot p + E^2)/3 + Ea\cdot p$ for a generic (non-null) p, no equations of motion: zero at orders 1, 2, 3.

C2-2 Radiation identities (14) from $\hat{\nabla}_\mu \hat{T}^{\mu\nu} = 0$ alone (solved order by order for $\partial_\eta \delta_{T,n}$, $\partial_\eta v_n$): zero at orders 1, 2, 3.

C2-3 Integrand-level comparison of route (i) $[\ln(\hat{E}_o/\hat{E}_e) = -\int \Gamma^0_{kk} \mathrm{d}\eta + [\ln(-u\cdot k)]^o_e]$ with route (ii) $[\ln(\hat{T}_o/\hat{T}_e) - S]$ along an arbitrary perturbed ray $k^i = -n^i + \epsilon k^i_1 + \ldots$ with the non-affine geodesic equation substituted: residual $R \equiv 0$ at orders 1, 2 and 3 modulo conservation (14/70/340 terms before reduction).

C2-4 $-I_1 = -D[\delta_{T,1} + \Psi_1 - \hat{\mathbf{n}}\cdot v_1] + \Phi_1' + \Psi_1'$ with $I_1 = \sigma_{1,kk}$: exact (SPEC check (a)).

C2-5 Canonical triad (Gram–Schmidt w.r.t. $\hat{g}$, orthogonal to $\hat{u}$): $\hat{g}(e_a, e_b) = \delta_{ab}$, $\hat{g}(e_a, \hat{u}) = 0$ through order 3.

C2-6 Direction map (19): $k^\mu(\eta_o)$ null through order 3 (with $|\hat{\mathbf{n}}| = 1$); first-order geodesic right-hand side $G_1^i = -\partial_i(\Phi + \Psi) + 2n^i\, \hat{\mathbf{n}}\cdot\nabla(\Phi + \Psi) - n^i(\Phi' + \Psi')$ consistent with the null condition $\hat{\mathbf{n}}\cdot k_1 = -(\Phi + \Psi)$.

Einstein–fluid equations (agent C1; Secs. 3–4)

C1-1 Engine: $g\, g^{-1} = 1$ and $g_{\mu\nu}u^\mu u^\nu = -1$ to $O(\epsilon^3)$; background part of every $E_{\mu\nu}$ and of $\nabla_\mu T^{\mu\nu}$ vanishes identically.

C1-2 First-order equations reproduce the standard linearised Poisson-gauge equations (39); $\Psi_1 = \Phi_1$; $v_1^T = 0$; $\mathrm{curl}\, v_1 = 0$ identically.

C1-3 $\Phi_1'' + 4\Phi_1'/\eta + k^2\Phi_1/3 = 0$ verified; transfer functions (41) verified by substitution; conservation equations ($\nu = 0..3$) vanish identically on the solution; agreement with the lead's independent linear check (T_Φ, $\delta_T \to \zeta/3$, $V \to \eta\zeta/3$).

C1-4 Second-order SVT structure (44) verified component by component ($S_{0y} = S_{yz} = S_{xy} = 0$ by reflection symmetry; vector/tensor kernels free of $\sqrt{1-\mu^2}$ after projection).

C1-5 Contracted Bianchi identity $g^{\mu\alpha}\nabla_\alpha E_{\mu\nu} + g_{\nu\beta}\nabla_\mu T^{\mu\beta} \equiv 0$ to $O(\epsilon^3)$ (hence the constraints propagate at every order).

C1-6 Exact dilation family (36) solves $E_{\mu\nu} = 0$ to $O(\epsilon^3)$ for arbitrary A, B; $k = 0$ second-order equations fix only $\delta_2 = -2\Psi_2 + \dfrac{32}{9}c^2$; angle-averaged $k \to 0$ limit of the regular solution lies on the family.

C1-7 Vector sector: $\mathrm{curl}\, \hat{u}_i \neq 0$ at second order; relativistic Kelvin (45) holds and the algebraic ω_2 satisfies the ij-vector ODE identically.

C1-8 $\zeta^{(2)}$, $\zeta^{(3)}$ expansions (29)–(30); $\lim \eta^2 K_{S_{00}} = 32/3$; $K_Q(0)$ and $K^0_{\Phi_2}$ (31)–(32).

C1-9 Tensor source K_h identical to Kohri–Terada (8) / Baumann et al. (11)–(12) with $w = 1/3$, $\Phi = \Psi$: numerically exact.

C1-10 Superhorizon Φ_2, Ψ_2 (33)–(34) = Pitrou–Uzan–Bernardeau (3.12b,c) with $\mathcal{R}^{(2)} = -2\mathcal{R}^{(1)2}$: exact.

C1-11 Bartolo–Matarrese–Riotto II (7.30): local and gradient terms agree; velocity-quadrupole term differs ($27F$ vs our $3F$) because BMR use the late-time velocity; ours = PUB.

C1-12 Green's functions (48): $\mathcal{L}G = 0$, $G(\eta,\eta) = 0$, $\partial_\eta G|_{\eta'=\eta} = 1$ (re-checked for the rewritten form of G_Φ in this write-up).

C1-13 K_{S_Φ} and K_{3c,S_Φ} finite as $\eta \to 0$ (small-η expansions), so particular integrals vanish at $\eta = 0$; contracted vector/tensor sources vanish after angular integration.

C1-14 Squeezed limit $C_3(q \to 0) = -32/9$ equals the prediction of the exact dilation family: pass.

Harmonic machinery (agent C3; Sec. 5)

C3-1 Rule (A-s), $s = 0..3$; rule (A-λ), $\lambda = \pm1, \pm2$, in the k-frame and in a random frame (via D-matrices and spin-weighted harmonics): 10^{-14}.

C3-2 $C^{(11)}$ normalisation $2/\pi$ via the real-space route (monochromatic $\xi = \sigma^2|f|^2 j_0(k_0\chi|\hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2|) \Rightarrow C_\ell = 4\pi\sigma^2|f|^2 j_\ell^2$): 10^{-14}.

C3-3 (B1), both pairings, plain and gradient weights, correlated maps, vs exact real-space Wick + Legendre transform: 2.6×10^{-12}; Monte Carlo (400 Gaussian realisations, exact band-limited SHT): plain product 6.6%, gradient product 4.1%, $\langle X\rangle_{\rm sky}$·map 4.5%, cross term (ii) 3.1% (statistical, excludes any factor-2 error).

C3-4 Type-D Fourier $\leftrightarrow$ angular-gradient identity (two radii): 2.9×10^{-12}.

C3-5 Lensing control: $\frac{1}{4}C^{(22)} = \sum F^2/(2\ell+1)[C^\phi C^\Theta + C^{\phi\Theta}C^{\Theta\phi}]$ algebraically (2.5×10^{-16}); Monte Carlo of the Taylor-expanded remapping with correlated ϕ, Θ: (22) 2.3%, (13) $= -\ell(\ell+1)RC_\ell$ 4.6%; unlensed C_ℓ recovered 2.1%.

C3-6 SPEC prefactors re-derived: $C^{(22)}$ factor 2 and $C^{(13)}$ factor 3 confirmed; SPEC's $K_1^* K_3$ is the conjugate of $\langle a^{(1)} a^{(3)*}\rangle$ (irrelevant for Re).

C3-7 Type C: $K_{2,\ell m}$ finite $3j$ sums vs direct 2D quadrature (random k_1, k_2, two time nodes, degree 3): 1.1×10^{-14}; isotropy reduction random frame vs $\hat{\mathbf{K}} = \hat{z}$: 8.9×10^{-15}; (C1) $\equiv$ (B1) for product kernels (monochromatic): 3.7×10^{-15}; (C2) vs (B1): 3.6×10^{-15}.

C3-8 Rule (A-L): 3.2×10^{-14}; reduction (E1) with $b \le 3$, $a \le 2$, $\Delta\chi = 0$ and $\neq 0$: $\le 2.8 \times 10^{-15}$.

C3-9 Monochromatic normalisation $\langle\zeta^2\rangle = \sigma^2$: 5.9×10^{-7} (Gaussian approximation of the δ-function).

C3-10 Not verified numerically (formulas only, Wick counting consistent with the verified ones): (E2), (E3), $T_3^{(\rm ii)}$; the collapse of (E3) to (64)(ii) for a type-B m_2 passes analytically.

Non-perturbative arbiter (agent N1; planar symmetry)

N1-1 Exact planar GR + radiation solver (comoving-orthogonal gauge, lapse fixed algebraically by (14): $\alpha = e^{-\zeta}(AB^2)^{1/3}$, $T_{\rm phys} = 1/\alpha$), constraint violations $\sim 10^{-13}$: Theorem 2.3 verified non-perturbatively, $T_{\rm obs}/T_o = e^{-S}$ to 7×10^{-15} at $\epsilon = 0.1$ ($\Theta_{\rm rms} = 0.08$).

N1-2 Linear limit of the planar solver vs the Poisson-gauge transfer functions (41) through a derived gauge transformation: pass. Reference $a_{\ell 0}[\Theta^{(1,2,3)}]$ for a single plane wave are stored for the assembly check (Sec. 10).

Not yet checked / pending

Independent referee of the second- and third-order kernels (V3); dilation (constant-ζ) and gradient-mode invariance of the assembled map (SPEC 9(d)); closed forms of $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$ (C4); first-order $C_\ell^{(11)}$ closed form (C5); comparison of the assembled kernels with N1.

References

[1] R. K. Sachs and A. M. Wolfe, "Perturbations of a cosmological model and angular variations of the microwave background," Astrophys. J. **147**, 73 (1967).

[2] G. F. R. Ellis, "Relativistic cosmology," in *General Relativity and Cosmology*, Proc. Int. School of Physics "Enrico Fermi" XLVII, ed. R. K. Sachs (Academic Press, 1971); reprinted Gen. Rel. Grav. **41**, 581 (2009).

[3] D. S. Salopek and J. R. Bond, "Nonlinear evolution of long-wavelength metric fluctuations in inflationary models," Phys. Rev. D **42**, 3936 (1990).

[4] J. Maldacena, "Non-Gaussian features of primordial fluctuations in single field inflationary models," JHEP **0305**, 013 (2003), astro-ph/0210603.

[5] S. Weinberg, "Adiabatic modes in cosmology," Phys. Rev. D **67**, 123504 (2003), astro-ph/0302326.

[6] M. Mirbabayi and M. Zaldarriaga, "CMB anisotropies from a gradient mode," JCAP **1503**, 025 (2015), arXiv:1409.4777.

[7] P. Creminelli, C. Pitrou and F. Vernizzi, "The CMB bispectrum in the squeezed limit," JCAP **1111**, 025 (2011), arXiv:1109.1822.

[8] N. Bartolo, S. Matarrese and A. Riotto, "CMB anisotropies at second order I," JCAP **0606**, 024 (2006), astro-ph/0604416.

[9] N. Bartolo, S. Matarrese and A. Riotto, "CMB anisotropies at second-order II: analytical approach," JCAP **0701**, 019 (2007), astro-ph/0610110.

[10] C. Pitrou, J.-P. Uzan and F. Bernardeau, "The cosmic microwave background bispectrum from the non-linear evolution of the cosmological perturbations," JCAP **1007**, 003 (2010), arXiv:1003.0481.

[11] K. A. Malik and D. Wands, "Cosmological perturbations," Phys. Rept. **475**, 1 (2009), arXiv:0809.4944.

[12] K. Kohri and T. Terada, "Semianalytic calculation of gravitational wave spectrum nonlinearly induced from primordial curvature perturbations," Phys. Rev. D **97**, 123532 (2018), arXiv:1804.08577.

[13] D. Baumann, P. Steinhardt, K. Takahashi and K. Ichiki, "Gravitational wave spectrum induced by primordial scalar perturbations," Phys. Rev. D **76**, 084019 (2007), hep-th/0703290.

[14] J. R. Espinosa, D. Racco and A. Riotto, "A cosmological signature of the SM Higgs instability: gravitational waves," JCAP **1809**, 012 (2018), arXiv:1804.07732.

[15] T. H.-C. Lu, K. Ananda, C. Clarkson and R. Maartens, "The cosmological background of vector modes," JCAP **0902**, 023 (2009), arXiv:0812.1349.

[16] W. Hu, "Weak lensing of the CMB: a harmonic approach," Phys. Rev. D **62**, 043007 (2000), astro-ph/0001303.

[17] A. Lewis and A. Challinor, "Weak gravitational lensing of the CMB," Phys. Rept. **429**, 1 (2006), astro-ph/0601594.

[18] L. Boubekeur, P. Creminelli, G. D'Amico, J. Noreña and F. Vernizzi, "Sachs–Wolfe at second order: the CMB bispectrum on large angular scales," JCAP **0908**, 029 (2009), arXiv:0906.0980.

[19] Z. Huang and F. Vernizzi, "The full CMB temperature bispectrum from single-field inflation," Phys. Rev. D **89**, 021302 (2014), arXiv:1311.6105.

[20] G. W. Pettinari, C. Fidler, R. Crittenden, K. Koyama and D. Wands, "The intrinsic bispectrum of the cosmic microwave background," JCAP **1304**, 003 (2013), arXiv:1302.0832.

[21] C. Fidler, G. W. Pettinari, M. Beneke, R. Crittenden, K. Koyama and D. Wands, "The intrinsic B-mode polarisation of the cosmic microwave background," JCAP **1407**, 011 (2014), arXiv:1401.3296.

[22] C. Fidler, K. Koyama and G. W. Pettinari, "A new line-of-sight approach to the non-linear cosmic microwave background," JCAP **1504**, 037 (2015), arXiv:1409.2461.

[23] K. Tomita, "Second-order power spectra of CMB anisotropies due to primordial random perturbations in flat cosmological models," Phys. Rev. D **77**, 103521 (2008), arXiv:0712.2511.

[24] G. D'Amico, N. Bartolo, S. Matarrese and A. Riotto, "CMB temperature anisotropies from third order gravitational perturbations," JCAP **0801**, 005 (2008), arXiv:0707.2894.

[25] X. Gao, "On non-linear CMB temperature anisotropy from gravitational perturbations," arXiv:1005.1219.